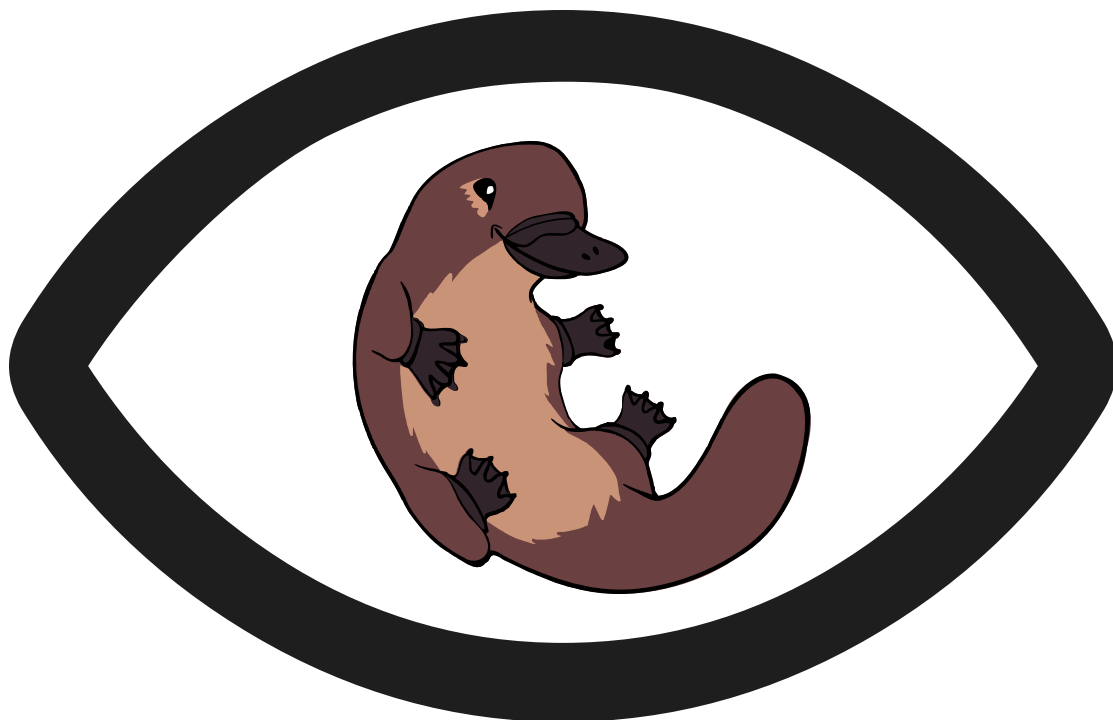




BORIS user guide

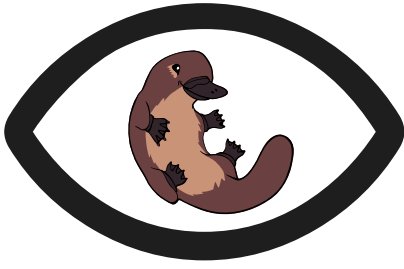
v. 9.14

Olivier Friard and Marco Gamba

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1. User Guide for BORIS, the Behavioral Observation Research Interactive Software

BORIS is a user-friendly application designed for event logging during video/audio coding and live observations. It is free and open source, and it runs on GNU/Linux, Microsoft-Windows and macOS with some limitations.

The official BORIS website is <https://www.boris.unito.it>.

This user guide applies to BORIS version **9.14**.

A [PDF version](#) of this guide is also available.

2. User guide

2.1 Installation

BORIS can be installed by following the instructions in the [download section](#) of the BORIS website.

All previous versions of BORIS are available in the [Releases section of the GitHub repository](#).

2.1.1 GNU / Linux

BORIS runs on several Linux distributions, including Ubuntu, Debian, Raspberry Pi OS, and ChromeOS.

See the [BORIS for Linux page](#) to install BORIS for Linux.

2.1.2 Microsoft-Windows

See the [BORIS for Microsoft-Windows page](#) to install BORIS for Windows.

Two versions are available: BORIS Portable and BORIS Setup.

2.1.3 macOS

BORIS for macOS is currently experimental. Please report any issues on GitHub.

See the [BORIS for macOS page](#) to install BORIS for macOS.

2.2 Starting BORIS

Once BORIS is installed, it can be launched by clicking on its icon.

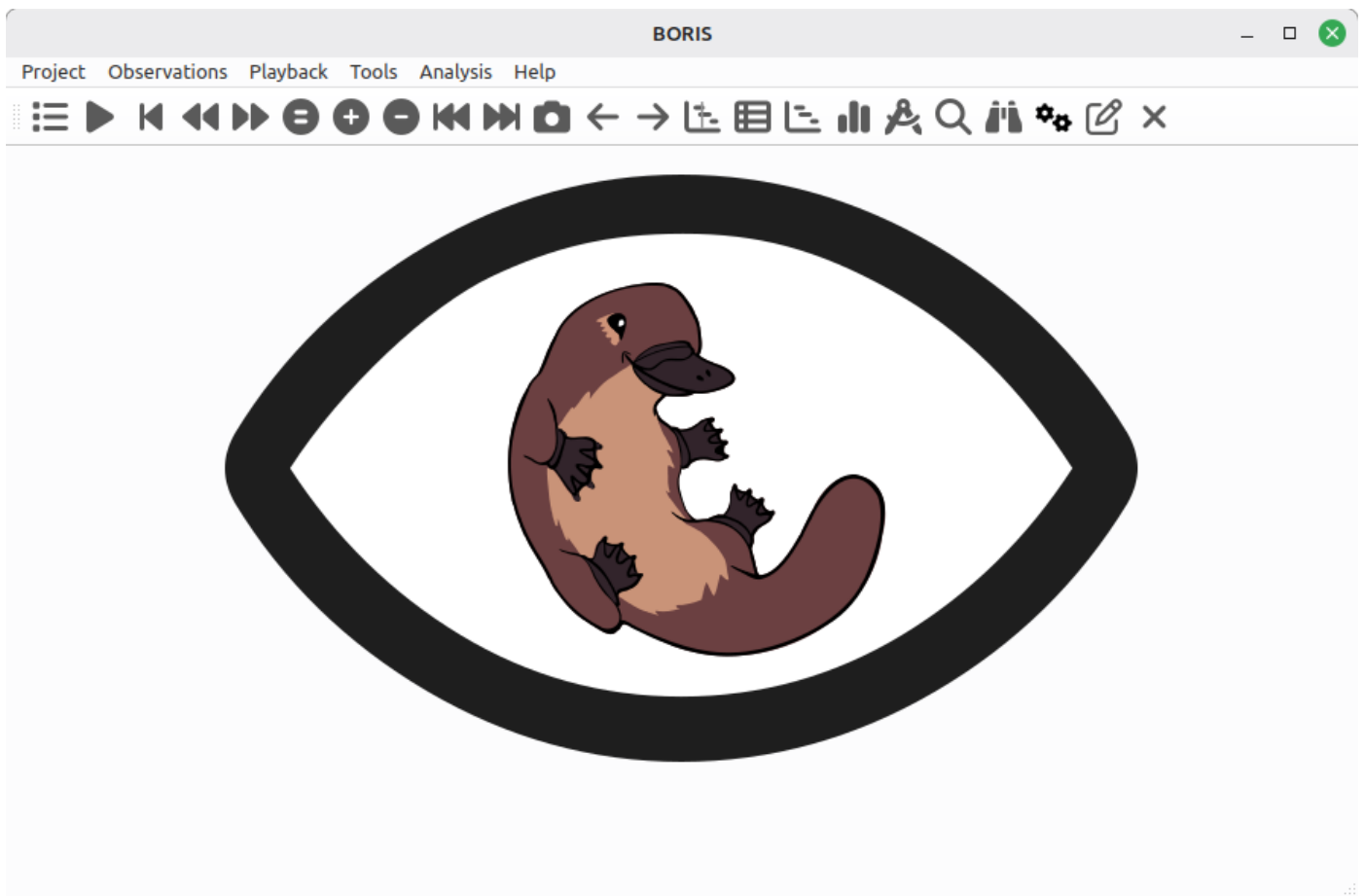
Warning for Windows users

BORIS does not yet use signed binaries, so you will need to allow the downloaded executable to run. If the option is not immediately visible, click "More info" in the warning dialog and then click "Run anyway".

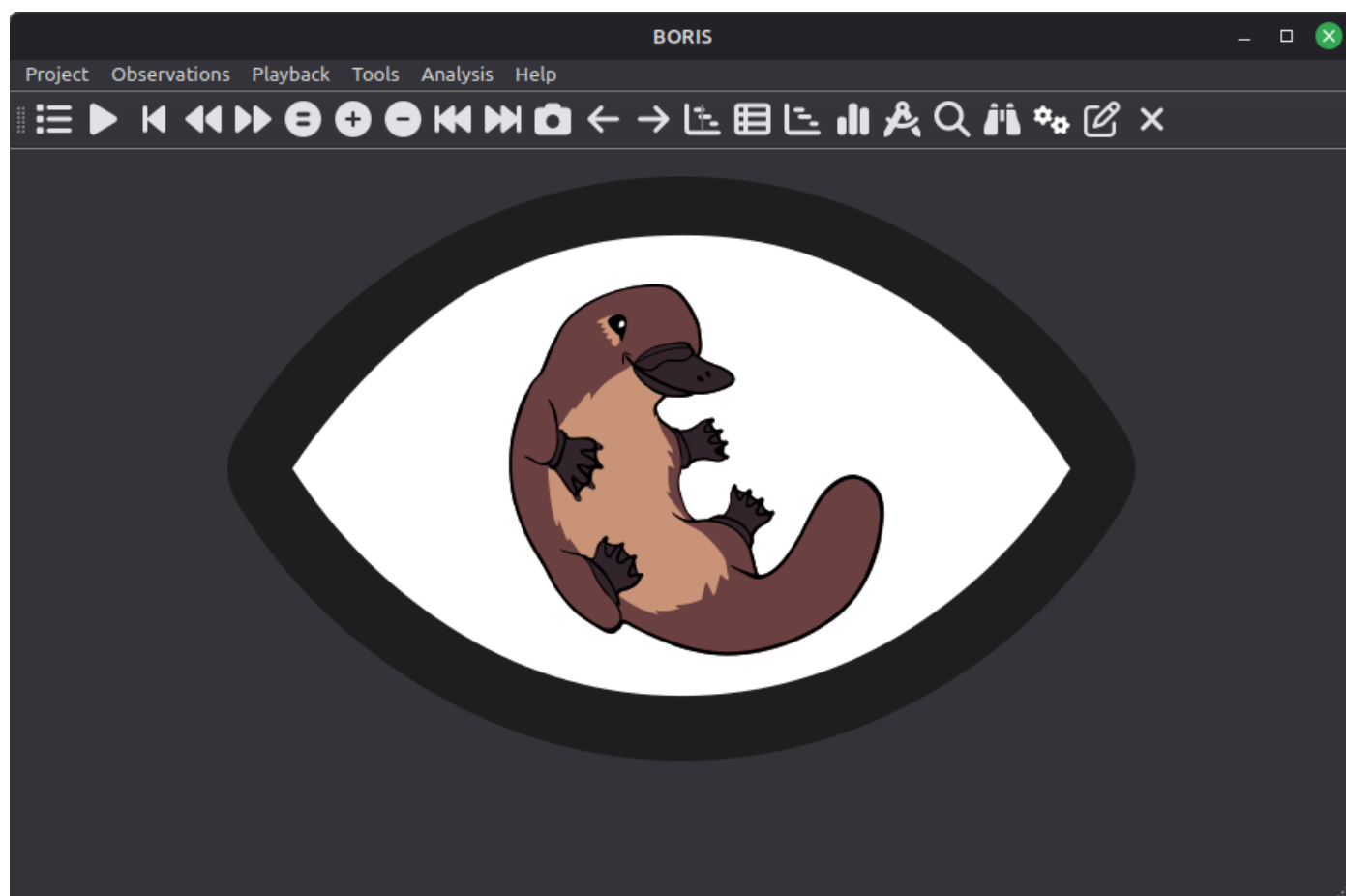
First launch

BORIS may take a little time to open the first time. Please be patient.

The BORIS main window will appear. At this stage, all toolbar commands are disabled except the Preferences button.



The BORIS main window



The BORIS main window - Dark theme

To launch BORIS from source code, refer to the [Run BORIS from source code](#) section.

2.3 Create a project

The BORIS project file serves as a container for all project-related information, excluding media files. It includes the **ethogram**, **independent variables**, **subject definitions**, **behavioral coding maps**, **converters**, and **observation** data. To save the project to your local file system, use **File > Save Project** or **Save Project As**

Additionally, you can activate the automatic backup feature in the [Preferences](#) section.

Very important

It is **EXTREMELY IMPORTANT** to perform regular backups of your project files to prevent the loss of data. While software can be reinstalled, your data might be irretrievably lost. Consider using an external drive and/or a cloud service for secure backup.

BORIS lets you create an unlimited number of projects, but only one project can be open at a time.

A video tutorial about creating a project is available at [this link](#).

To create a new project, select **File > New project**.

Enter the project name in the **Project name** field in the **Information** tab. Once the project has been saved, **Project file path** will show the full path to the project file.

Date is automatically set to the current date and time, but you can change it to match your media date and time, or any other value you prefer.

Description can contain any relevant information about your project, or it can be left empty.

Time format can be set to either **seconds** or **hh:mm:ss.mss**. This setting can be changed at any time under **File > Preferences** or by using the **Preferences** button in the toolbar.

new project

Information | Ethogram | Subjects | Independent variables | Behaviors coding map | Converters

Project name

Project date and time 2023-03-17 17:52:0 ▾

Project description

Project time format ☒ seconds ☐ hh:mm:ss.mss

Project format version: 7.0

Cancel OK

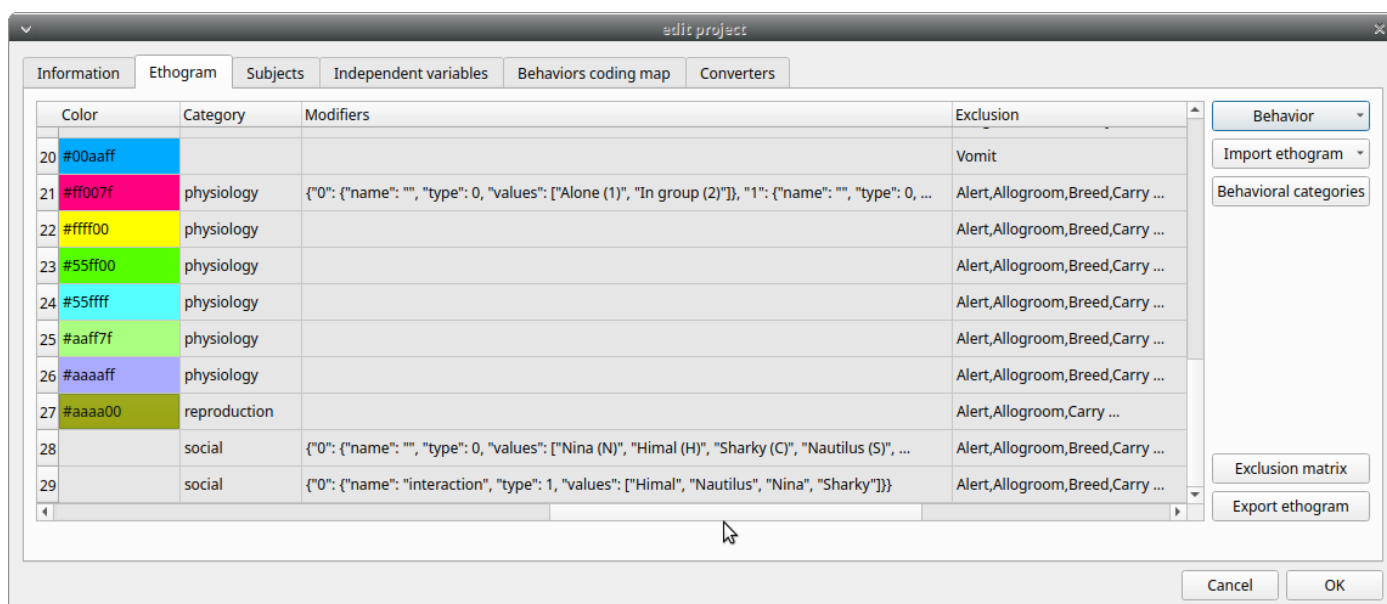
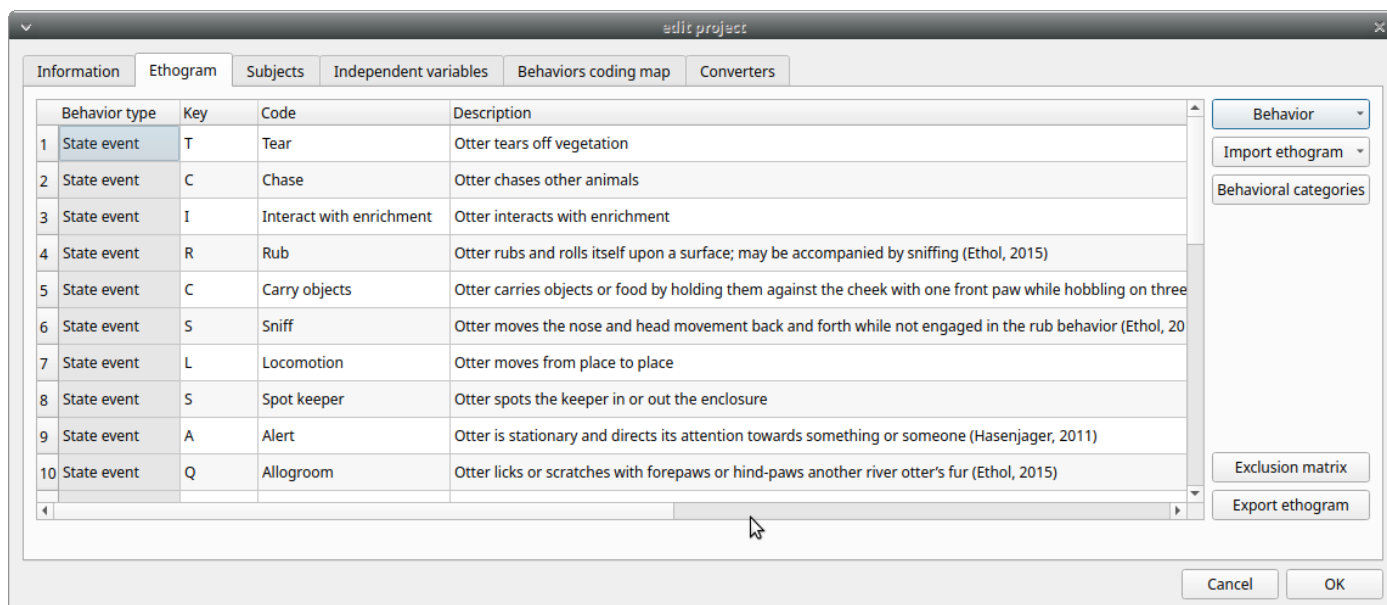
BORIS main window

2.3.1 Set an ethogram

See the [Wikipedia ethogram definition](#).

In the **Ethogram** tab, you can:

- create an ethogram from scratch;
- import an existing ethogram from another BORIS project;
- import an ethogram from a JWatcher global definition file (`.gdf`);
- import an ethogram from a plain text file or a spreadsheet file (`.xlsx` or `.ods`).



Set your ethogram from scratch

ADD A BEHAVIOR

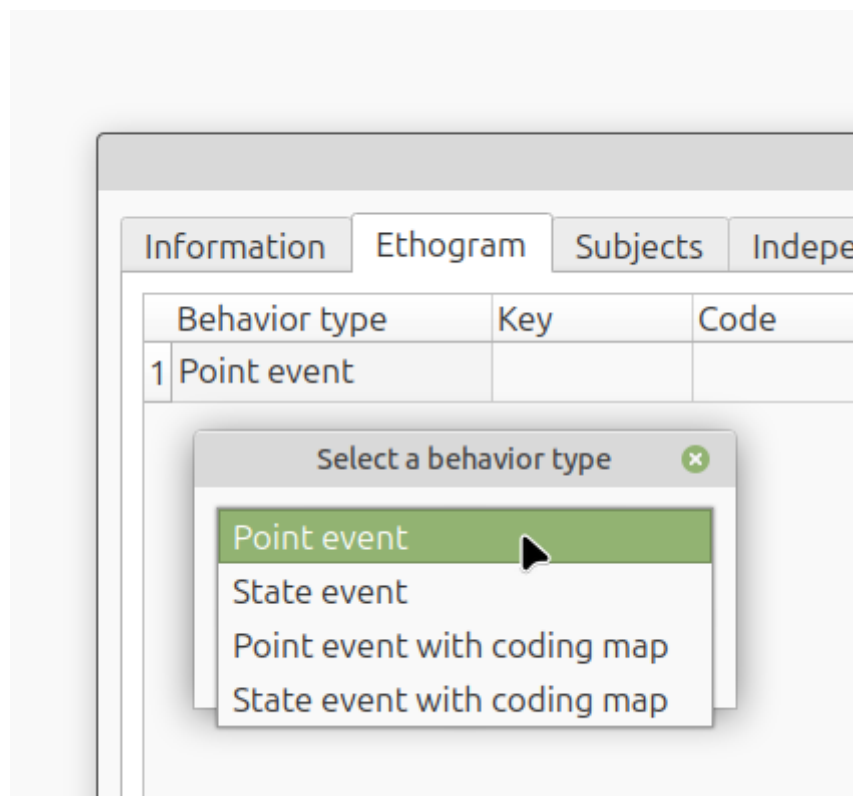
Click **Behavior > Add behavior** to add a new row to the **Ethogram** table. The behavior type is automatically set to **Point event**.

Cells with a gray background cannot be edited directly. You must double-click on them and then select a value.

The order of behaviors in the table can be changed by right-clicking the row of the behavior you want to move up or down.

SET THE TYPE OF BEHAVIOR

You can define **2 types** of behavior. Double-click the cell and select the behavior type:



- **Point event** behavior when the behavior has **no duration**.

The behavior is coded by pressing the assigned keyboard key (see below) or by double-clicking the corresponding row in the Ethogram table.

- **State event** behavior when the behavior has a **duration**.

The start and stop of the behavior are coded by pressing the assigned keyboard key (see below) or by double-clicking the corresponding row in the Ethogram table. These behaviors **must** have both a start event and a stop event; otherwise, an **UNPAIRED events** warning will appear when you close the observation or run an analysis.

- **Point event with a coding map**

a **Point event** that can be coded using a **coding map**.

- **State event with a coding map**

a **State event** that can be coded using a **coding map**.

You can switch between behavior types at any time by double-clicking the **Behavior type** cell. You can also add a **Coding map** to either a **State event** (**State event with coding map**) or a **Point event** (**Point event with coding map**). See the **Coding map** section for details.

An existing behavior can be duplicated using the **Clone behavior** button. Its code must then be changed. To remove a selected behavior, click **Remove behavior**. The **Remove all behaviors** button clears the **Ethogram** table. Both operations must be confirmed.

Behaviors can be sorted by clicking the Ethogram table headers. They cannot be sorted manually.

SET A KEY FOR THE BEHAVIOR (OPTIONAL)

For each behavior, you can assign a keyboard key in the **Key** column; this key will be used to code the corresponding events. You can choose whether you want to set a unique key for each behavior or use the same key for more than one behavior. If you assign the same key to more than one behavior, BORIS will pause coding and ask which behavior you want to record. The keys are **case-sensitive**.

Note

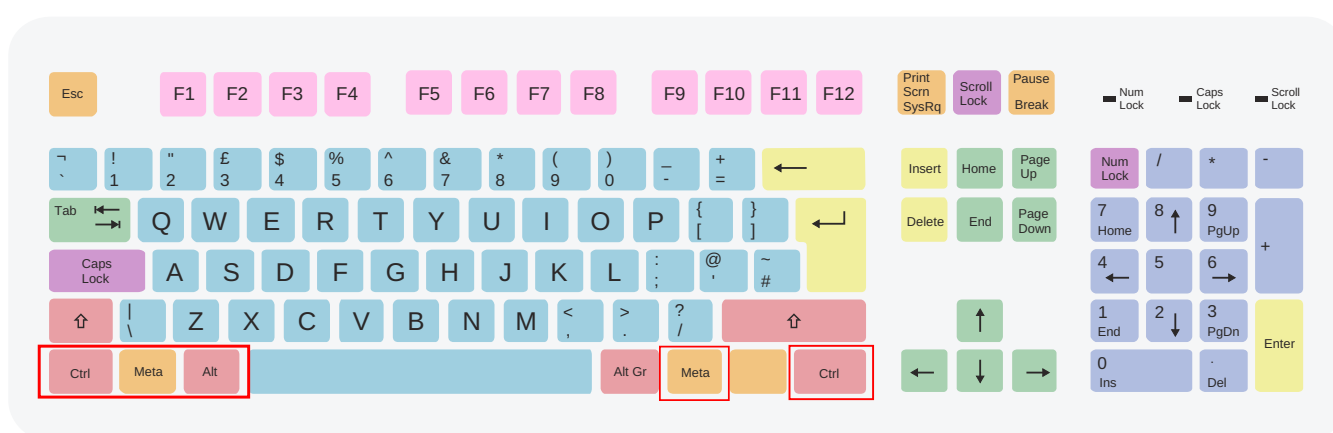
If your project was created with an older version of BORIS (< v.7), you can use **Convert keys to lower case** to convert all keys to lowercase. Otherwise, you will need to code the observation using uppercase keys.

If you open a project file created with a version earlier than v.7, BORIS will ask you to convert uppercase behavior and subject keys to lowercase.

Important

Do not use the / and * keys! They are reserved for the frame-by-frame mode.

From **version 9.9** onward, you can assign key combinations to trigger a behavior, including **Ctrl**, **Alt**, and **Meta** (also known as the "Windows key") in combination with a single key.

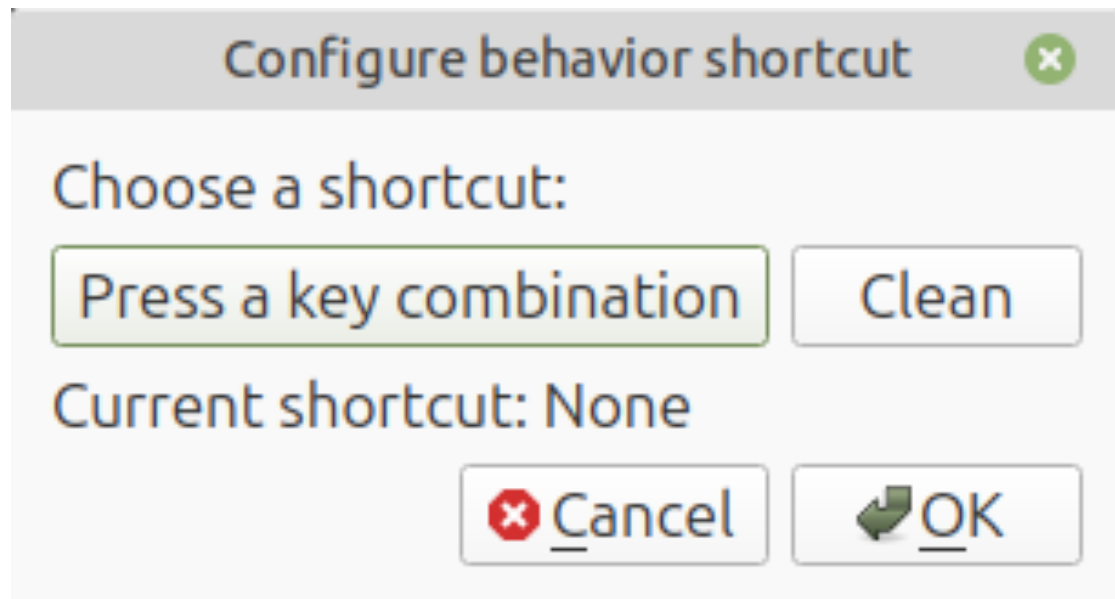


- | | | | |
|---|---|--|---|
| Character keys | Enter and editing keys | Navigation keys | Numeric keypad |
| Modifier keys | System and GUI keys | Function keys | Lock keys |

ISO keyboard with 105 keys

For the **Meta** key (orange in the keyboard diagram), see https://en.wikipedia.org/wiki/Windows_key.

To set a key combination, double-click the **Key** cell. The following window will appear:



Press the key combination you want to assign. A single modifier key, such as , , , or , is not a valid key combination.

ctrl
ctrl shift

alt
ctrl shift meta

Meta
shift meta

You can also use the shortcut tool to select a single key.

Examples of combinations:

 + ,  + ,  + ,  +  + ,  +  + 
 + 

Function keys (pink in the keyboard diagram above) can also be assigned to a behavior from  to .

Examples of combinations including function keys:

 + ,  +  + ,  + 

Note

BORIS uses keyboard shortcuts for several functions, such as:

- saving a project ( + )
- creating a new observation ( + )
- starting an observation ( + )
- close the current observation (++)ctrl+q+)
- editing an observation ( + )
- opening the list of observations ( + )
- Add an event ( + )
- jumping playback ( + )
- jumping backward ( + )
- jumping to a specific time ( + )
- ...

None of these shortcuts can be used to code behaviors.

Important

Consider that assigning a key combination to a behavior makes the project unusable in BORIS versions earlier than **9.9**.

Note

On MacOS on the ethogram table:

- The **control** key will be indicated as **Meta**
- The **option** key will be indicated as **Alt**
- The **command** key will be indicated as **Ctrl**



ISO keyboard with 105 keys

SET A CODE FOR THE BEHAVIOR (MANDATORY)

In the **Code** column, you have to add a unique code for each behavior. Duplicated codes are not accepted and BORIS will warn in red about duplicates on the bottom left of the **Ethogram** tab. The code can be an alphanumeric string (which must not include the pipe character |).

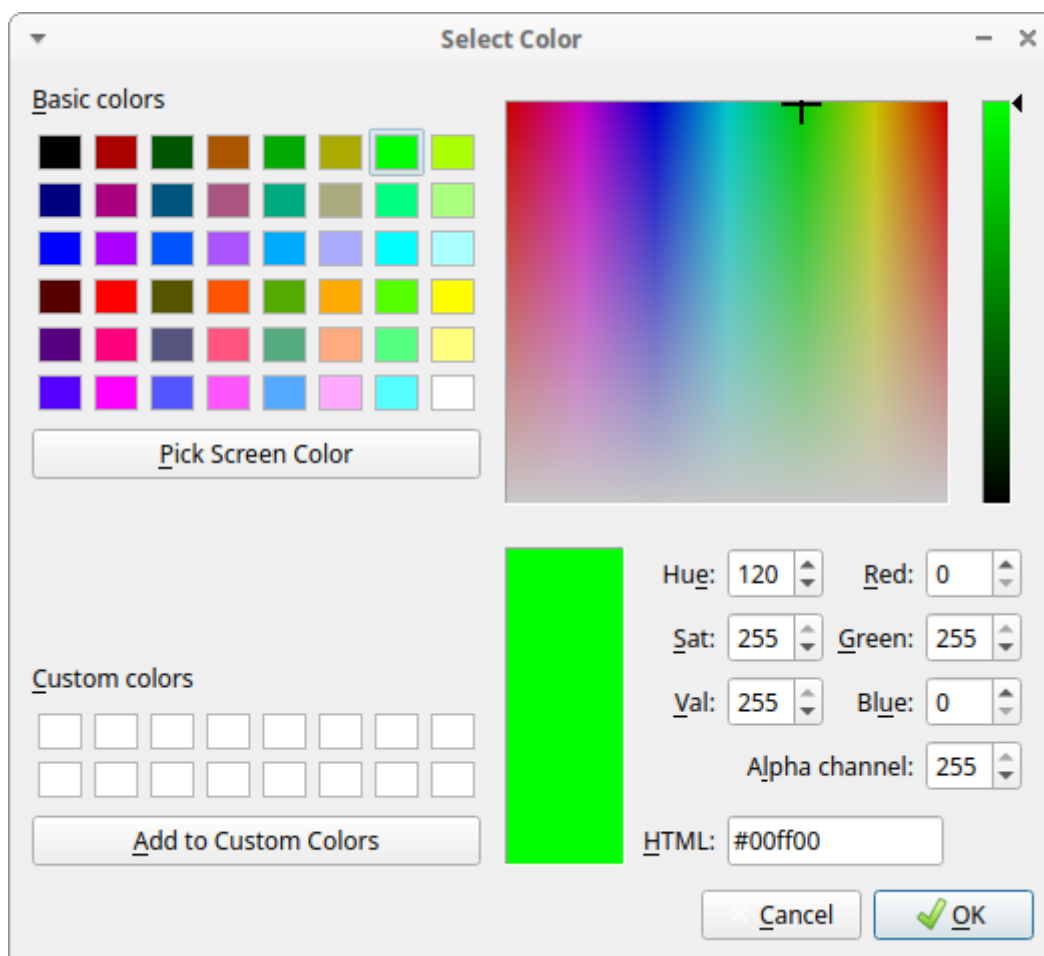
GIVE A DESCRIPTION FOR THE BEHAVIOR (OPTIONAL)

The **Description** of your behavior is optional. The **Description** column can be useful to add information about a specific behavior, its characteristics (e.g. to standardise observation between different users) or to refer to external information (e.g. reference to a previous ethogram).

The columns with a grey background (**Behavior type**, **Color**, **Category**, **Modifiers**, **Exclusion**, **Modifiers coding map**) cannot be edited directly.

SELECT A COLOR FOR THE BEHAVIOR (OPTIONAL)

The **Color** column allow to select a color for the behavior. This color will be used for plotting events. Double-click on the cell and select the color you want to associate to the behavior.



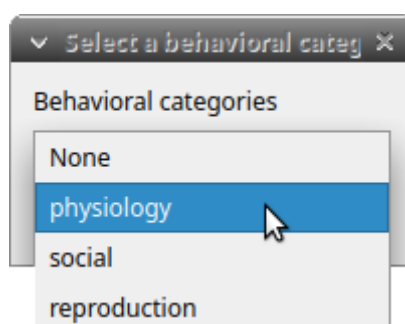
Select the color to associate to the behavior

CATEGORIES OF BEHAVIORS

Defining categories of behaviors can be useful for the analysis of coded events (for example the [time budget analysis](#)).

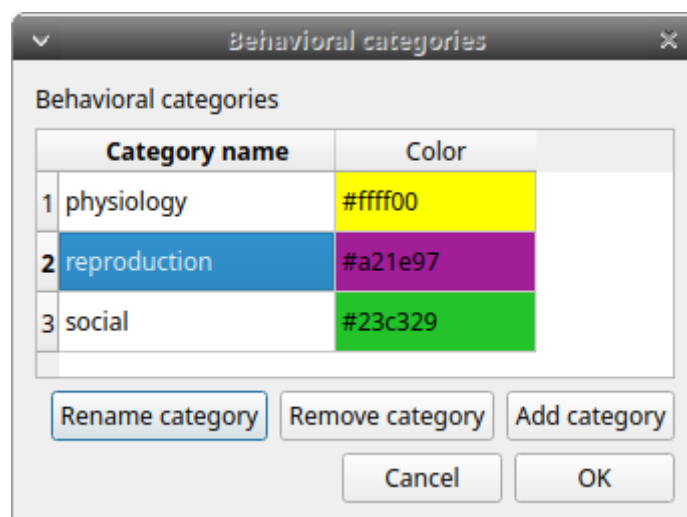
The **Category** column allow you to include the behavior to a predefined behavioral category.

Double-click on the cell and select the behavioral category for the behavior.



Choose a behavioral category for the behavior

To add, remove or rename a behavioral category, click the **Behavioral categories** button. A color can also be associated to a behavioral category.



Behavioral categories manager

SET THE MODIFIERS

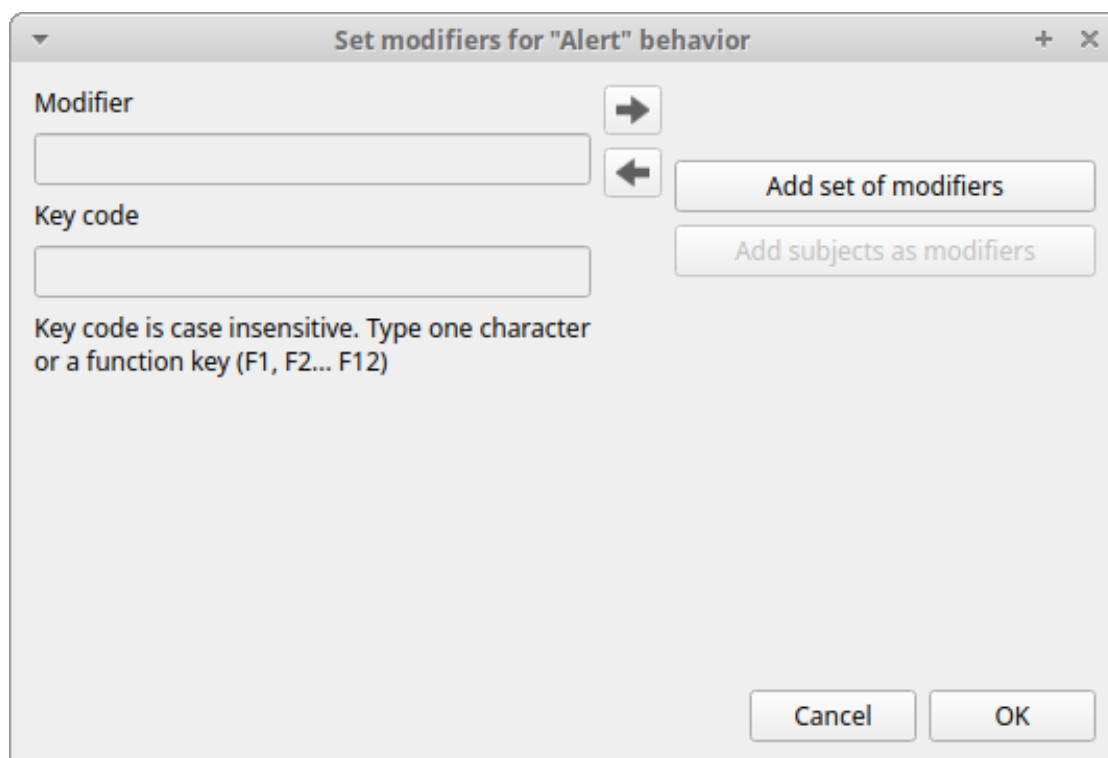
Modifiers can be used to add attributes to a behavior. A single behavior can have two or more modifiers attached (e.g. the behavior **play** may have **solitary** or **social** as modifiers). The use of modifiers can be convenient to significantly reduce the number of keys and simplify the behavioral coding.

4 types of modifiers are available: **Single selection**, **Multiple selection**, **Numeric** and **Value from external data file**:

- the **Single selection** type will allow the observer to select only **one** modifier for the current behavior.
- the **Multiple selection** type will allow the observer to select one or more modifiers for the current behavior.
- the **Numeric** type will allow the observer to input a number. For example a distance of interaction.
- the **Value from external data file** type will save the value of a variable from an external data file.

In BORIS modifiers can also be added in different modifier sets (e.g. **play social** may have a modifier set (#1) for **brothers** and another (#2) for **sisters**). In the case of using sets of modifiers, you can select one/more modifier for each set.

To add modifiers to a behavior, you need to double-click the **Modifiers** cell corresponding to the behavior you want to add the modifiers to. The following window will show up:



The dialog box is titled "Set modifiers for 'Alert' behavior" and has standard window controls (minimize, maximize, close) in the top right corner. It contains two input fields: "Modifier" and "Key code". Between these fields are two arrow buttons: a right-pointing arrow above a left-pointing arrow. To the right of the "Key code" field are two buttons: "Add set of modifiers" and "Add subjects as modifiers". At the bottom right are "Cancel" and "OK" buttons. A note below the "Key code" field states: "Key code is case insensitive. Type one character or a function key (F1, F2... F12)".

Set modifiers for "Alert" behavior

Modifier

Key code

Key code is case insensitive. Type one character or a function key (F1, F2... F12)

Add set of modifiers

Add subjects as modifiers

Cancel OK

Modifiers configuration

Click the **Add a set of modifiers** button:

Modifiers configuration

Select the modifier type using the **Modifier type** combo box. You have to choose between **Single selection**, **Multiple selection Numeric** and **Value from external data file**.

Single selection and Multiple selection modifiers

Set a name for the new modifiers set by typing it in the **Set name** edit box. Setting a modifiers\' set name is not mandatory.

Within a set of modifiers, you can add a modifier by writing the modifier in the **Modifier** edit box. You can choose a shortcut (one character - case sensitive) to this modifier (optional). Then press the **right-arrow** button to add the new modifiers to the set.

Set modifiers for "Carry objects" behavior

Modifier: Stones

Key code: 1

Key code is case insensitive. Type one character or a function key (F1, F2... F12)

Set #1

Set name: Type of object

Modifier type: Single selection

Values:

Move modifier up, Move modifier down, Remove modifier, Add set of modifiers, Remove set of modifiers, Move set left, Move set right, Add subjects as modifiers, Cancel, OK

Modifiers configuration

To modify a modifier, select it and press the **left-arrow** button, edit the modifier and press the **right-arrow** button.

A modifier can be removed by pressing the **Remove modifier** button.

After adding all modifiers the window will appear like this:

Modifiers configuration

All defined subjects can be added as modifiers using the **Add subjects as modifiers** button. This can help in case of coding the interactions between subjects for example.

The modifiers can be loaded from a plain text file Use the **Load modifiers from file** button.

The modifier position into the modifiers\' set can be manually set using the **Move modifier up** and **Move modifier down** buttons. The modifiers can be sorted alphabetically (use the **Sort modifiers** button).

You can add and/or remove sets using the buttons **Add set of modifiers** and **Remove set of modifiers**.

The position of a modifiers\' set can be customized (using the **Move set left** and **Move set right** buttons)

Modifiers cannot contain the following characters: (), ` ~ !

Example of a **multiple selection** modifiers set:

Set modifiers for "Play on the ground" behavior

Modifier → Set #1

Key code ← Set name

interaction

Modifier type

Multiple selection

Values

Nina (N)
Himal (H)
Sharky (C)
Nautilus (S)

Move modifier up Move modifier down

Remove modifier

Add set of modifiers Remove set of modifiers

Move set left Move set right

Cancel OK

Modifiers configuration

Many values can be selected together.

Example of 2 sets of modifiers:

Set modifiers for "Eat" behavior

Modifier Set #1 Set #2

Key code Set name

Key code is case insensitive. Type one character or a function key (F1, F2... F12)

Modifier type
Single selection

Values
Alone (1)
In group (2)

Move modifier up Move modifier down

Remove modifier

Add set of modifiers Remove set of modifiers

Move set left Move set right

Cancel OK

Modifiers configuration

*Modifiers configuration***Numeric modifier**

Set a name for the new set by typing it in the **Set name** edit box. Setting a modifiers' set name is not mandatory.

When a **Numeric** modifier will trigger, BORIS will ask the observer for a numeric value.

Value from external data file modifier

This modifier can be used to record the value of a variable coming from an external data file (defined during the creation of the observation).

You have to define the variable name in the **Variable name** edit box. This is mandatory and the name of the variable **must** be the same than the variable defined in the observation.

See [External data files](#)

modifier value from external data file

Click **OK** to save modifiers in the **Ethogram** table.

Set the exclusion matrix

The occurrence of an event (State or Point) can exclude the occurrence of a state event. This can be set using the **Exclusion matrix** window, which can be opened clicking on the **Exclusion matrix** button. BORIS will ask for including **Point events** or not and a new **Exclusion matrix** window will open.

Exclusive behavior may be selected by checking on the corresponding checkbox in the automatically-generated matrix. We suggest to work on the **Exclusion matrix** when all the behaviors have been added to your ethogram.

All behaviors can be excluded by a particular behavior by selecting the corresponding entire row (click on the row header of the behavior) and by clicking on the **Check selected** button. You can also uncheck all behaviors by selecting the **Uncheck selected** button.

[illegible]

Example of an exclusion matrix

For example in the previous figure, the **Alert** behavior will exclude the following behaviors: **Allogroom**, **Breed**, **Carry objects**, **Chase** ...

During the observation, the excluding event will stop all the current excluded state events one millisecond before the occurrence of the event.

Set the Modifiers coding map

If the behavior is defined as a **Point event with coding map** or a **State event with coding map** you can associate a **Modifiers coding map** to select the modifiers from a map.

Import an ethogram from an existing project

Behaviors within an ethogram can be imported from an existing BORIS project (.boris) using the **Import ethogram > from a BORIS project** button. BORIS will ask to select a BORIS project file and whether imported behaviors should replace or be appended to the **Ethogram** table. Imported behaviors will retain all the previously defined behavior parameters (namely Behavior type, Key, Code, Description, Modifiers and Exclusion information).

Import an ethogram from a spreadsheet file

Behaviors can be imported from a spreadsheet file using the **Import ethogram > from spreadsheet file (XLSX/ODS)** button.

The first row of your spreadsheet (header) must contain the following labels. The order is not mandatory:

- Behavior code
- Behavior type
- Description
- Key
- Behavioral category
- Excluded behaviors

Behavior code is mandatory, the others fields can be empty.

Optional fields can be added:

- Color
- Modifiers (JSON)

BORIS will ask to select a spreadsheet file (by default: .xlsx or .ods) and whether imported behaviors should replace or be appended to the **Ethogram** table. The missing information for the imported behaviours have to be redefined.

Import an ethogram from a plain text file

Behaviors can be imported from a plain text file using the **Import ethogram > from text file** button. The fields must be separated by TAB, comma (,) or semicolon (;). All rows must contain the same number of fields.

The first row of your plain text file must contain the following labels. The order is not mandatory but respect the case:

- Behavior code
- Behavior type
- Description
- Key
- Behavioral category
- Excluded behaviors

Behavior code is mandatory, the others fields can be empty.

Example of a plain text ethogram definition:

```
Behavior type,Behavior code,Key,Behavioral category,Description,Excluded behaviors
state event,Play,p,,Play on the garden,s
point event,Sleep,s,,Subject is sleeping,p
```

BORIS will ask to select a plain text file (by default: *.txt *.csv *.tsv) and whether imported behaviors should replace or be appended to the **Ethogram** table. The missing information for the behaviours imported from text file have to be redefined.

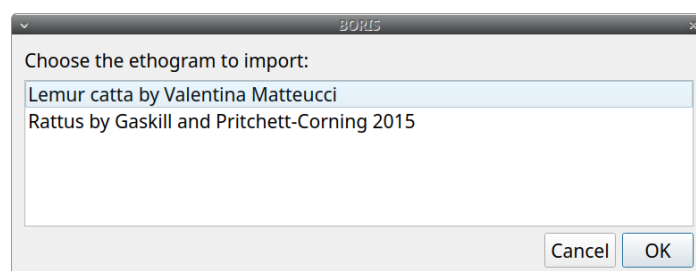
Import an ethogram from a JWatcher global definition file (.gdf)

Behaviors can be imported from a JWatcher global definition file (.gdf) using the **Import ethogram > from JWatcher** button. BORIS will ask to select a JWatcher file (.gdf) and whether imported behaviors should replace or be appended to the **Ethogram** table. Behavior type and exclusion information for the behaviours imported from JWatcher have to be redefined.

Access to the BORIS ethogram repository

This function can be activated by clicking the **Import ethogram > from the BORIS repository** button.

A list of available ethograms will open and an ethogram can be loaded in the current project.

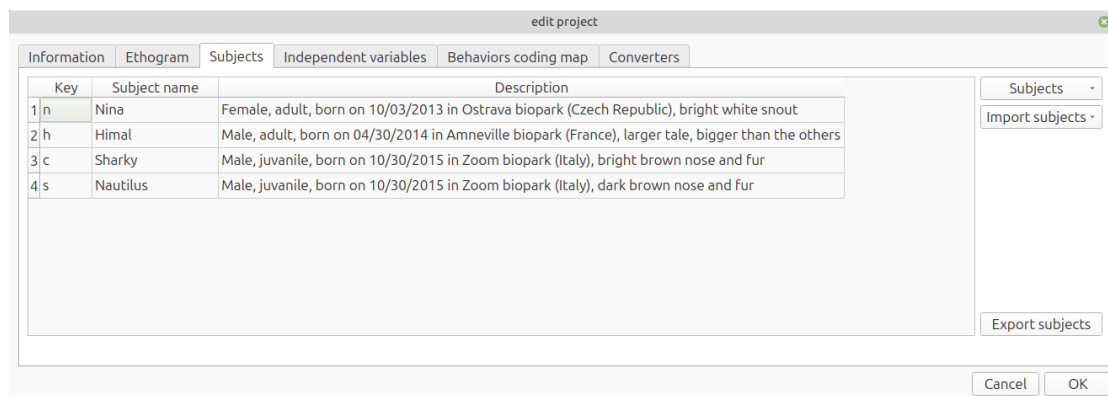


BORIS ethogram repository

Export the ethogram

The entire ethogram can be exported in various formats (TSV, CSV, XLSX, ODS, HTML). See **File > Edit project > Ethogram tab > Export ethogram**

2.3.2 Define the subjects



configuration of subjects

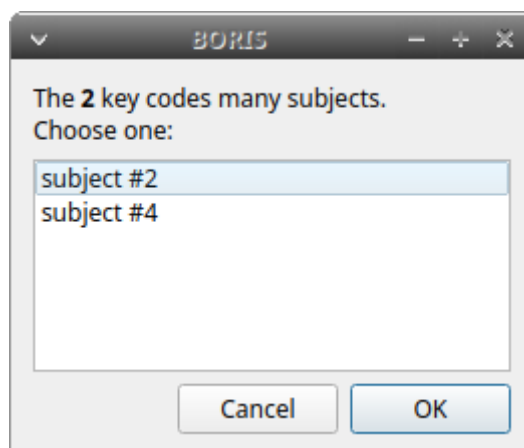
BORIS allows coding behaviors for different subjects within a single observation. The **Subject** table allows the selection of subjects using a **Key** of the keyboard. , **Subject name** (e.g., **Kanzi**), and **Description** (e.g., male, born on October 28, 1980).

Set a key for the subject (optional)

With the subjects defined in the previous figure, pressing **n** will set **Nina** as the focal subject for behavioral coding. Pressing **n** again will deselect **Nina** and set the focal subject to **No focal subject**.

The key definition is not mandatory. In this case, you will have to select the current subject from the subjects list with a double-click.

The keys are **case-sensitive** and the same key can be used to select more than one subject. In this case a dialog will show up and will allow to select



Choose a subject

The definition of one or more subjects is not mandatory. Addition, removal and sorting of the subjects follows the same logic of the **Ethogram** table (see [Set your ethogram from scratch](#) for info).

From **version 9.9** onward, it is possible to assign a combination of keyboard keys to select a subject, allowing the use of **Ctrl**, **Alt** and **Meta** (also known as "Windows key") keys in combination with a single key.

Refer to [Set a key for the behavior](#) for details.

Important

Consider that assigning a key combination to a subject makes the project unusable in BORIS versions earlier than **9.9**.

Note

If your project was created with a version of BORIS earlier than 7, you can use **Convert keys to lower case** to convert all keys to lowercase. Otherwise, you will need to code your observations using uppercase keys.

The subjects can also be imported from an existing BORIS project: use the **Import Subjects from a BORIS project** button.

Import subject from a spreadsheet

The subjects can be imported from a spreadsheet (Google spreadsheet, Microsoft-Excel, LibreOffice Calc).

The spreadsheet must contain one subject by row and have to be organized as above:

- 1st column: Subject key (One character - Case sensitive - Optional)
- 2nd column: Subject name (mandatory)
- 3rd column: Description of subject (optional)

Select all cells of your spreadsheet (**Ctrl** + **A**), copy to clipboard (**Ctrl** + **C**). Click the **Import from clipboard** button.

Note

If you open a project file created with a version older than v.7 BORIS will ask you to convert the upper case behavior and subject keys to lower case.

2.3.3 Define the Independent variables

Label	Description	Type	Predefined value	Set of values
1 Location	Location where observations where made	text	44° 55' 59" N - 7° 25' 18" E	
2 Weather	Meteorological conditions	value from set	sun	sun,rain,clouds
3 Temperature	Average temperature of the day (°C)	numeric		
4 Visitors	Visitors per day	numeric		

Label:

Description:

Type:

Predefined value:

Buttons: Add variable, Remove variable, Import variables from a BORIS project, Cancel, OK

Independent variables

BORIS allows adding information about the observation using **Independent variables**. This can be used to specify factors that may influence the behaviors (e.g. group composition, temperature, weather conditions) but will not change during a single observation within a project. Each independent variable can be defined by a **Label** (e.g. weather), a **Description** (e.g. weather conditions), a **Type** (*text*, *numeric*, *value from set* or *timestamp*).

The values of a set are defined in the **Set of values** column separating the available values with a comma (,). Please note that the first value of the set will be selected by default. It should be useful to define a NA value as first value of every set.

The values for the independent variables will be asked when creating a new observation. Addition, removal and sorting of the independent variables follows the same logic of the **Ethogram** table (see **Set your ethogram from scratch** for info). The independent variables can also be imported from an existing BORIS project using the **Import Variables from a BORIS project**.

Label	Description	Type	Predefined value	Set of values
1 Location	Location where observations were made	text	44° 55' 59" N - 7° 25' 18" E	
2 Weather	Meteorological conditions	value from set	sun	sun,rain,clouds
3 Temperature	Average temperature of the day (°C)	numeric		
4 Visitors	Visitors per day	numeric		

Label:
 Description:
 Type:
 Predefined value:
 Set of values (separated by comma):

Example of an independent variable (Weather) defined as "set of values"

The predefined value must be contained in the set of value.

2.3.4 Converters' table

Converters are used for plotting external data when the timestamp values are not expressed in seconds. Converters can be written by the user, loaded from file or loaded from the repository of the BORIS web site (<http://www.boris.unito.it/static/converters.json>).

Name	Description	Code

Name:
 Description:
 Python code:

Converters tab

Add new converter

Converters can be written using the Python 3 programming language.

The **INPUT** variable will be loaded with the original value of the external data file (for example 01:22:32).

The **OUTPUT** variable must contain the converted value in seconds (the dot must be used for decimal separator).

Example code to convert HH:MM:SS format to seconds:

```
h, m, s = INPUT.split(':')
OUTPUT = int(h) * 3600 + int(m) * 60 + int(s)
```

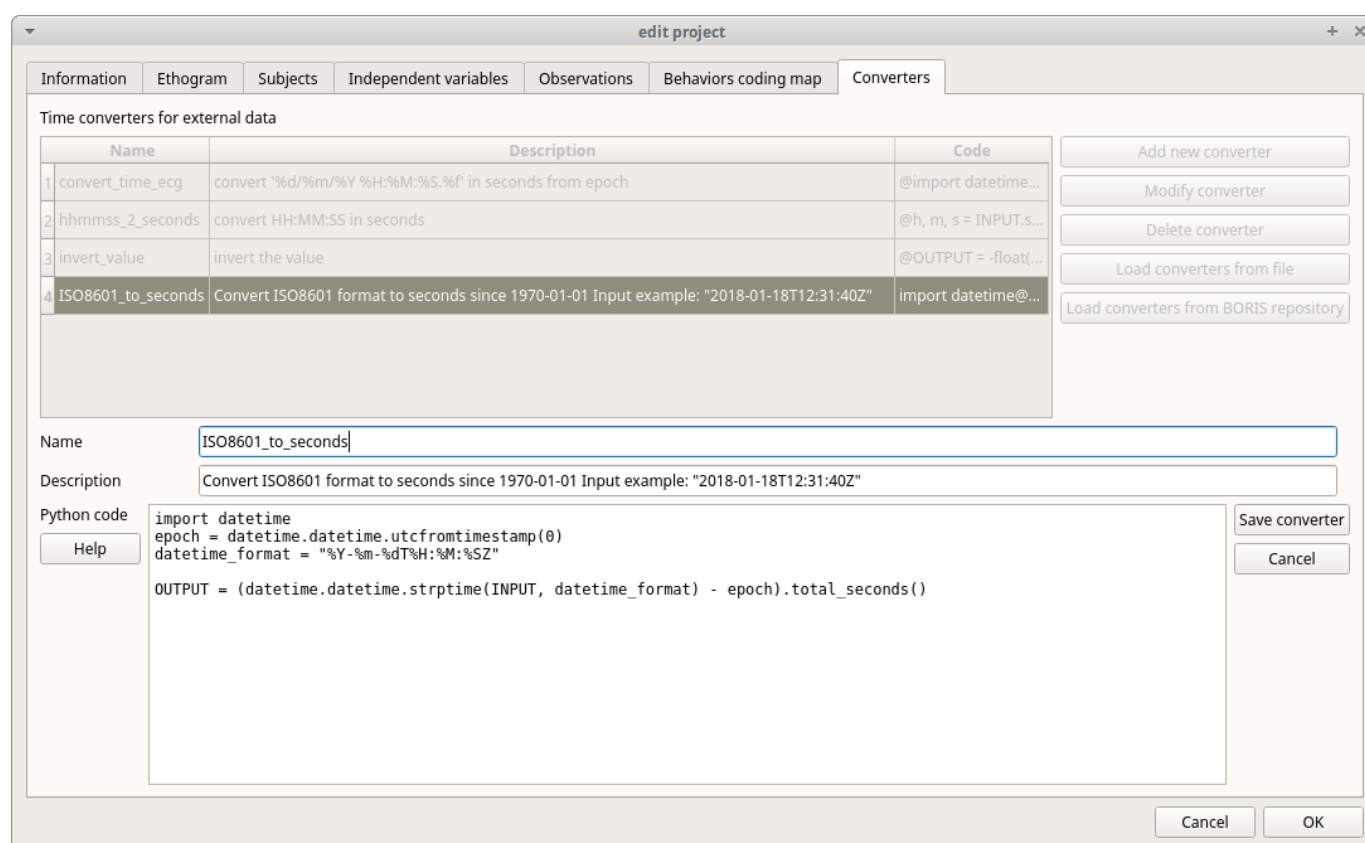
The Python function **strptime()** from the **datetime** module can be useful for converting time values: <https://docs.python.org/3/library/datetime.html#strptime-strptime-behavior>

Example code to convert a date in ISO 8601 format to seconds using the **strptime()** function:

```
import datetime
epoch = datetime.datetime.utcfromtimestamp(0)
datetime_format = "%Y-%m-%dT%H:%M:%SZ"

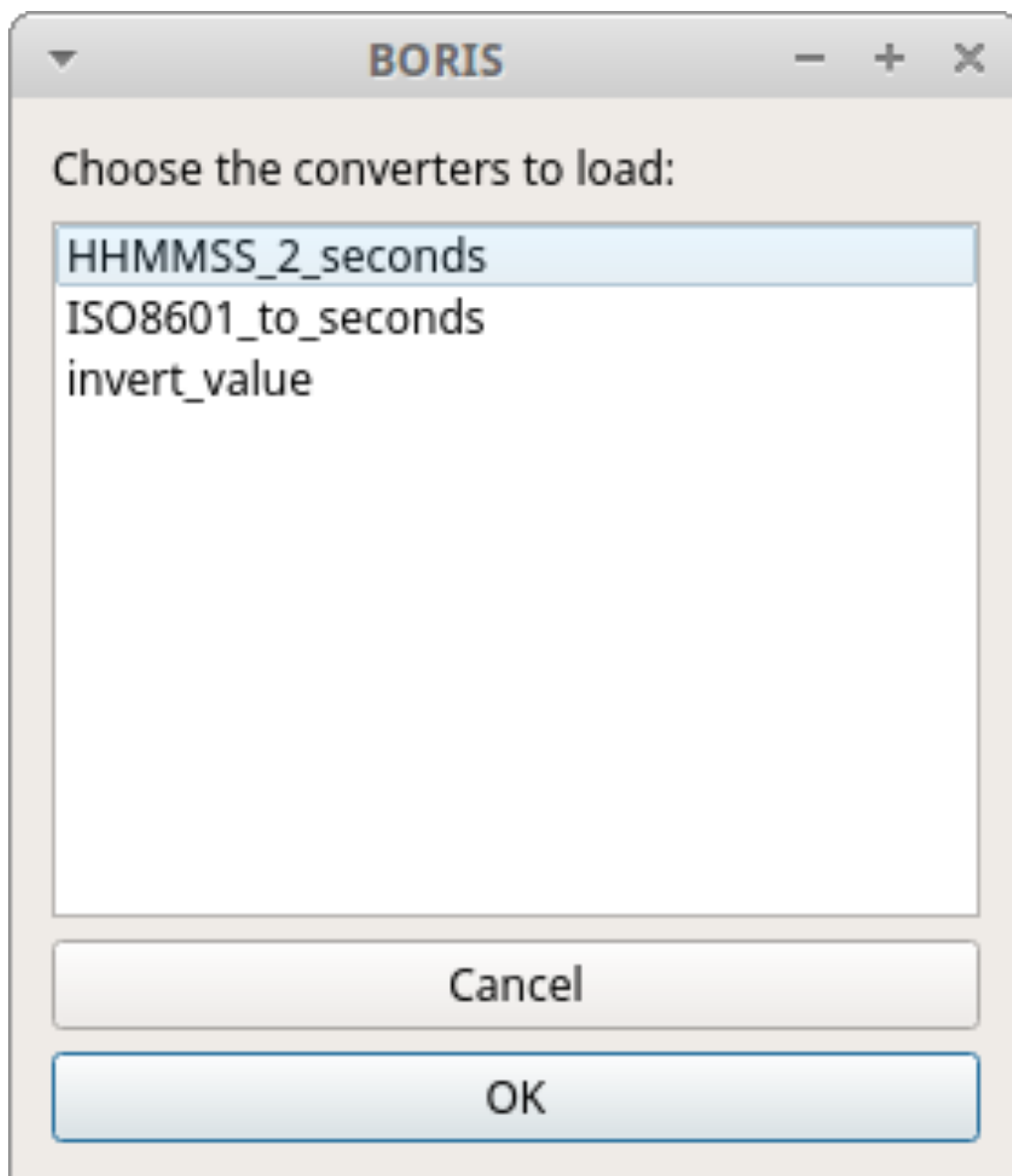
OUTPUT = (datetime.datetime.strptime(INPUT, datetime_format) - epoch).total_seconds()
```

File > Edit project > Converters

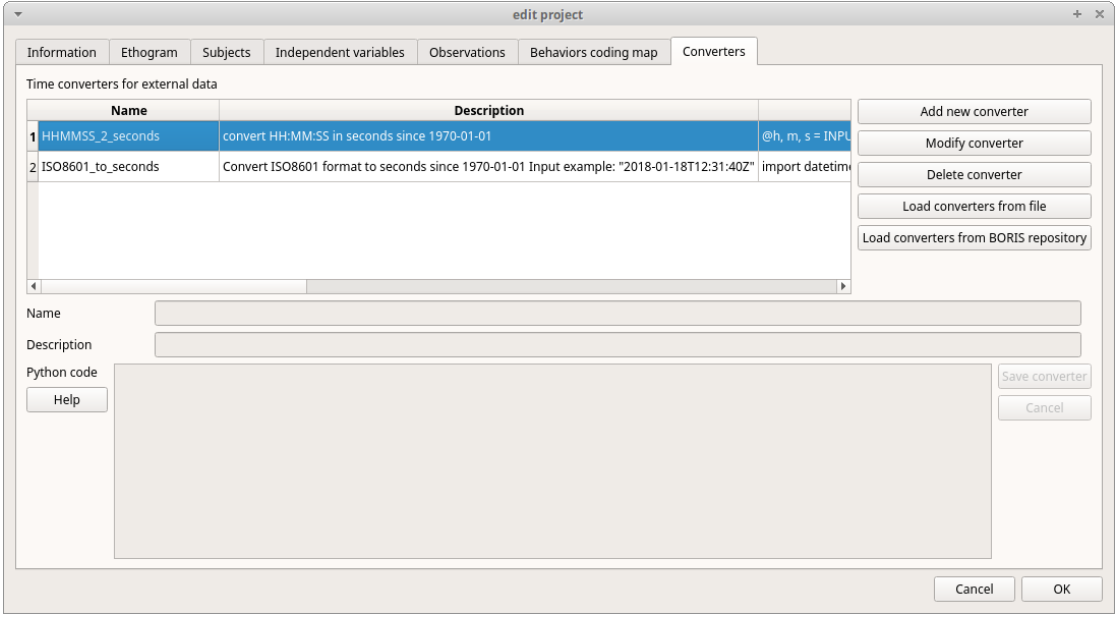


Load converters from BORIS web site

Click **Load converters from BORIS repository** and select the converters to be added to your project.



Converters selection from repository



Converters tab with 2 converters defined

Writing a converter

See [Converters for external data values](#)

The converters loaded in your project can then be selected for converting timestamps (or other values) in external data files.

See [Converters](#)

2.4 Create a new observation

A video tutorial about making an observation is available at [this link](#).

To create a new observation, you must first [create a new BORIS project](#) or open an existing one.

Click **Observations > New observation** to open the **New observation** window.

New observation window

This window lets you enter several observation details:

- a mandatory **Observation id** (must be unique across all observations in the open project);
- **Date**, which is automatically set to the current date and time, but can be changed to match your media date and time or any other value you prefer.
- **Description**, which can contain any relevant information about your observation, or it can be left empty.
- **Independent variables** (e.g. to specify factors that may influence the behaviors but will not change during the observation within a project). See the [independent variables](#) section for details.
- **Time offset**. BORIS allows you to specify a time offset that can be added to or subtracted from the media timecode.
- The **Limit observation to a time interval** option can be used to limit the observation to an arbitrary time interval.

You must then indicate whether you want to create an observation based on **pre-recorded media (audio / video)** or a **live observation**.

2.4.1 Live observation

During a live observation, BORIS shows a timer that is used to record event times.

Click on the **Live observation** radio button to create a live observation.

New live observation

Start from current time

If you want the timer to start from the current time, select the **Start from current time** checkbox.

Set a live observation to start from current time

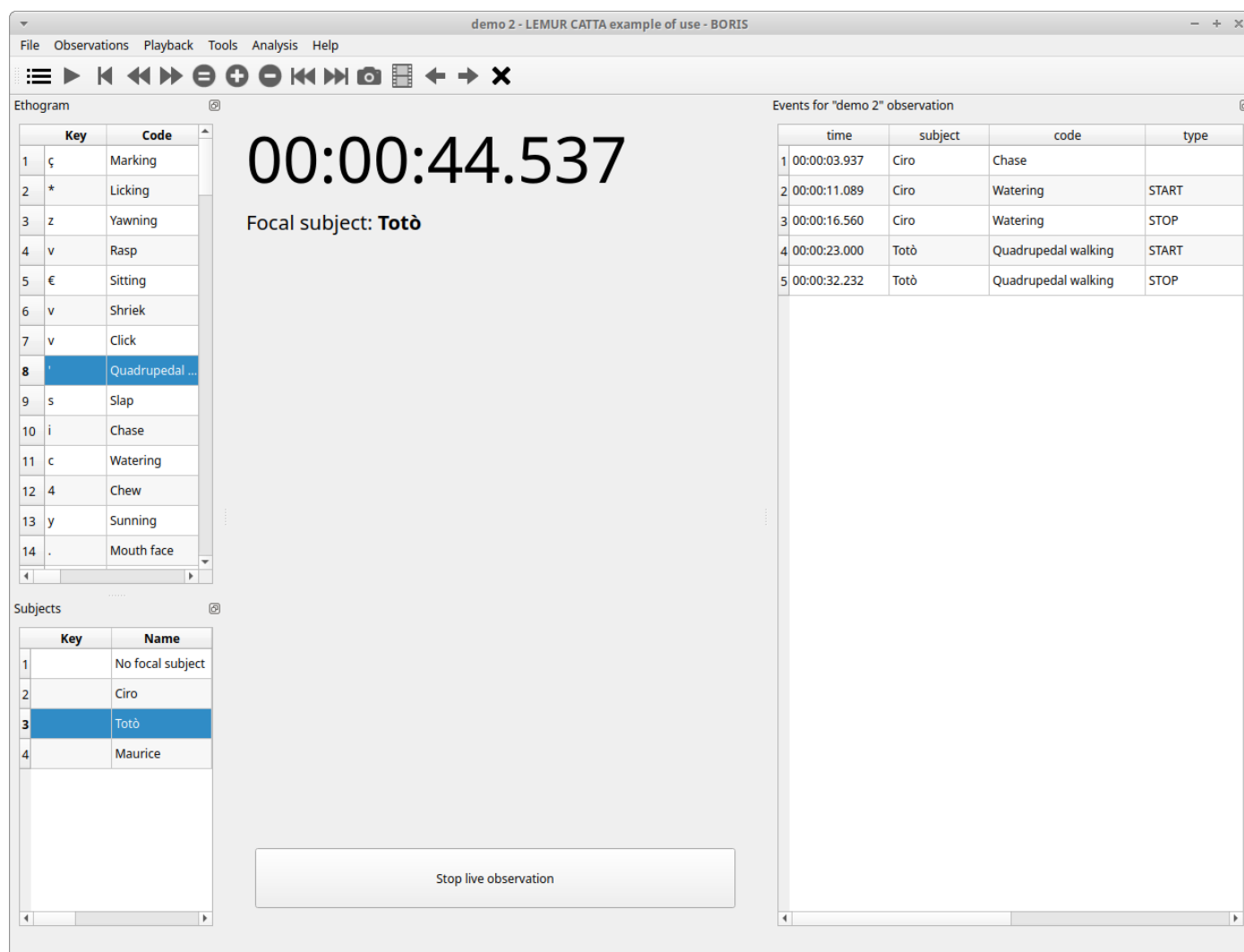
If the **Day time** option is checked, the start time will be the computer's current time when you press the **Start** button.

If the **Epoch time** option is checked, the start time will be the number of seconds since January 1, 1970 (1970-01-01). See [Unix time](#) for details. This option is useful for long observations lasting several days or for observations that start before midnight and end after it.

Start the observation

Click the **Start** button to begin the live observation or **Save** to save it in the [Observations list](#).

The main window during a live observation will look like this:



The main window during a live observation

See the [Live observations](#) section to start coding.

2.4.2 Observation from media file(s)

Select **Observation from media file(s)** to create an observation based on one or more media files.

Observation from media files

The **Observation from media file(s)** section contains 2 tabs: **Media files** and **Data files**.

Click the **Media files** tab and add one or more media files using the **Add media** button. You have 4 options:

- **with absolute path**: the full media file path will be recorded in the project
- **with relative path**: the media file path will be recorded relative to the BORIS project file location. This option is useful if you need to move your BORIS project file to another computer.
- **from directory with absolute path**: all media files found in the directory will be added to the playlist (the full media file path will be recorded in the project)
- **from directory with relative path**: all media files found in the directory will be added to the playlist (the relative media file path will be recorded in the project)

Information about the selected media file will be extracted and displayed in the media list: media file path, media duration, number of frames per second (FPS), the presence of a video stream, and the presence of an audio stream.

Player	Offset (seconds)	Display	Path	Duration	FPS	Video	Audio
1	0	Nothing	lemurs_catta.mp4	00:01:00.327	29.97	True	True

Media files

You can choose to use the media file name as **Observation id** by clicking the **Use media file name as observation id** button.

The drop-down list in the first column lets you choose a player (up to a maximum of 8). If you want to observe more media files simultaneously, you must use consecutive players starting from 1. See the example below:

Media files		Data files					
	Player	Offset (seconds)	Path	Duration	FPS	Video	Audio
1	1	0	video1.mp4	00:02:59.960	25	True	False
2	2	0	video2.mp4	00:02:59.960	25	True	False
3	3	0	video3.mp4	00:02:59.960	25	True	False

Media files tab

If you need to synchronize 2 or more videos, you can use the **Offset** column to indicate when the second player should start. For example, if the video loaded in the second player starts 15 seconds after the first video, enter **15** in the **Offset** cell. If the second video starts before the first player, you can enter a negative value in the **Offset** cell.

If you want to play several videos sequentially, select the same player (#1) for all loaded videos. This means that an event occurring at time $t \sim x \sim$ in the media file queued as second (e.g. second_video.mp4) in the playlist will be recorded at time $t \sim 1 \sim + t \sim x \sim$ (where $t \sim 1 \sim$ is the duration of the first media file, e.g. first_video.mp4).

The **Remove selected media** button can be used to remove all the selected media files.

BORIS can play all media types supported by the MPV player.

The **Use media file name as observation id** button sets the first media file name as the **observation id**.

Spectrogram and waveform visualization

BORIS can display audio information — such as the sound spectrogram and/or waveform — during media coding. In the **Display** column, choose the type of plot you want to visualize.

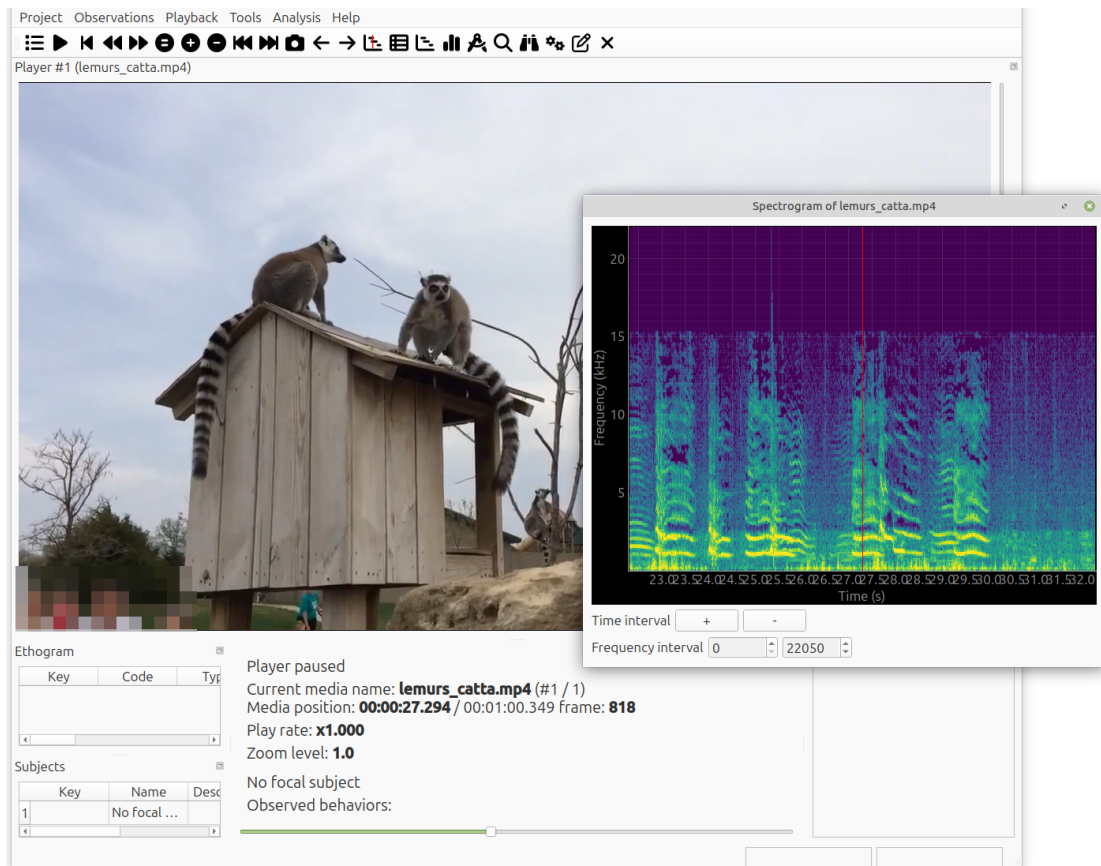
Media files

Data files

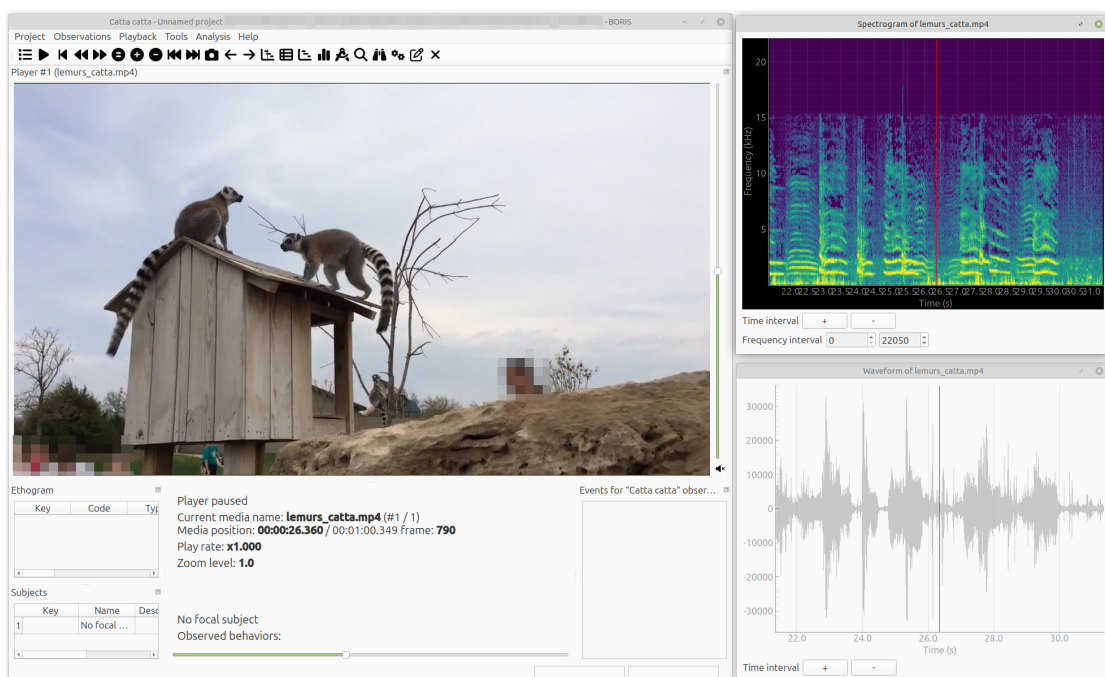
Player	Offset (seconds)	Display	Path	Duration	FPS	Video	Audio	
1	1	0	Nothing	lemurs_catta.mp4	00:01:00.327	29.97	True	True
		spectrogram						
		waveform						
		spectrogram,waveform						

Select display

The spectrogram visualization will be synchronized to the media position during the observation.



Spectrogram visualization



Spectrogram and waveform visualization

Stop all state events between media files

If your media files are not contiguous, you can enable the **Stop ongoing events between successive media files** option to automatically stop state events between media files. This option can be useful, for example, when coding videos from camera-trap systems.

Add media ▾
Remove selected media
Use media file name as observation id

Scan sampling every (s) 0
Image display duration (s) 1

☐ Stop ongoing state events between successive media files

Stop state events between media files

External data files


Note

At this time, only 2 external data series can be plotted with your media file

You can select one or more external data files to be plotted synchronously with your media. Click the **Data files** tab and use the **Add data file** button to select a data file.

Media filesData files

Data files to plot

	Path	Columns to plot	Plot title	Variable name	Converters	Time interval (s)	Start position (s)	Subtract first value	Color
1	submarine_7col_colum.csv	4,5				60	0	True	b-

External data file table

The data files must be plain text files with at least **2 columns** separated by a comma or a TAB character. One column must contain a timestamp that will be used to synchronize the plot with the media. The sampling rate can be variable.

Example of a plain text data file with 5 columns separated by comma (,):

```
Display,X Pos,Y Pos,Start Time (secs),Pupil Diameter
1,864,509,549.233,0.00295773451216519
1,863,505,549.25,0.00281810853630304
1,863,503,549.266,0.00287826382555068
1,861,502,549.283,0.0030536837875843
1,858,501,549.3,0.00308083021081984
1,856,499,549.316,0.00306266942061484
1,854,499,549.333,0.00305776367895305
[...]
```

In the above example the 4th column contains the timestamp and the 5th the value to be plotted.

Input the index of the column containing the timestamp and the index of the column containing the value to be plotted. The two indices must be separated by a comma (,). Click **OK** to close the window.

View data

View first and last rows of **submarine_7col_column.csv** file

1	2	3	4	5	6	7
latitude	longitude	mission_msecs	time	depth	altitude	total_depth
		38:58.1	14:38:58	0.049345195	12.4144278	12.463773
		38:59.9	14:39:00	0.05376504	12.4467316	12.50049664
		39:01.9	14:39:02	0.038770307	12.3420744	12.38084471
		39:03.8	14:39:04	0.123303935	12.3966513	12.51995524
		...	...	...	...	...
		28:23.9	20:28:24	0.106934205	13.6745434	13.78147761

Descriptive statistics

	latitude	longitude	depth	altitude	total_depth
count	10658.000000	10658.000000	10658.000000	10658.000000	10658.000000
mean			5.933995	9.564539	15.498534
std			4.145813	4.814589	2.888231
min			-0.714470	0.462451	7.194569
25%			2.528735	5.489861	14.128838
50%			5.578776	9.264291	15.267252
75%			9.220541	13.094842	16.482292
max			18.622009	28.562168	30.686549

Enter the column indices to plot (time, value) separated by comma (,)

4,5

CancelOK

Selection of columns (time, value)

A new row will be added in the data files table.

Data files to plot									
	Path	Columns to plot	Plot title	Variable name	Converters	Time interval (s)	Start position (s)	ubstract first valu	Color
1	eye_tracker.tsv	4,5	Eye tracker	Pupil diameter		60	0	True	b-

Data file table

You can modify/complete the following parameters by directly typing in the table cells:

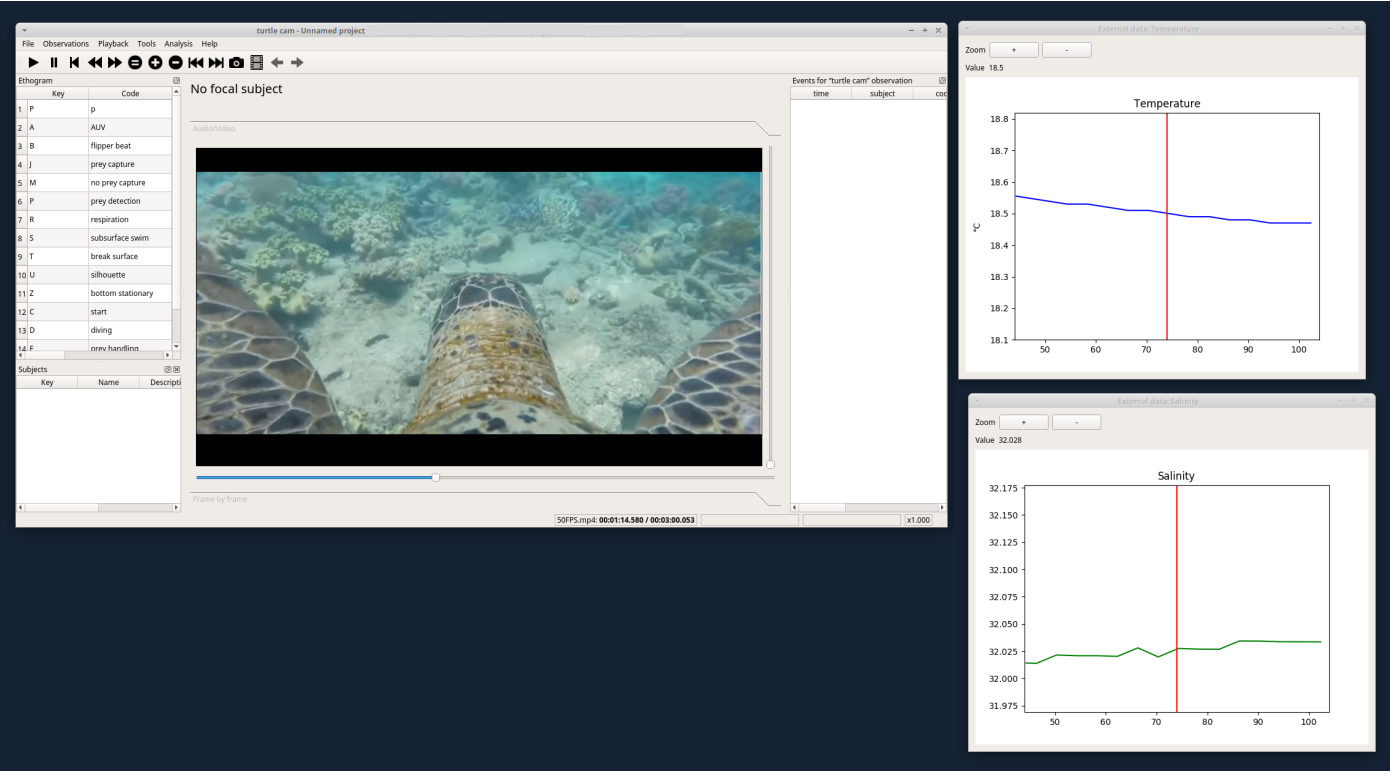
- Columns to plot
- Plot title: the title of the plot
- Variable name
- Converters: Used if the timestamp is not expressed in seconds (see below for details)
- Time interval: The time interval that will be plotted (in seconds)
- Start position: the start position of data for synchronisation with the media (in seconds)
- Substract first value: if the timestamp does not start with a 0 value you can choose to substract the first value to all timestamp values.
- Color: the color of the color

NOTE : if you want to record the value of the plotted variable in a modifier of a behavior (see [Value from external data file modifier](#)) the modifier must have the same **variable name**.

You can check if the data from file can be correctly plotted by using the **Show plot** button. If the data are compatible you will see a plot otherwise you will obtain a message with an explanation.

For now only 2 values can be plotted synchronously with your media file. The values can come from the same file or from two different files.

During the observation, the values you selected in the external data files will be plotted synchronously with the media file.



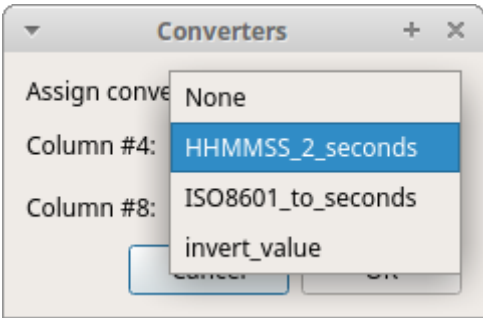
Observation with 2 values plotted from external data files: Temperature and salinity

CONVERTERS

If the values in the timestamp column are not expressed in seconds (e.g., 12.45) but in another format (such as HH:MM:SS, MM:SS, or ISO 8601 like 2018-01-18T12:31:40Z), you must use a converter to transform the current format into seconds.

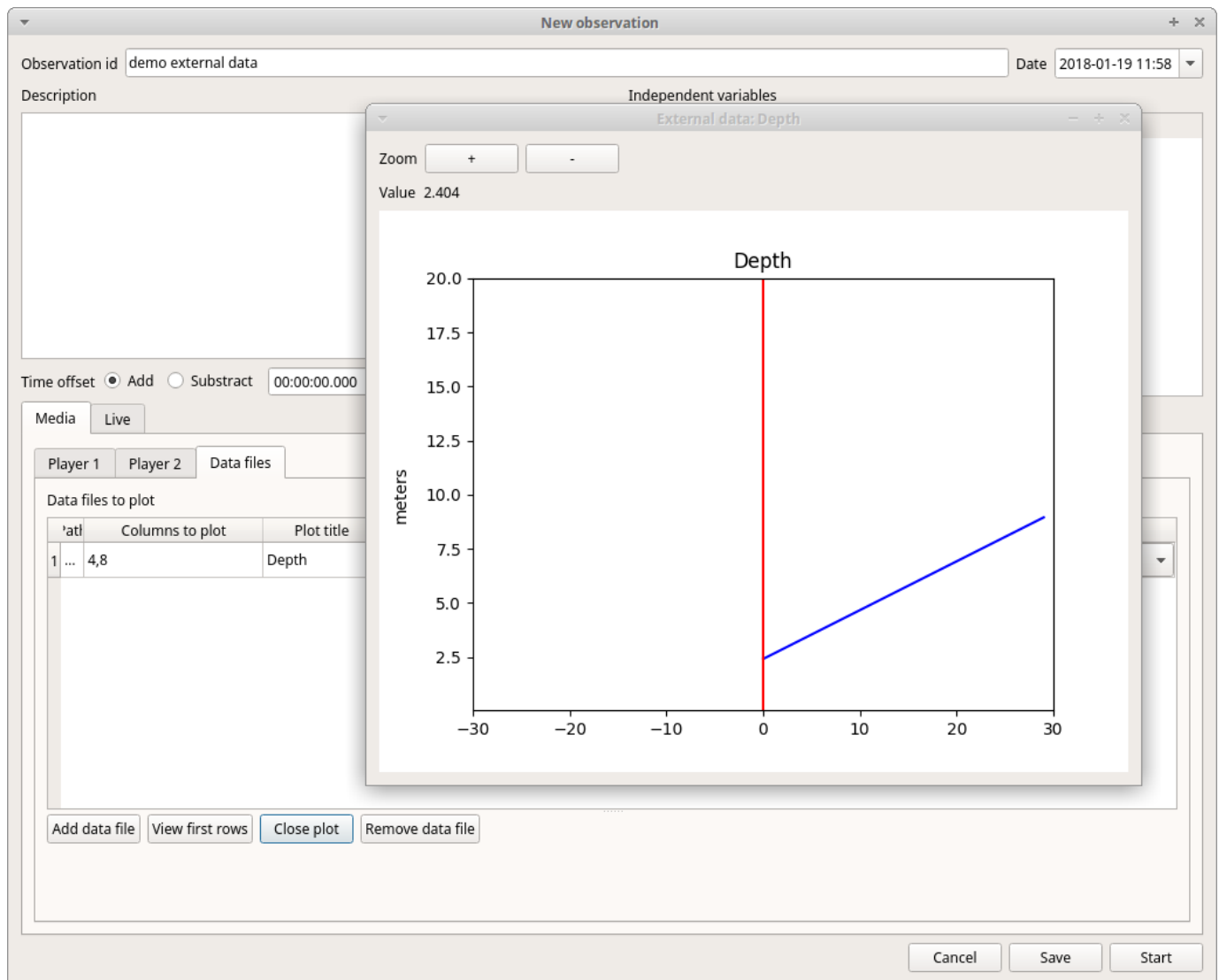
See the [Converters' table](#) in the project configuration.

A **double-click** on the converters cell will allow you to select a converter for each column to be plotted



Data files to plot							
	Path	Columns to plot	Plot title	Variable name	Converters	Time interval (s)	Start position (s)
1	/home/user/external_data2.csv	4,8	Depth	meters	{'4': 'HHMMSS_2_seconds'}	60	0

Use the **Show plot** button to verify if your external data can be plotted without problem. The **Close plot** button will close the plot window.



Converters can also be used to convert values that are not time value.

Example of a converter for inverting value:

```
OUTPUT = - float(INPUT)
```

Start the observation

Click the **Start** button to start coding. The **Observation** window will be closed and you'll be transferred to the main **BORIS** window. If you do not want to start the observation click the **Save** button. The observation will be saved in the [observations list](#).

The main window during the observation of a single media file will look like this:

Ethogram

Key	Code	Type	Description
10	v	Purr	Point event
11	v	Chirp	Point event
12	1	Approach	Point event
13	2	Arm-over	Point event
14	l	Touch	Point event
15	è	Licked by conspecific	State event

Subjects

Key	Name	Description	Current state(s)
1	No focal ...		
2	Ciro	Lemur catta - Male	Huddling
3	Totò	Lemur catta - Male	Huddling
4	Maurice	Lemur catta - Male	

Player paused
 videoS1.mp4: 00:02:00.600 / 00:06:36.066 frame: 3016
 media #1 / 1
 No focal subject
 Observed behaviors:

Events for "DEMO1" observation

Time	Subject	Code	Type	Modifier	Comment
27 00:01:21.439	Totò	Head ...			
28 00:01:27.919	Totò	Hanging			
29 00:01:38.000	Maurice	Wave tale		Male vs male	
30 00:01:48.199	Totò	Grooming	START		
31 00:01:50.359	Ciro	Grooming	START		
32 00:01:53.546	Ciro	Grooming	STOP		
33 00:01:53.546	Totò	Grooming	STOP		
34 00:01:56.120	Maurice	Anoint tale		Male vs male	
35 00:02:00.559	Ciro	Look away			
36 00:02:00.559	Totò	Look away			
37 00:02:13.640	Maurice	Anoint tale			
38 00:02:13.640	Maurice	Wave tale		Male vs male	
39 00:02:15.200	Totò	Threat			
40 00:02:15.320	Maurice	Flee			
41 00:02:18.199	Ciro	Huddling	STOP		
42 00:02:18.199	Totò	Huddling	STOP		

Player rate: x1.000

The main window during the observation of one video

See the [Media based coding](#) section to start coding.

2.4.3 Observation from pictures

Click on the **Observation from pictures** radio button to create an observation based on pictures.

New observation

Observation id * Date and time 2022-09-05 16:38:40

Description

Time offset 0.000 ☐ hh:mm:ss ☒ seconds

☐ Limit observation to a time interval

Observation type

☐ Observation from media file(s) ☐ Live observation ☒ Observation from pictures

Time

☒ No time ☐ Use the EXIF DateTimeOriginal tag ☐ Time lapse (s) 0.000

Variable	Type	Value
1 Location	text	44° 55' 59" N - 7° 25' 18" E
2 Weather	value from set	sun
3 Temperature	numeric	
4 Visitors	numeric	

Observation from pictures tab

Use the **Add directory** to select a directory containing the pictures you want to code. You can select many directories, in this case the pictures will be browsed in the order of the directories were added.

The **Use the pictures directory as observation id** button will set the directory name as **observation id**

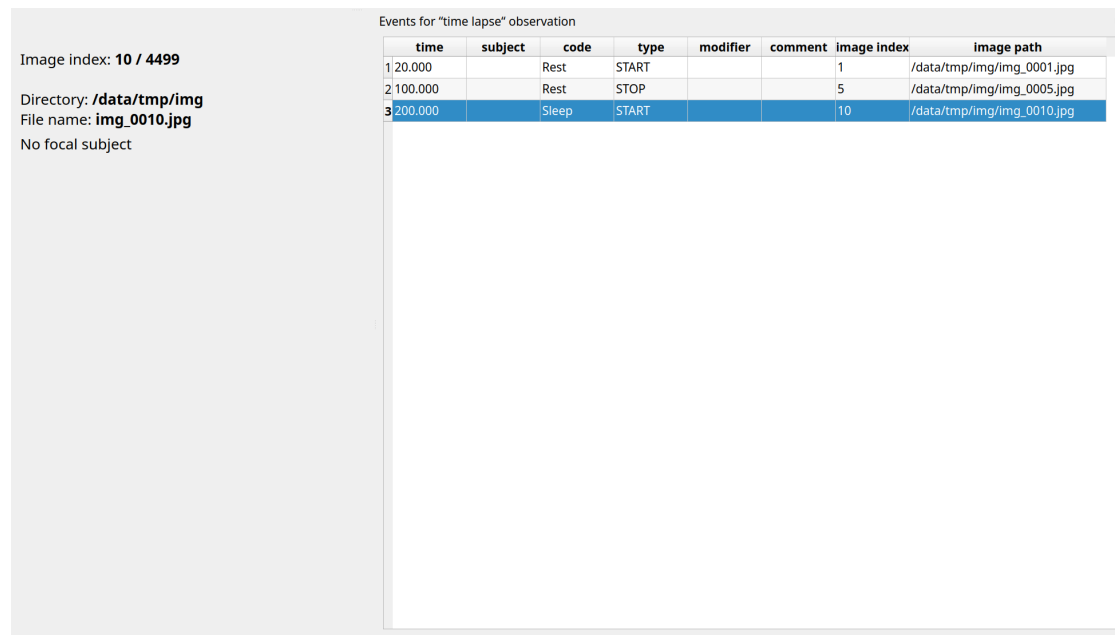
Time

You have 3 option for the coding time:

- No time: no time will be recorded. The image index (the position of image in the directory) and the image file path will be recorded.
- Use the EXIF DateTimeOriginal tag: the time will be extracted from the EXIF tag of the picture file (if any).
- Time lapse: this option will let you define the time interval between the pictures.

Start the observation

Click the **Start** button to start coding. The **Observation** window will be closed and you'll be transferred to the main **BORIS** window. If you do not want to start the observation click the **Save** button. The observation will be saved in the [observations list](#).



The main window during the coding of a picture directory

See the [Media based coding](#) section to start coding.

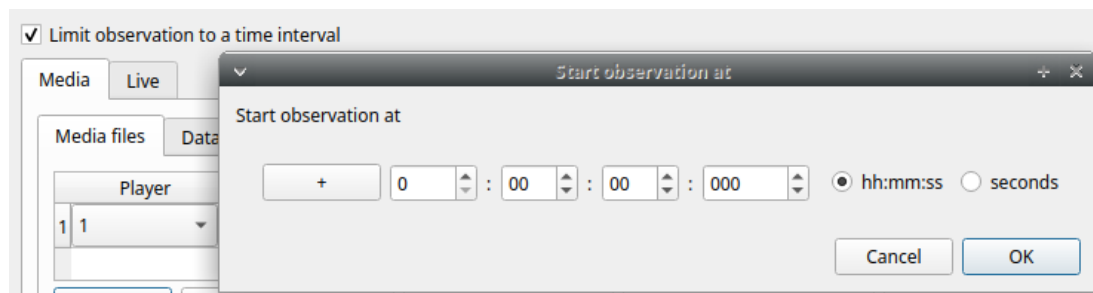
2.4.4 Various options

Scan sampling (Live and media observation)

You can select a time for **Scan sampling** observation. In this case the timer will stop at every time offset you indicated and all the coded events will have the same time value.

Limit observation to a time interval (Live and media observation)

This option can be used to limit the observation to a time interval for live or media based observations.



2.5 Managing observations

2.5.1 Observations list

Observations > **Observations list** shows all the observations contained in the current BORIS project.

The following values are displayed:

- the observation id (**id**)
- the **description** of the observation
- the coded subjects (**subjects**)
- the **observation duration** (as the difference between the last recorded event and the first one)
- the percent of **exhaustivity** of the coding (as the sum of the length of the coded events divided by the observation duration)
- the **media** file path, **LIVE** for a live observation, or the pictures directory path for an observation based on pictures
- the values of the independent variables (if defined)

Select observations for plotting events

1443 observations

id contains

	id	date	description	subjects	observation duration	exhaustivity %	media	Location	Weather	temperature	Visitors
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Himal, Nautilus	34.470	89.6	#1: 20160517162540.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Himal, Nautilus, Nina	22.500	87.1	#1: 20160517162641.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
5	0004	2016-05-17 00:00:59.000	Central trunks	Sharky, Himal, Nautilus, Nina	50.697	72.9	#1: video1.mp4	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	40.770	100.0	#1: 20160517163131.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3	#1: 20160517163231.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Sharky, Himal, Nina	124.986	41.3	#1: 20160517163347.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Himal, Nautilus	64.800	85.1	#1: 20160517163743.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Himal, Nina	46.277	84.6	#1: 20160517163927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Himal, Nautilus	16.779	81.1	#1: 20160517164021.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Himal, Nautilus, Nina	25.101	49.1	#1: 20160517164106.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nautilus, Nina	78.210	62.4	#1: 20160517164204.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Himal, Nautilus, Nina	63.631	66.2	#1: 20160517164715.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
15	0014	2016-05-17 00:00:52	In the pool	Sharky, Himal, Nautilus, Nina	242.596	53.5	#1: 20160517164927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Himal, Nautilus, Nina	111.302	49.7	#1: 20160517165631.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046

Select all Unselect all Cancel OK

The observations can be sorted by clicking the desired column header (alphabetic order ascending or descending).

Checking the observations

The observation status is displayed in the first column (**id**). If the background of this column is **red**, the observation has one or more UNPAIRED state events. These UNPAIRED observations will not be analyzed. See [Fix unpaired state events](#) for details.

Observations list - BORIS

1443 observations

id contains

	id	date	description	subjects	observation duration	exhaustivity %
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5
3	0002	2016-05-17 00:00:24	Vegetation	Nautilus, Himal, Sharky	34.470	89.6
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Nina, Himal, Nautilus	22.500	87.1
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Himal, Nautilus	50.697	72.9
6	0005	2016-05-17 00:00:49	In the pool	Nautilus, Sharky	40.770	100.0
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3
8	0007	2016-05-17 00:00:13	In the pool	Nina, Himal, Sharky	124.986	43.5
9	0008	2016-05-17 00:00:17	In the pool	Nautilus, Himal, Sharky	64.800	85.1
10	0009	2016-05-17 00:00:10	In the pool	Nina, Himal, Sharky	46.277	84.6
11	0010	2016-05-17 00:00:57	Area near the glass window	Nautilus, Himal, Sharky	16.779	81.1
12	0011	2016-05-17 00:00:50	Area near the glass window	Nina, Nautilus, Himal, Sharky	25.101	49.1
13	0012	2016-05-17 00:00:45	Area near the glass window	Nina, Nautilus, Sharky	78.210	62.4
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Nina, Himal, Nautilus	63.631	66.2
15	0014	2016-05-17 00:00:52	In the pool	Nina, Nautilus, Himal, Sharky	242.596	53.5
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Nina, Himal, Nautilus	111.302	49.7

Cancel Start View Edit

Filtering the observations

The observations list can be filtered by selecting a field and a condition from the drop-down lists.

In the following example observations are filtered: only observations with **description** containing the **In the pool** subject are shown:

Observations list - BORIS

93 observations

description contains in the pool

	id	date	description	subjects	media	Location	Weather	Temperature
1	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: 20160517163131.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24
2	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: 20160517163231.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24
3	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: 20160517163347.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
4	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: 20160517163743.m2ts	44° 55' 59" N - 7° 25' 18" E		
5	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: 20160517163927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
6	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160517164927.m2ts	44° 55' 59" N - 7° 25' 18" E		
7	0185	2016-05-28 00:00:23	In the pool	Nina, Sharky, Nautilus	#1: 20160528124413.m2ts	44° 55' 59" N - 7° 25' 18" E		
8	0212	2016-05-31 00:00:53	In the pool	Sharky	#1: 20160531104533.m2ts	44° 55' 59" N - 7° 25' 18" E		
9	0217	2016-05-31 00:00:49	In the pool	Nina, Himal	#1: 20160531145324.m2ts	44° 55' 59" N - 7° 25' 18" E		
10	0220	2016-05-31 00:00:36	In the pool	Sharky, Nina, Nautilus, Himal	#1: 20160531153747.m2ts	44° 55' 59" N - 7° 25' 18" E		
11	0359	2016-06-06 00:00:54	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160606163758.m2ts	44° 55' 59" N - 7° 25' 18" E		
12	0411	2016-06-07 00:00:53	In the pool	Nina, Nautilus	#1: 20160607103354.m2ts	44° 55' 59" N - 7° 25' 18" E		
13	0412	2016-06-07 00:00:21	In the pool	Nautilus	#1: 20160607103630.m2ts	44° 55' 59" N - 7° 25' 18" E		

Cancel Start View Edit

Observations list

Observations can be filtered with **Independent variables** values.

The following example displays only the observations that do not contain "Sunny" in the **Weather** independent variable :

Observations list - BORIS

1430 observations

Weather

	id	date	description	subjects	media	Location	Weather	Temperature
1	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: 20160517163743.m2ts	44° 55' 59" N - 7° 25' 18" E		
2	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Nautilus, Himal	#1: 20160517164021.m2ts	44° 55' 59" N - 7° 25' 18" E		
3	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Nina, Nautilus, Himal	#1: 20160517164106.m2ts	44° 55' 59" N - 7° 25' 18" E		
4	0013	2016-05-17 00:00:25	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517164715.m2ts	44° 55' 59" N - 7° 25' 18" E		
5	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: 20160517164927.m2ts	44° 55' 59" N - 7° 25' 18" E		
6	0015	2016-05-17 00:00:18	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517165631.m2ts	44° 55' 59" N - 7° 25' 18" E		
7	0016	2016-05-17 00:00:58	Central trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517165923.m2ts	44° 55' 59" N - 7° 25' 18" E		
8	0017	2016-05-17 00:00:37	Suspended trunks	Nina, Sharky, Nautilus, Himal	#1: 20160517170018.m2ts	44° 55' 59" N - 7° 25' 18" E		
9	0018	2016-05-17 00:00:18	Central trunks	Nina, Himal	#1: 20160517170259.m2ts	44° 55' 59" N - 7° 25' 18" E		
10	0019	2016-05-17 00:00:23	Central trunks	Sharky, Nina, Nautilus, Himal	#1: 20160517170519.m2ts	44° 55' 59" N - 7° 25' 18" E		
11	0020	2016-05-25 00:00:24	Indoor entrance	Nina, Himal	#1: 20160525145403.m2ts	44° 55' 59" N - 7° 25' 18" E		
12	0021	2016-05-25 00:00:47	Indoor entrance	Sharky, Himal	#1: 20160525145814.m2ts	44° 55' 59" N - 7° 25' 18" E		
13	0022	2016-05-25 00:00:30	Indoor entrance	Nina, Himal	#1: 20160525150240.m2ts	44° 55' 59" N - 7° 25' 18" E		

Cancel Start View Edit

Observations list

Observations with a value of **Temperature** independent variable between 18 and 22:

Observations list - BORIS

10 observations

Temperature

	id	date	description	subjects	media	Location	Weather	Temperature
1	0001_a	2016-05-17 00:00:31	Vegetation	Nautilus, Himal	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
2	0001_b	2016-05-17 00:00:31	Vegetation	Nautilus, Himal	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Nautilus, Himal	#1: 20160517162540.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24
4	0003	2016-05-17 00:00:05	Vegetation	Nina, Sharky, Nautilus, Himal	#1: 20160517162641.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Nautilus, Himal	#1: 20160517162952.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: 20160517163131.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: 20160517163231.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24
8	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: 20160517163347.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
9	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: 20160517163927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25
10	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nina, Nautilus	#1: 20160517164204.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26

Cancel Start View Edit

Observations list

Observations with a value of **Visitors** independent variable greater than 1000:

Observations list - BORIS

825 observations

Visitors > 1000

	id	date	description	subjects	hedi	Location	Weather	Temperature	Visitors
4	0003	2016-05-17 00:00:05	Vegetation	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
5	0004	2016-05-17 00:00:59	Central trunks	Sharky, Nina, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Nina, Sharky, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Nina, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Nina, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nina, Nautilus	#1: ...	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
15	0014	2016-05-17 00:00:52	In the pool	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
16	0015	2016-05-17 00:00:18	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046
17	0016	2016-05-17 00:00:58	Central trunks	Nina, Sharky, Nautilus, Himal	#1: ...	44° 55' 59" N - 7° 25' 18" E			1046

Cancel Start View Edit

Observations list

2.5.2 Delete observations from project

Observations can be deleted from the project using the following procedure:

Observations > Remove observations

Select the observations you want to delete.

Click the **OK** button and confirm the deletion.

Deletion is irreversible, and deleted observations cannot be restored.

It's a good idea to back up your project before proceeding with removing observations.

2.5.3 Create observations in bulk

Observations from media files can be created from a directory of media files:

Observations > Create observations

Choose the directory.

Select the parameters.

Observation parameters ✕

Set the following observation parameters

- ☐ Recurse the subdirectories
- ☐ Visualize spectrogram
- ☐ Visualize waveform
- ☐ Media creation date as offset
- ☐ Close behaviors between videos

Time offset (in seconds)

0.000

Media scan sampling duration (in seconds)

0.000

Cancel OK

Parameters for creating observations

The ID of each created observation will be the path to the media file.

2.5.4 Import observations

The **Observations > Import observations** option allows you to import observations. Two formats are available for importing observations:

From a BORIS project file

Choose the BORIS project file and then the observations to import. BORIS will check whether observations with the same ID already exist in the current project. BORIS will also check if behaviors and/or subjects used in the imported observations are not defined in the current project.

From a spreadsheet file

Observations can be imported from:

- OpenDocument (ODS)
- Microsoft-Excel (XLSX)

Choose the spreadsheet file

2.5.5 Export a list of observations

This option allows you to export the selected observations as a list in various formats (CSV, TSV, ODS, XLSX, HTML):

Observations > Export observations list

The data frame will contain the following columns:

- Observation id
- Date
- Description
- Subjects
- Media files/Live observation
- independent variables

2.6 Coding

When looking at the BORIS main window, the window title bar shows the **Observation id - Project name - BORIS**. The media (the first in the queue) will be loaded in the media player and paused.

On macOS, the video is not embedded in the BORIS main window, so it is **VERY IMPORTANT** to click on the **BORIS main window** before coding as usual.

2.6.1 Media based coding

The toolbar



The BORIS toolbar

List of observations

Play (becomes Pause when media is playing)

Rewind resets the media to the beginning

Fast backward jumps back by n seconds in the media (see [preferences](#))

Fast forward jumps forward by n seconds in the media (see [preferences](#))

Set the playback speed to 1x

Increase the playback speed (See [preferences](#))

Decrease the playback speed (See [preferences](#))

Jump to the previous media file

Jump to the next media file

Take a snapshot of current video or frame

Move one frame back

Move one frame forward

Real time plot of events

Time budget of the current observation

Plot events of the current observation

Plot the time budget of the current observation

Geometric measurements

Find in events

Explore project

Preferences

Edit current observation

Close current observation

The toolbar can be resized in **Preferences > Interface**.

Media can also be controlled with dedicated keyboard shortcuts:

Page Up Switch to the next media file

Page Down Switch to the previous media file

↑ Up Jump forward in the current media

↓ Down | Jump backward in the current media

[↑ Home](#) Increase the playback speed (See [Observations preferences](#) to set the step value)

End Decrease the playback speed (See [Observations preferences](#) to set the step value)

⏮ Backspace Set the playback speed to 1x

← Left Go to the previous frame

→ Right Go to the next frame

2.6.2 Live observations

During a live observation, the media control toolbar is disabled.

Press the **Start live observation** button to start your observation. If some events have already been coded, BORIS will ask whether they should be deleted.

A timer will be displayed. Events will be recorded in the events widget.

Project

Observations

Playback

Tools

Analysis

Help

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Ethogram

	Key	Code	Type	Description
1	T	Tear	State event	Otter tears off ...
2	C	Chase	State event	Otter chases ...
3	I	Interact with ...	State event	Otter interacts...
4	R	Rub	State event	Otter rubs and...
5	C	Carry objects	State event	Otter carries ...
6	S	Sniff	State event	Otter moves t...
7	L	Locomotion	State event	Otter moves ...
8	S	Spot keeper	State event	Otter spots th...

Subjects

	Key	Name	Description	Current state(s)
1		No focal subject		
2	N	Nina	Female, adult, born on ...	
3	H	Himal	Male, adult, born on ...	Chase (None)
4	C	Sharky	Male, juvenile, born on ...	
5	S	Nautilus	Male, juvenile, born on ...	

live observation - Test project* - BORIS

00:00:35.893

Focal subject: **Himal**

Observed behaviors: Chase (None)

Stop live observation

Events for "live observation" observation

	Time	Subject	Code	Type	Modifier	Comment
1	00:00:00.000	Nina	Tear	START	None	
2	00:00:08.387	Nina	Tear	STOP	None	
3	00:00:13.859	Himal	Chase	START	None	

2.6.3 Ethogram table in the main window

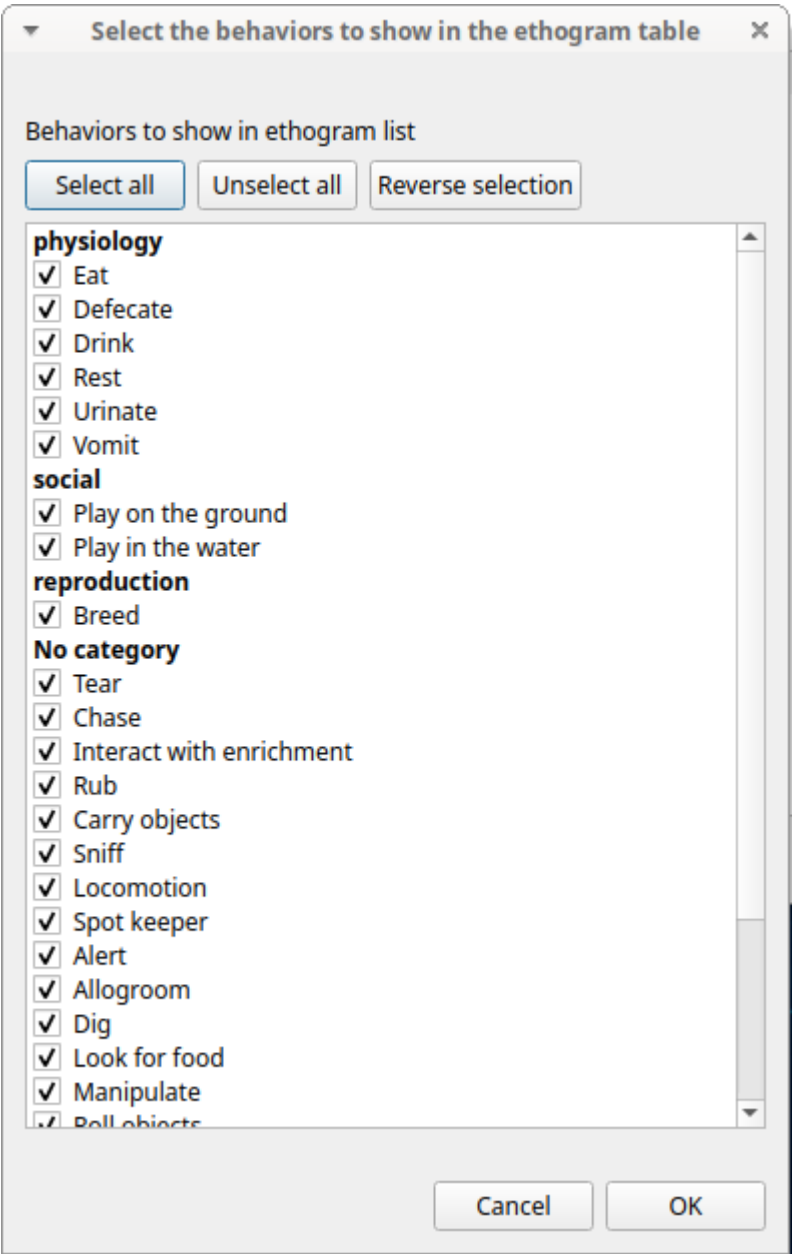
	Key	Code	Type	Description	Color	Category	Modifiers	Excluded
1	T	Tear	State event	Otter tears off vegetation			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
2	C	Chase	State event	Otter chases other animals			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
3	I	Interact with ...	State event	Otter interacts with enrichment			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
4	R	Rub	State event	Otter rubs and rolls itself upon a surface; may be ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
5	C	Carry objects	State event	Otter carries objects or food by holding them ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Chase,Defecate,Dig,Drink,Eat,In...
6	S	Sniff	State event	Otter moves the nose and head movement back ...			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
7	L	Locomotion	State event	Otter moves from place to place			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
8	S	Spot keeper	State event	Otter spots the keeper in or out the enclosure			{'0': {'name': '', 'type': 0, ...	Alert,Allogroom,Breed,Carry ...
9	A	Alert	State event	Otter is stationary and directs its attention toward...				Allogroom,Breed,Carry ...
10	Q	Allogroom	State event	Otter licks or scratches with forepaws or hind-paw...				Alert,Breed,Carry ...
11	D	Dig	State event	Otter uses front legs to move sand, stones on the ...				Alert,Allogroom,Breed,Carry ...

The **Ethogram** table provides the list of behaviors defined in the **Ethogram**. It can be used to record an event by double-clicking the corresponding row. The **Key** column indicates the keyboard shortcut assigned to each behavior, if any. Pressing a key records the corresponding behavior, which then appears in the **Events** table.

The behaviors shown in the Ethogram widget can be filtered:

Right-click on ethogram widget > **Filter behaviors**

Check or uncheck individual behaviors, or double-click a behavioral category.



2.6.4 Subjects table in the main window

Subjects

Key	Name	Description	Current state(s)
1		No focal subject	
2	N	Nina	Female, adult, born on 10/03/2013 in Ostrava biopark (Czech Republic), bright white snout
3	H	Himal	Male, adult, born on 04/30/2014 in Amneville biopark (France), larger tale, bigger than the others
4	C	Sharky	Male, juvenile, born on 10/30/2015 in Zoom biopark (Italy), bright brown nose and fur
5	S	Nautilus	Male, juvenile, born on 10/30/2015 in Zoom biopark (Italy), dark brown nose and fur

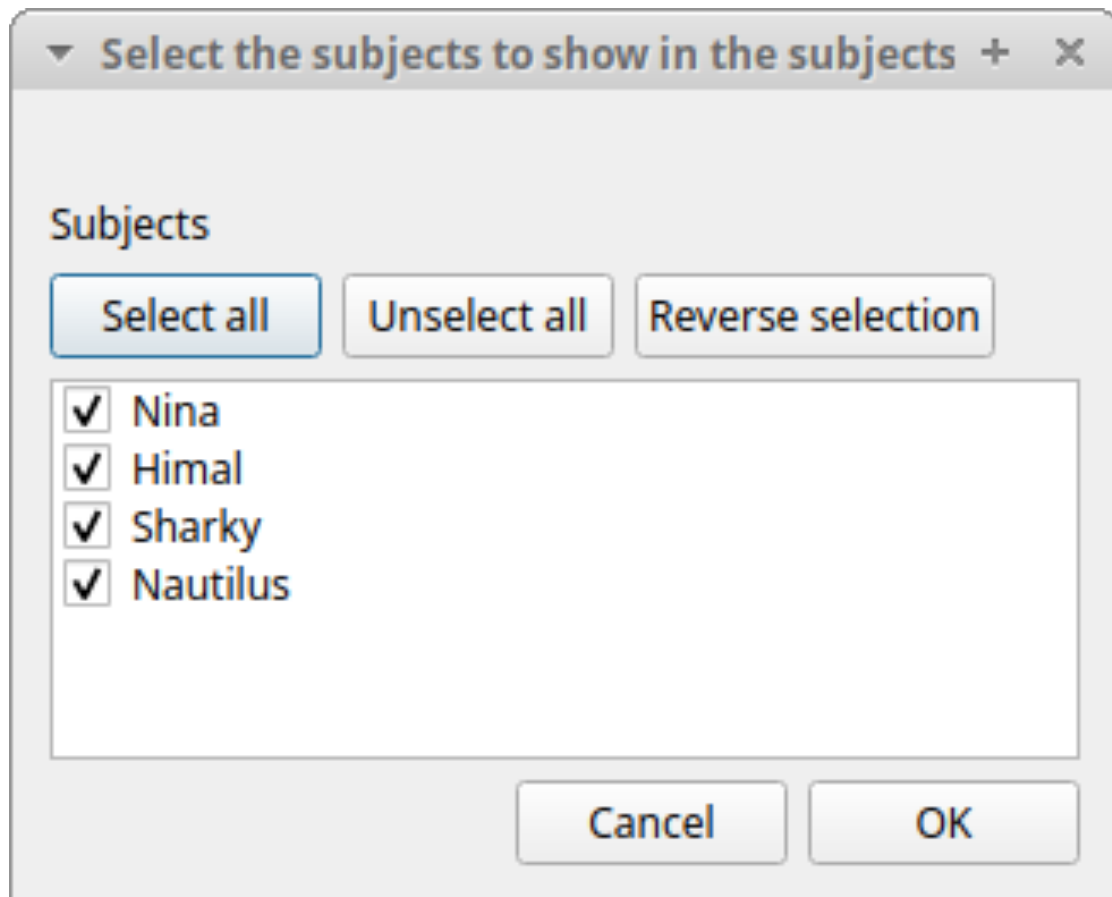
The subjects table in main window

The **Subjects** widget provides the list of subjects defined in the **Subject** tab of the **Project** window. It can be used to assign a focal subject to recorded behaviors by double-clicking the corresponding row. When a subject is selected, the name appears above the media player. The **Key** column indicates the keyboard shortcut assigned to each subject, if any.

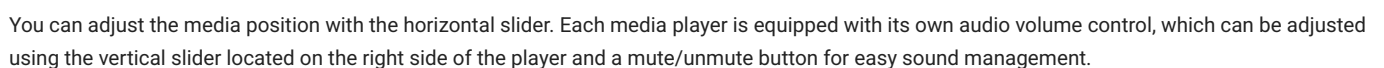
The subjects shown in the Subjects widget can be filtered:

Right-click on subjects widget > **Filter subjects**

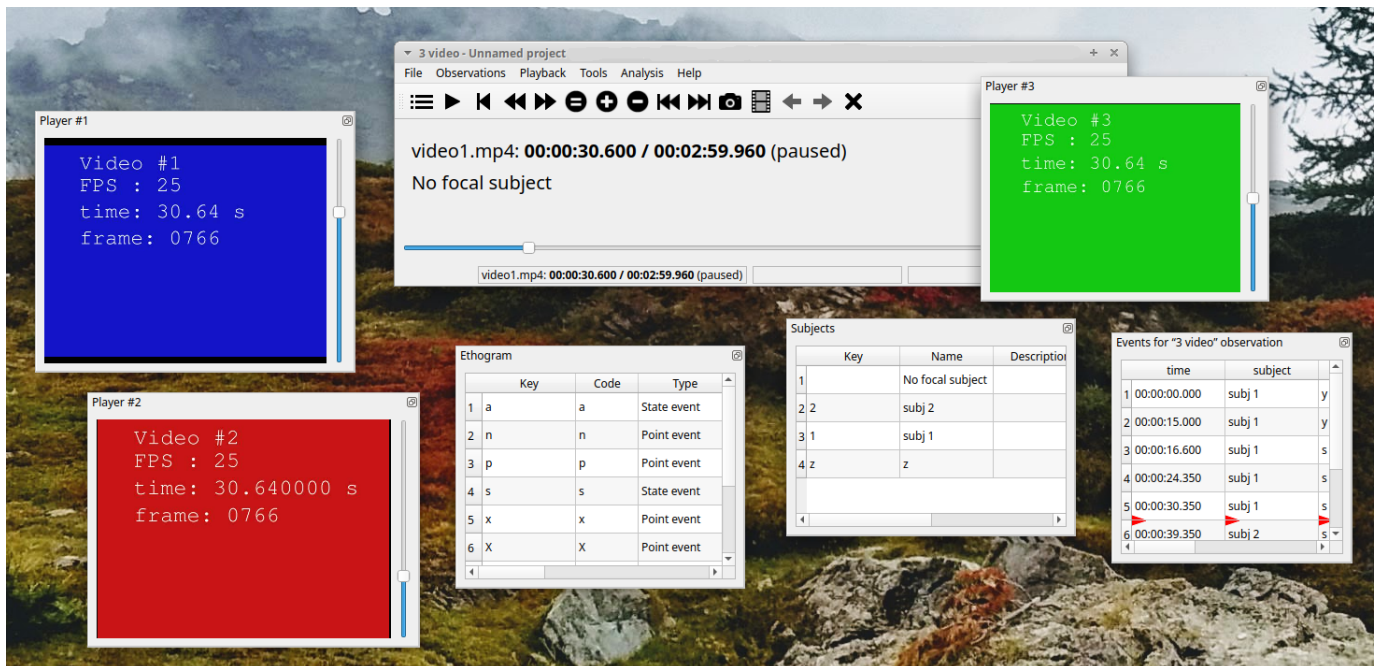
Check or uncheck subjects to show or hide them in the Subjects table.



Filter subjects in subjects table



The arrangement of different widgets can be personalized to suit your preferences. In the screenshot provided, all widgets are detached from the main program window. This feature is particularly useful when using a multi-monitor setup.



2.6.6 The events table

The **events table** shows all the recorded behaviors (events).

The displayed parameters, organized in columns, depend on the observation type:

Observation from media file

Events for "0004" observation							
	Time	Frame index	Subject	Code	Type	Modifier	Comment
1	00:00:08.253	207	Himal	Locomotion	START	Run	
2	00:00:08.329	209	Nautilus	Locomotion	START	Run	
3	00:00:10.400	260	Sharky	Swim	START		
4	00:00:11.779	295	Himal	Locomotion	STOP	Run	
5	00:00:12.778	320	Nina	Alert	START		
6	00:00:13.789	345	Nautilus	Locomotion	STOP	Run	
7	00:00:13.790	345	Nautilus	Locomotion	START	Walk	
8	00:00:14.348	359	Himal	Locomotion	START	Jump	
9	00:00:14.660	367	Nina	Alert	STOP		
10	00:00:14.865	372	Nautilus	Locomotion	STOP	Walk	
11	00:00:14.866	372	Nautilus	Locomotion	START	Jump	
12	00:00:15.001	375	Nina	Rest	START		
13	00:00:16.467	412	Himal	Locomotion	STOP	Jump	

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Frame index** the frame index corresponding to the event;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, for a **state event**, indicates whether the time corresponds to the start or the stop. It is empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, is an open field where the user can add notes.

A tracking cursor (red triangle) marks the current event. This cursor can be positioned above the current event; see [tracking cursor position](#) in the **Preferences** window.

A double-click on a row will reposition the media player to the moment of the corresponding event. See [Time offset for media reposition](#) in **Preferences** window to customize the time offset for media repositioning.

Live observation

Events for "live observation" observation

	Time	Subject	Code	Type	Modifier	Comment
1	00:00:01.248	Nina	Alert	START		
2	00:00:15.664	Himal	Manipulate	START		vegetables
3	00:00:22.857	Himal	Manipulate	STOP		
4	00:00:24.841	Himal	Roll objects	START		
5	00:00:29.985	Himal	Roll objects	STOP		
6	00:00:38.009	Nina	Alert	STOP		
7	00:00:41.577	Sharky	Stomp	START		
8	00:00:49.105	Sharky	Stomp	STOP		

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, in case of a **state event** indicates whether the time corresponds to the start or to the stop. Empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, is an open field where the user can add notes.

Observation from pictures

Events for "cametra trap" observation

	Time	Subject	Code	Type	Modifier	Comment	Image index	Image path
1	NA		Coleoptera		None Bostrichoidea Cerambycidae Lagriinae Anthaxia None Cetonia aurata None None None		1	/data/08010001.JPG
2	NA		Diptera		Brachycera None Conopidae Eristalinae Chrysotoxum None Cheilosia impressa None None None		2	/data/08010002.JPG
3	NA		Hymenoptera		None Apoidea Cephidae Arginae Anthidiellum Bombus s.s. Bombus distinguendus None None None		3	/data/08010003.JPG
4	NA		Lepidoptera		Macroheterocera None Geometridae Dismorphiinae Albulina None Anarta melanopa None None None		4	/data/08010004.JPG
5	NA		Others	flower_on			5	/data/08010005.JPG
6	NA		Hymenoptera		None Cephioidea Argidae Cephinae Andrena Ashtoniopsisithyrus Bombus bohemicus None None None		6	/data/08010006.JPG

The following parameters are displayed:

- **Time**, the time at which the event occurred;
- **Subject**, the focal subject (if any);
- **Code**, the behavior code;
- **Type**, in case of a **state event** indicates whether the time corresponds to the start or to the stop. Empty for a **point event**;
- **Modifier**, indicates the modifier(s) that was/were selected (if any);
- **Comment**, is an open field where the user can add notes;
- **Image index**, the image index (in the directory) corresponding to the event,
- **Image path**, the path of the image corresponding to the event (can be relative or absolute).

To simplify the **events table** the relevant behaviors and subjects can be filtered see [Filter events](#)

2.6.7 Events

Recording an event

An event is a unique combination of a **time**, a **subject** and a **behavior**. If the subject is not set it will be **No focal subject**.

Once ready to begin your coding, you can start the media player using the **Play** button or the **Space bar**.

An **event** can be recorded by:

- pressing the predefined **key** of the keyboard corresponding to the behavior to record.
- double-clicking to the corresponding row in the **Ethogram** table.
- using the **Coding pad** (See [coding pad](#)).

The **focal subject** can be selected by:

- pressing the predefined **key** of the keyboard corresponding to the subject to select.
- double-clicking to the corresponding row in the **Subjects** table.
- using the **Subject pad** (See [subject pad](#)).

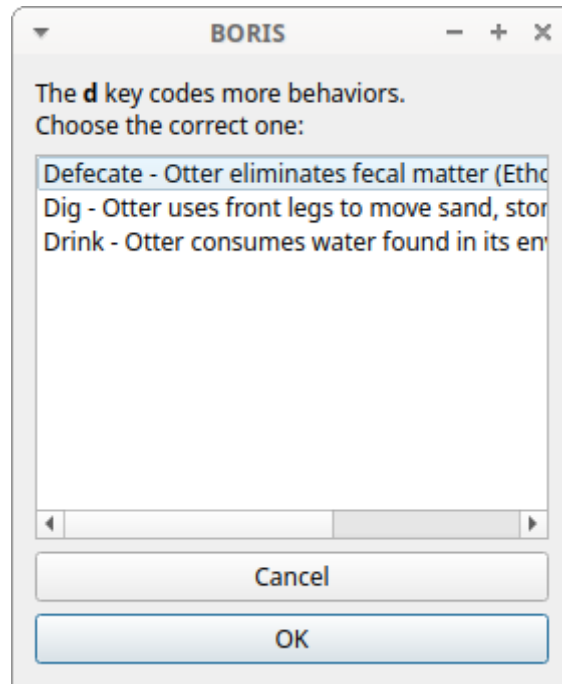
The screenshot displays the BORIS software interface. At the top is a menu bar with 'Project', 'Observations', 'Playback', 'Tools', 'Analysis', and 'Help'. Below the menu is a toolbar with various icons for navigation and analysis. The main area is divided into two panels. The top panel, titled 'Ethogram', contains a table with 12 rows and 9 columns: Key, Code, Type, Description, Color, Category, Modifiers, and Excluded. The bottom panel, titled 'Subjects', contains a table with 5 rows and 4 columns: Key, Name, Description, and Current state(s).

	Key	Code	Type	Description	Color	Category	Modifiers	Excluded
1	t	Tear	State event	Otter tears off vegetation			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
2	c	Chase	State event	Otter chases other animals			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
3	i	Interact with ...	State event	Otter interacts with enrichment			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
4	r	Rub	State event	Otter rubs and rolls itself upon...			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
5	c	Carry objects	State event	Otter carries objects or food b...			{'0': {'name': ", ...	Alert,Allogroom,Breed,Chase,Defecate,Dig...
6	s	Sniff	State event	Otter moves the nose and hea...			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
7	l	Locomotion	State event	Otter moves from place to place			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
8	s	Spot keeper	State event	Otter spots the keeper in or o...			{'0': {'name': ", ...	Alert,Allogroom,Breed,Carry ...
9	a	Alert	State event	Otter is stationary and directs ...				Allogroom,Breed,Carry ...
10	q	Allogroom	State event	Otter licks or scratches with ...				Alert,Breed,Carry ...
11	d	Dig	State event	Otter uses front legs to move ...				Alert,Allogroom,Breed,Carry ...
12	f	Look for food	State event	Otter looks for food in the ...				Alert,Allogroom,Breed,Carry ...

	Key	Name	Description	Current state(s)
1		No focal subject		
2	n	Nina	Female, adult, born on 10/03/2013 in Ostrava biopark (Czech Republic), bright white snout	
3	h	Himal	Male, adult, born on 04/30/2014 in Amneville biopark (France), larger tale, bigger than the others	
4	c	Sharky	Male, juvenile, born on 10/30/2015 in Zoom biopark (Italy), bright brown nose and fur	
5	s	Nautilus	Male, juvenile, born on 10/30/2015 in Zoom biopark (Italy), dark brown nose and fur	

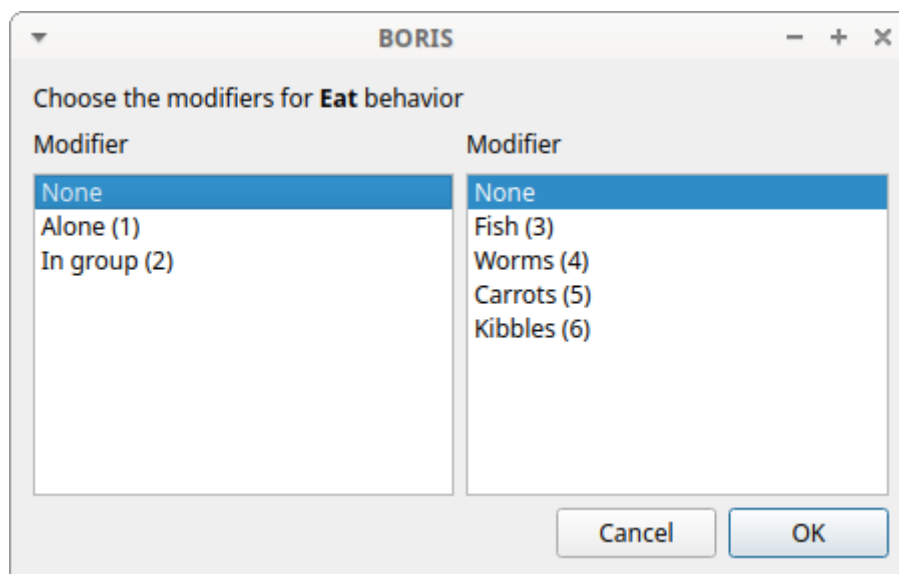
Ethogram and subjects widgets

If the pressed key defines a single event, the corresponding event will be recorded directly in the **Events** table. In the case you have specified the same key for two (or more) events (e.g. key **d** in the figure below), BORIS will prompt you for the desired behavior.



Ask for a behavior

In the case you have specified modifiers (one or more sets), BORIS will prompt you for the desired modifier(s) if any (e.g. **ball** or **opponent** in the figure below). You can select the modifiers using the mouse or the keyboard (**1**, **2**, **3**, **4**, **5** or **6** key)



Ask for modifiers

If no keys are defined for the modifier selection, you can type the first character of the modifier and use the **Up arrow** and **Down arrow** keyboard keys to select the correct modifier.

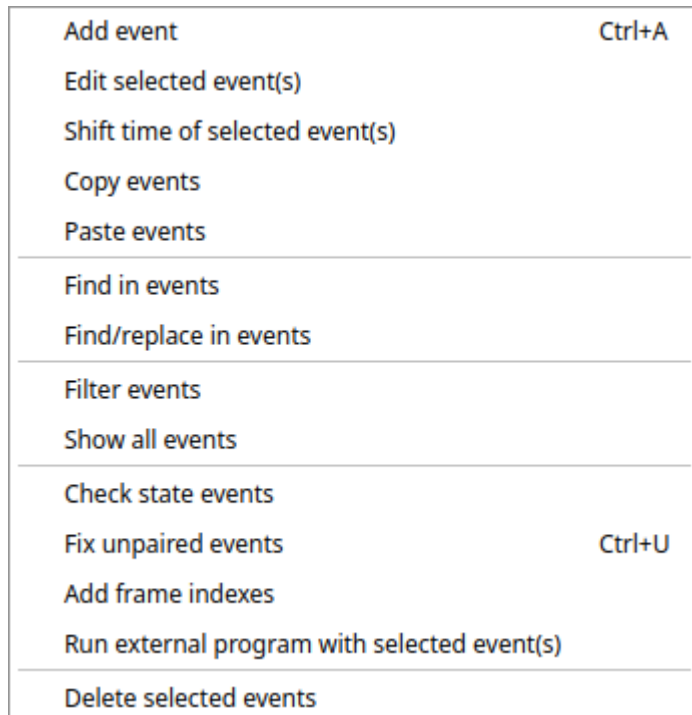
In the case your behavior type is a **Point event with coding map** or a **State event with coding map**, BORIS will show the **Coding map** window and will allow selecting the desired area(s). In case you click a part of the map in which two (or more) areas overlap, the corresponding codes will be recorded.

A recorded event can be edited (once selected) using the **Observations > Edit event** menu option. The resulting *Edit event parameters* allows modifying every parameter (e.g. time, subject, code, modifiers, and comment).

The **Observations > Add event** menu option allows adding a new event by specifying its time and the other parameters.

The Events table context menu

Some functions are available in the Events table context menu. Right-click on the Events table and the menu will pop-up.



The various functions available in the menu are described below. The same functions and others are available in the **Observations** menu.

Undo an even recording

A wrong event can be removed from the events list using the **Undo** function (**Ctrl** + **Z**). You can go back till 25 events recorded events.

Add event

This option allows adding a new event by specifying its time and the other parameters.

Add a new event

☐ Seconds
 ☐ hh:mm:ss
 ☐ Date time

Time seconds Set to current time

+

Subject

Code

Comment

Cancel OK

Add a new event

Select a time format and input the time value.

Select the **subject** from the drop-down menu or leave empty for **No focal subject**.

Select the **behavior** from the drop-down menu.

Edit selected event(s)

This option allows to edit the selected event(s). When many events are selected you have to choose the field to edit between **Subject**, **Behavior** and **Comment**. In this case the new value will apply to all selected events.

	time	subject	code	type	modifier	comment
1	0.000	Himal	Tear	START	Branches	
2	0.000	Nautilus	Tear	START	Branches	
3	30.199	Himal	Tear	STOP	Branches	
4	30.200	Himal	Locomotion	START	Walk	
5	32.400	Himal	Locomotion	STOP	Walk	
6	32.400	Nautilus				

Edit selected events

☒ Subject

☐ Behavior

New value

Nina

 Himal

 Sharky

 Nautilus

☐ Comment

New comment

Cancel

OK

Edit time of selected event(s)

This option allows to add or subtract a time value (in seconds) to all selected events. For subtracting a value use a negative value.

	time	subject	code	type	modifier	comment
1	0.000	Himal	Tear	START	Branches	
2	0.000	Nautilus	Tear	START	Branches	
3	30.199	Himal	Tear	STOP	Branches	
4	30.200	Himal	Locomotion	START	Walk	
5	32.400	Himal	Locomotion	STOP	Walk	
6	32.400	Nautilus	Tear	STOP	Branches	

Time value

Value to add or subtract (use negative value):

-10.200

Cancel

OK

Copy events

This option allows to copy the selected events in the clipboard. The clipboard will contain the values of the selected events (except the **type** field) separated by a **<TAB>** character. The copied values are: **Time, Subject, Behavior, Modifier(s), Frame index**

Example of clipboard content:

```

8.253   Himal   Locomotion  Run    207
8.329   Nautilus Locomotion  Run    209
10.400  Sharky  Swim        260
11.778  Himal   Locomotion  Run    295
12.778  Nina    Alert        320
13.788  Nautilus Locomotion  Run    345
13.789  Nautilus Locomotion  Walk    345
14.348  Himal   Locomotion  Jump    359
14.660  Nina    Alert        367
14.865  Nautilus Locomotion  Walk    372
14.865  Nautilus Locomotion  Jump    372
15.000  Nina    Rest        375
16.466  Himal   Locomotion  Jump    412
16.467  Himal   Alert        412
23.600  Nautilus Locomotion  Jump    590
23.600  Nautilus Rest        590
24.228  Nautilus Rest        606
24.407  Himal   Alert        611
24.917  Himal   Locomotion  Walk    623
39.682  Nautilus Locomotion  Run    992
40.549  Nina    Rest        1014
42.313  Nautilus Locomotion  Run    1058
42.314  Nautilus Rest        1058
44.759  Himal   Locomotion  Walk    1119
44.761  Himal   Allogroom    1119
48.219  Nautilus Rest        1206
48.363  Himal   Allogroom    1209
48.365  Himal   Locomotion  Walk    1209
49.274  Himal   Locomotion  Walk    1232
49.274  Himal   Drink        1232
50.408  Himal   Drink        1261
50.408  Himal   Swim        1261
50.851  Sharky  Swim        1472
58.950  Himal   Swim        1474

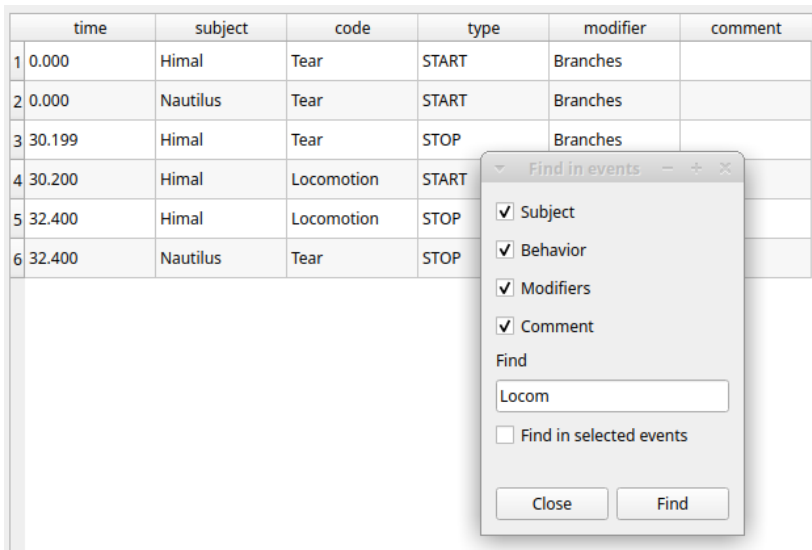
```

Paste events

This option allows to paste the clipboard content into the events table. The clipboard must respect the format described in the previous section: 5 columns separated by a **<TAB>** character.

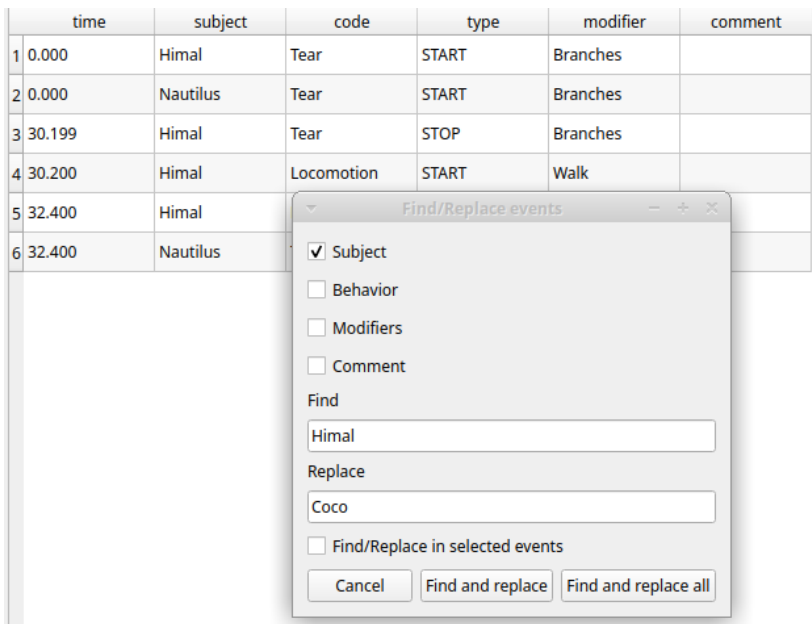
Find in events

This option allows to search for a string in the various field of events. Select the fields to be searched. The find/replace operation can be restricted to the selected events.



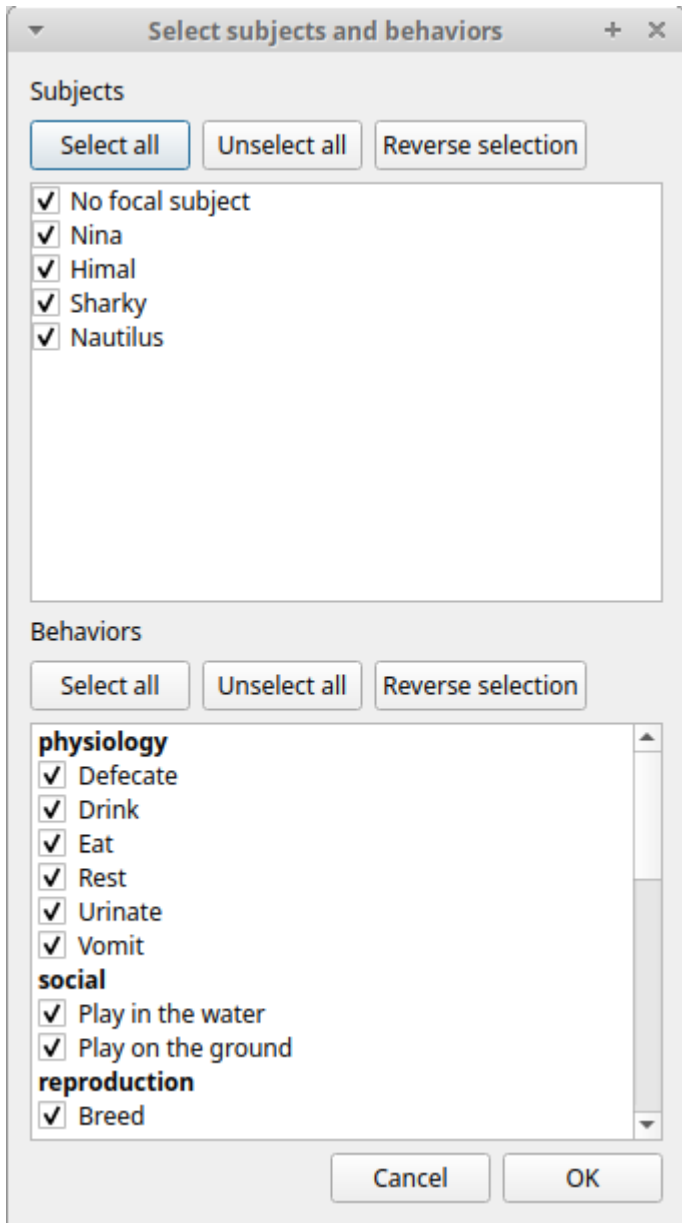
Find/Replace in events

This option allows to search for a string and replace it by a new value in the various field of events. Select the fields to be searched. The find operation can be restricted to the selected events.



Filter events

This option allows to filter the events by field value (Subject and Behavior).



Show all events

This option reverts the previous one and allows to visualize all coded events.

Check state events

This option allows to check if the **state events** are **PAIRED**, if they have a **START** and a **STOP** occurrence.

Delete selected events

This option allows to delete the selected events. This operation is irreversible!

Delete all events

This option is not present in the context menu but only in the main menu (**Observations > Delete all events**).

This option allows to delete all the events in the current observation. This operation is irreversible!

Fix unpaired state

You can use the **Fix unpaired events** function to fix the **state events** without a STOP event.

Observations > Fix unpaired events (keyboard shortcut: ⌘ + U)

The program will ask for a time at which insert the STOP events for all unpaired **state events**

This function can be run on a set of selected observations (when no observation is open). In this case the STOP events will be inserted at the end of observation.

Add frame indexes

This function can be used for the observations from a video. The frame index corresponding to the coded events will be added in the events table.

Run external program with selected events

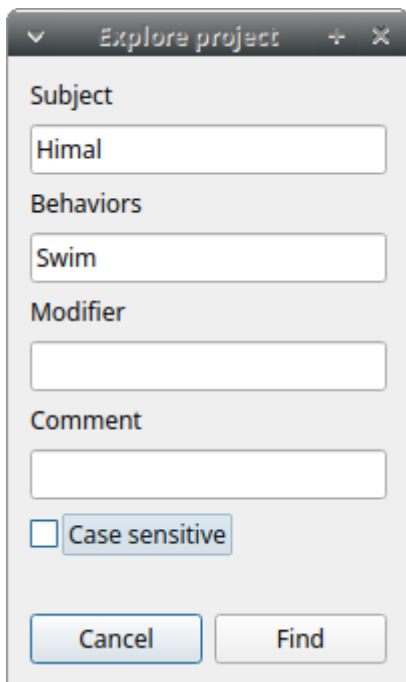
This function is not yet implemented.

Explore project

You can search information in various fields in all observations in the current project (Observations > Explore project).

The searchable fields are: **subject**, **behavior**, **modifier** and **comment**.

If more than one field is searched a logic **AND** will apply.



The events that were found are listed in a table. By double-clicking on the row the corresponding observation will be opened and the visualization will be scrolled to the row corresponding to the event.

The screenshot shows the BORIS software interface. The main window displays a table of events for observation '0004'. A dialog box titled 'BORIS' is open, showing 148 results in a table with columns 'Observation id' and 'row index'.

Key	Code	Type
1 A	Alert	State event
2 Q	Allogroom	State event
3 B	Breed	
4 C	Carry object	
5 C	Chase	
6 D	Defecate	
7 D	Dig	
8 D	Drink	
9 E	Eat	
10 I	Interact with	
11 L	Locomotion	
12 F	Look for food	
13 M	Manipulate	
14 P	Play in the water	
15 P	Play on the ground	

time	subject	code	type
14 00:00:16.468	Himal	Alert	START
15 00:00:23.600	Nautilus	Locomotion	STOP
16 00:00:23.601	Nautilus	Rest	START
17 00:00:24.228	Nautilus	Rest	STOP
18 00:00:24.407	Himal	Alert	STOP
19 00:00:24.917	Himal	Locomotion	START
20 00:00:39.683	Nautilus	Locomotion	START
21 00:00:40.550	Nina	Rest	STOP
22 00:00:42.313	Nautilus	Locomotion	STOP
23 00:00:42.314	Nautilus	Rest	START
24 00:00:44.760	Himal	Locomotion	STOP
25 00:00:44.761	Himal	Allogroom	START
26 00:00:48.219	Nautilus	Rest	STOP
27 00:00:48.364	Himal	Allogroom	STOP
28 00:00:48.365	Himal	Locomotion	START
29 00:00:49.274	Himal	Locomotion	STOP
30 00:00:49.275	Himal	Drink	START
31 00:00:50.408	Himal	Drink	STOP
32 00:00:50.409	Himal	Swim	START
33 00:00:58.852	Sharky	Swim	STOP

Key	Name	Sex	Age
1	No focal subject		
2 N	Nina	Female	adult
3 H	Himal	Male	adult

Observation id	row index
1 0004	32
2 0004	34
3 0006	7
4 0006	9
5 0006	12
6 0006	15
7 0007	8
8 0007	11
9 0007	18
10 0007	19
11 0007	20

Frame-by-frame mode

You can switch between the media player and the frame-by-frame mode using the **arrow buttons** in the toolbar:

In frame-by-frame mode the video will stop playing and the user will visualize the video frame by frame.

Note

Some video files should be re-encoded to be used in frame-by-frame mode. Otherwise the extracted frames are not reliable or it will not be possible to move backward.

You can move between frames by using the arrow keys in the toolbar (on the right) or by using keyboard special keys:

- ← Left Go to the **previous frame**
- Right Go to the **next frame**
- ⌂ Page Up Switch to the **next media**
- ⌂ Page Up Switch to the **previous media**
- ↑ Up **Jump forward** in the current media
- ↓ Down **Jump backward** in the current media

If you have a numeric keypad you can use the following keys in alternative:

- The key / will allow you to view the previous frame
- The key * will allow you to view the next frame

To return in the media player mode press the **Play**  button in the toolbar.

2.7 Export events

Coded events can be exported in various formats.

2.7.1 Export events in tabular format

Observations > Export events > Tabular events

This function exports the events from selected observations to one or more files. Various formats are available:

- Plain text in tabular format
 - **Tab Separated Values** (TSV)
 - **Comma Separated Values** (CSV)
 - **Hyper Text Markup language** (HTML)
- Spreadsheet files
 - **OpenDocument** (ODS)
 - **Microsoft Excel** (XLSX, XLS)
- **Pandas dataframe** (to be loaded in Python with the [pickle module](#))
- **R dataframe** (to be loaded in R with [readRDS function](#))

If many observations are selected, BORIS will ask you to choose a directory in which to save the files. For spreadsheet formats (XLSX and ODS), the events can be exported to multiple worksheets within a single workbook. All these formats are suitable for further analysis.

Select the subjects, the behaviors and the time interval.

Set the time interval to **Observed events**.

Select subjects and behaviors

Subjects

Select all Unselect all Reverse selection

- ☒ Nina
- ☒ Himal
- ☒ Sharky
- ☒ Nautilus

Behaviors

Select all Unselect all Reverse selection

physiology

- ☐ Eat
- ☐ Defecate
- ☒ Drink
- ☒ Rest
- ☐ Urinate
- ☐ Vomit

Time interval

☒ Observed events ☐ User defined ☐ Media file(s) duration

Cancel OK

Select a **User defined** time interval.

Select subjects and behaviors

Subjects

Select all

Unselect all

Reverse selection

☒ Nina
☒ Himal
☒ Sharky
☒ Nautilus

Behaviors

Select all

Unselect all

Reverse selection

physiology

☐ Eat
☐ Defecate
☒ Drink
☒ Rest

Time interval

☐ Observed events
☒ User defined
☐ Media file(s) duration

Start time

+

0

:

00

:

10

:

000

☒ HH:MM:SS:MS
☐ seconds

End time

+

0

:

00

:

50

:

000

☒ HH:MM:SS:MS
☐ seconds

Cancel

OK

Example output of tabular events:

Observation id	Observation date	Description	Observation duration	Observation type	Source	Media duration (s)	FPS	Location	Weather	Temperature	Visitors
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046
0002	2016-05-17 00:00:24	Vegetation	34.47	Media file(s)	player #1:20160517162540.m2ts	36.000	25.000	4° 55' 59" N - 7° 25' 18"	sun	24	1046

Subject	Behavior	Behavioral category	Modifier #1	Behavior type	Time	Media file name	Image index	Image file path	Comment
Himal	Tear		Branches	START	0.000	20160517162540.m2ts	NA	NA	
Sharky	Tear		Branches	START	0.000	20160517162540.m2ts	NA	NA	
Nautilus	Tear		Branches	START	1.359	20160517162540.m2ts	NA	NA	
Nautilus	Tear		Branches	STOP	25.776	20160517162540.m2ts	NA	NA	
Nautilus	Carry objects		Branches	START	25.777	20160517162540.m2ts	NA	NA	
Nautilus	Carry objects		Branches	STOP	27.732	20160517162540.m2ts	NA	NA	
Sharky	Tear		Branches	STOP	30.688	20160517162540.m2ts	NA	NA	
Sharky	Locomotion		Walk	START	30.689	20160517162540.m2ts	NA	NA	
Sharky	Locomotion		Walk	STOP	31.819	20160517162540.m2ts	NA	NA	
Himal	Tear		Branches	STOP	33.898	20160517162540.m2ts	NA	NA	
Himal	Locomotion			START	33.899	20160517162540.m2ts	NA	NA	
Himal	Locomotion			STOP	34.470	20160517162540.m2ts	NA	NA	

2.7.2 Export aggregated events

Observations > Export events Aggregated events

This function exports the events corresponding to the selected subjects and behaviors from the selected observations.

Various formats are available:

- Plain text in tabular format
 - **Tab Separated Values** (TSV)
 - **Comma Separated Values** (CSV)
 - **Hyper Text Markup language** (HTML)
- Spreadsheet files
 - **OpenDocument** (ODS)
 - **Microsoft Excel** (XLSX, XLS)
- **SQL format** for populating a SQL database
- **SDIS format** for analysis with the GSEQ program available at <http://www2.gsu.edu/~psyrab/gseq>
- **Pandas dataframe** (to be loaded in Python with the [pickle module](#))
- **R dataframe** (to be loaded in R with [readRDS function](#))

If two or more observations are selected, you can choose to group all results into one file. If you do not want to group the results, BORIS will ask you to choose a directory in which to save the files (the observation ID will be used as the file name).

The **State events** are paired and in this case the event duration is available.

An arbitrary time interval can be selected by checking the **Limit to time interval** option. In this case, ongoing events will be started at the start time and stopped at the end time in the export file.

The following fields are available in the output:

- Observation id
- Observation date and time
- Observation description
- Observation type (Media file / Live / Pictures)
- Source (for media file and pictures)
- Total duration (in seconds, the duration of the observation based on the selected time interval)
- Media duration(s) (in seconds, for media file observation)
- FPS (frame/s, for video file, number of images per second)
- Independent variables (one column per variable)
- Subject name
- Observation duration by subject
- Behavior
- Behavioral category (if any)
- Behavior modifier(s) (one column per modifier)
- Behavior type (STATE / POINT)
- Start (seconds)
- Stop (seconds)
- Duration (seconds, duration of the event for STATE events)
- Media file name (for media file observation, media in which the event occurs)
- Image index start (for observations from pictures, index of the image where the event starts)
- Image index stop (for observations from pictures, index of the image where the event stops)
- Image file path start (for observations from pictures, path of the image where the event stops)
- Image file path stop (for observations from pictures, path of the image where the event stops)
- Comment start
- Comment stop

Example of aggregated event export (TSV, CSV, XLSX, ODS, HTML):

Observation id	Observation date	Description	Observation type	Source	Total duration	Media duration (s)	FPS (frame/s)	Location	Weather	Temperature	Visitors
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0004	2016-05-17 00:00:59	Central trunks	Media file	player #1:20160517162952.m2ts	50.697	59.040	25.000	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
0005	2016-05-17 00:00:49	In the pool	Media file	player #1:20160517163131.m2ts	40.770	40.800	25.000	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
0005	2016-05-17 00:00:49	In the pool	Media file	player #1:20160517163131.m2ts	40.770	40.800	25.000	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
0006	2016-05-17 00:00:42	In the pool	Media file	player #1:20160517163231.m2ts	61.830	61.920	25.000	44° 55' 59" N - 7° 25' 18" E	sun	24	1046

Subject	Observation duration by subject by observation	Behavior	Behavioral category	Modifier #1	Behavior type	Start (s)	Stop (s)	Duration (s)	Media file name	Image index start	Image index stop	Image file path start	Image file path stop	Comment start	Comment stop
Himal	47.613	Locomotion	Not defined	Run	STATE	8.253	11.779	3.526	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Run	STATE	8.329	13.789	5.460	20160517162952.m2ts	NA	NA				
Sharky	48.452	Swim	Not defined		STATE	10.400	58.852	48.452	20160517162952.m2ts	NA	NA				
Nina	27.431	Alert	Not defined		STATE	12.778	14.660	1.882	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Walk	STATE	13.790	14.865	1.075	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Jump	STATE	14.348	16.467	2.119	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Jump	STATE	14.866	23.600	8.734	20160517162952.m2ts	NA	NA				
Nina	27.431	Rest	physiology		STATE	15.001	40.550	25.549	20160517162952.m2ts	NA	NA				
Himal	47.613	Alert	Not defined		STATE	16.468	24.407	7.939	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Rest	physiology		STATE	23.601	24.228	0.627	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Walk	STATE	24.917	44.760	19.843	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Locomotion	Not defined	Run	STATE	39.683	42.313	2.630	20160517162952.m2ts	NA	NA				
Nautilus	24.431	Rest	physiology		STATE	42.314	48.219	5.905	20160517162952.m2ts	NA	NA				
Himal	47.613	Alligroom	Not defined		STATE	44.761	48.364	3.603	20160517162952.m2ts	NA	NA				
Himal	47.613	Locomotion	Not defined	Walk	STATE	48.365	49.274	0.909	20160517162952.m2ts	NA	NA				
Himal	47.613	Drink	physiology		STATE	49.275	50.408	1.133	20160517162952.m2ts	NA	NA				
Himal	47.613	Swim	Not defined		STATE	50.409	58.950	8.541	20160517162952.m2ts	NA	NA				
Sharky	40.77	Manipulate	Not defined		STATE	0.000	40.770	40.770	20160517163131.m2ts	NA	NA				
Nautilus	40.77	Manipulate	Not defined		STATE	0.000	40.770	40.770	20160517163131.m2ts	NA	NA				
Himal	44.368	Manipulate	Not defined		STATE	0.000	40.760	40.760	20160517163231.m2ts	NA	NA				
Nautilus	45.037	Manipulate	Not defined		STATE	0.000	17.448	17.448	20160517163231.m2ts	NA	NA				
Sharky	46.537	Manipulate	Not defined		STATE	1.824	48.361	46.537	20160517163231.m2ts	NA	NA				

Example of SQL export of aggregated events:

```
CREATE TABLE aggregated_events (id INTEGER PRIMARY KEY ASC, observation TEXT, subject TEXT, behavior TEXT, type TEXT, modifiers TEXT, start FLOAT, stop FLOAT,
comment TEXT, comment_stop TEXT, image_index_start INTEGER, image_index_stop INTEGER, image_path_start TEXT, image_path_stop TEXT);
INSERT INTO "aggregated_events" VALUES(1,'0001_a','Himal','Tear','STATE','Branches',0.0,30.199,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(2,'0001_a','Himal','Locomotion','STATE','Walk',30.2,32.4,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(3,'0001_a','Nautilus','Tear','STATE','Branches',0.0,32.4,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(4,'0001_b','Himal','Tear','STATE','Branches',0.0,30.199,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(5,'0001_b','Himal','Locomotion','STATE','Walk',30.2,32.4,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(6,'0001_b','Nautilus','Tear','STATE','Branches',0.0,31.4,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(7,'0002','Himal','Tear','STATE','Branches',0.0,33.898,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(8,'0002','Himal','Locomotion','STATE','','33.899,34.47','','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(9,'0002','Sharky','Tear','STATE','Branches',0.0,30.688,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(10,'0002','Sharky','Locomotion','STATE','Walk',30.689,31.819,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(11,'0002','Nautilus','Tear','STATE','Branches',1.359,25.776,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(12,'0002','Nautilus','Carry objects','STATE','Branches',25.777,27.732,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(13,'0003','Nina','Locomotion','STATE','Walk',21.626,22.5,'','','','NA','NA',NULL,NULL);
INSERT INTO "aggregated_events" VALUES(14,'0003','Nina','Manipulate','STATE','','0.0,21.625','','','','NA','NA',NULL,NULL);
```

2.7.3 Export events as behavioral sequences

Observations > Export events as behavioral sequences

Behavioral strings can be used with the **Behatrix** program: [Behatrix](#)

Example:

```
# observation id: demo#1
# observation description:
# Media file name: video1.mp4, video2.mp4
```

```
Subject #1:
eat|jump|eat|jump
```

```
Subject #2:
eat|rest|jump|eat|jump
```

2.7.4 Export events as Praat TextGrid

Observations > Export events as Praat TextGrid

Example:

```
File type = "ooTextFile"
Object class = "TextGrid"

xmin = 4.3
xmax = 113.988
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "Subject #1"
    xmin = 4.3
    xmax = 10.0
    intervals: size = 1
    intervals [1]:
      xmin = 4.3
      xmax = 10.0
      text = "eat"
  item [2]:
    class = "IntervalTier"
    name = "Subject #2"
    xmin = 26.6
    xmax = 113.988
```



```

intervals: size = 1
intervals [1]:
  xmin = 26.6
  xmax = 113.988
  text = "eat"

```

2.7.5 Export events for analysis with JWatcher

Observations > Export events for analysis with JWatcher

JWatcher is a powerful tool for the quantitative analysis of behavior.

Events coded with BORIS can be exported for analysis with JWatcher.

Click **Observations > Export events > for analysis with JWatcher** to export the coded events.

BORIS will ask you to select a directory. After that, for each combination of selected observation and selected subject, the following files will be created:

- the Focal Data File (.dat)
- the Focal Analysis Master File (.faf)
- the Focal Master File (.fmf)

These files can be used to analyze your observations with JWatcher.

2.7.6 Export events as Behaviors Binary Table

Observations > Export events as Behaviors Binary Table

A time interval will be requested (in seconds). The observation will be checked every n seconds and the presence (1, absence: 0) of the selected behaviors will be exported in a table for each selected subject.

Example for a time interval of 1 second:

time	Alert	Drink	Locomotion	Swim
0.0	0	1	0	0
1.0	0	1	0	0
2.0	0	1	0	0
3.0	0	1	0	0
4.0	0	1	0	0
5.0	0	1	0	0
6.0	0	1	0	0
7.0	0	1	0	0
8.0	0	1	0	0
9.0	1	0	0	0
10.0	1	0	0	0
11.0	0	0	1	0
12.0	1	0	0	0
13.0	1	0	0	0
14.0	1	0	0	0
15.0	1	0	0	0
16.0	1	0	0	0
17.0	1	0	0	0
18.0	0	0	1	0
19.0	0	0	1	0
20.0	0	0	1	0
21.0	0	0	1	0
22.0	1	0	0	0
23.0	0	0	0	0
24.0	0	0	0	0
25.0	0	0	0	0
26.0	0	0	0	0
27.0	0	0	0	0
28.0	0	0	0	0
29.0	0	0	0	0
30.0	0	0	0	0
31.0	0	0	0	0
32.0	0	0	0	0
33.0	0	0	0	1
34.0	0	0	0	1
35.0	0	0	0	1
36.0	0	0	0	1
37.0	0	0	0	1
38.0	0	0	0	1
39.0	0	0	0	1
40.0	0	0	0	1
41.0	0	0	0	1
42.0	0	0	0	1

43.0	0	0	0	1
44.0	0	0	0	1
45.0	0	0	0	1
46.0	0	0	0	1
47.0	0	0	0	1
48.0	0	0	0	1
49.0	0	0	0	1

2.7.7 Extract clips from media files corresponding to coded events

Sequences of media file corresponding to coded events can be extracted from media files:

1. Click on **Observations > Extract clips from media files** option.
2. Choose the observation(s).
3. Select the subjects and events to be extracted. Modifiers can be included
4. Select the time interval around the events (in seconds, the default value is 0).
5. Select the tracks to include in clips:
 - Video and audio
 - Only video
 - Only audio
6. Select a destination directory that will contain the extracted clips.

The time offset will be subtracted from the starting time of event and added to the stopping time. All the extracted clips will be saved in the selected directory following the file name format:

{observation id}_{player}_{subject}_{behavior}_{start time}-{stop time}

If modifiers are included:

{observation id}_{player}_{subject}_{behavior}_{start time}-{stop time}_{modifiers concatenated with +}

2.7.8 Extract frames corresponding to coded events

The frames corresponding to coded events can be extracted and saved as images.

1. Click on **Observations > Extract frames from media files** option.
2. Choose the observation(s).
3. Select the subjects and events to be extracted. Modifiers can be included
4. Select the time interval around the events (in seconds, the default value is 0).
5. Select the image format:
 - JPG - small size / low quality
 - PNG - big size / high quality
6. Select a destination directory that will contain the extracted frames.

2.7.9 Export transitions matrix

3 transitions matrix outputs are available: The matrix of frequencies of transitions, the matrix of frequencies of transition after each behavior and the matrix of number of transitions.

Matrix of frequencies of transitions

This matrix contains the frequencies of total transitions. The sum of all frequencies must be 1.

Example of frequencies of transitions matrix:

eat	sleep	walk	
eat	0.0	0.286	0.143
sleep	0.143	0.0	0.143
walk	0.286	0.0	0.0

In this matrix you can see that the **eat** behavior precedes the **sleep** behavior with a frequency of **0.286** of the total number of transitions.

Matrix of frequencies of transitions after behavior

This matrix contains the frequencies of transitions after each behavior. The sum of each row must be 1.

Example:

eat	sleep	walk	
eat	0.0	0.667	0.333
sleep	0.5	0.0	0.5
walk	1.0	0.0	0.0

In this example you can see that **sleep** follows **eat** with a frequency of **0.667** and **walk** follows with a frequency of **0.333**.

Matrix of number of transitions

This matrix contains the number of transitions after each behavior.

Example:

eat	sleep	walk	
eat	0	2	1
sleep	1	0	1
walk	2	0	0

2.8 Playback menu

2.8.1 Jump

Jump forward

Allows you to jump forward in the current media file. See **File > Preferences** for setting the jump value.

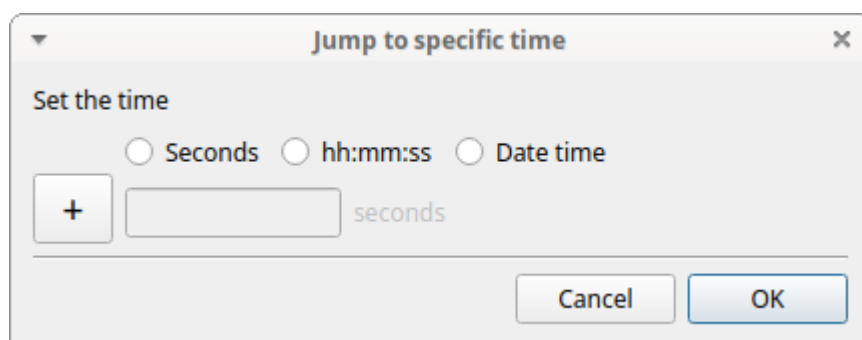
Jump backward

Allows you to jump backward in the current media file. See **File > Preferences** for setting the jump value.

Jump to specific time

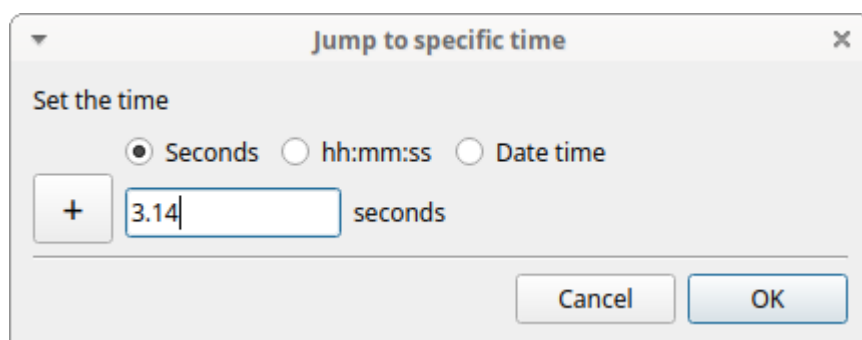
Allows you to go to a specific time in the current media file.

The time selection widget will pop-up:



3 formats are available to select the time:

- Decimal seconds:



- HH:MM:SS:ZZZ format (ZZZ indicates the milliseconds):

• A date-time format (YYYY-MM-DD hh:mm:ss.zzz):

2.8.2 Change time offset of players

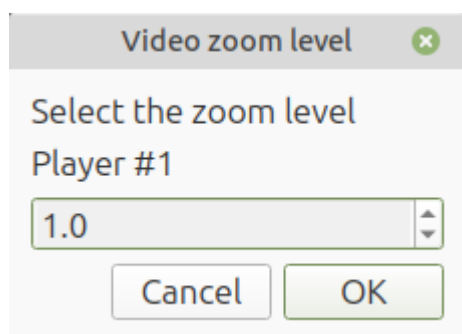
Allows you to change the time offset for each player (in seconds)

2.8.3 Deinterlace

Check this option if your video is interlaced

2.8.4 Zoom level

The zoom level can also be set using the menu **Playback > Zoom level**



Using the keyboard

Click the media player you want to set the zoom level.

Zoom in + Or +

Zoom out + Or +

Reset zoom level +

Important

On Microsoft Windows, only the and keys on the numeric keypad can be used.

Using the mouse

Click the media player you want to set the zoom level.

Zoom in + **Mouse Wheel Up** or Double click on left mouse button

Zoom out + **Mouse Wheel Down** or Double click on right mouse button

Reset zoom level Click the **right mouse button** on the video.

Important

On Microsoft Windows, the mouse cannot be used to zoom the video

2.8.5 Pan video

Click the media player you want to pan.

Using the keyboard

Pan Up +

Pan Down +

Pan Left +

Pan Right +

Using the mouse

Pan Up: Mouse Wheel Up (the video moves down)

Pan Down: Mouse Wheel Down (the video moves up)

Pan Left:  + **Mouse Wheel Up** (the video moves to the right)

Pan Right:  + **Mouse Wheel Down** (the video moves to the left)

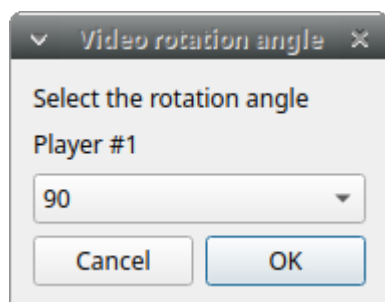
Reset Pan and zoom:  + **Left mouse button**

Important

On Microsoft Windows, the mouse cannot be used to pan the video

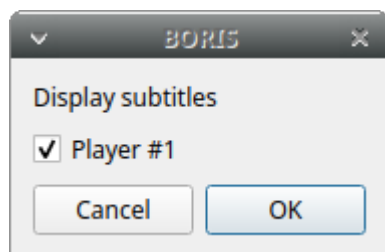
2.8.6 Rotate video

Select the video rotation angle for each player using the menu **Playback > Rotate video**. The available rotation angles are: 0, 90, 180 and 270.



2.8.7 Display subtitles

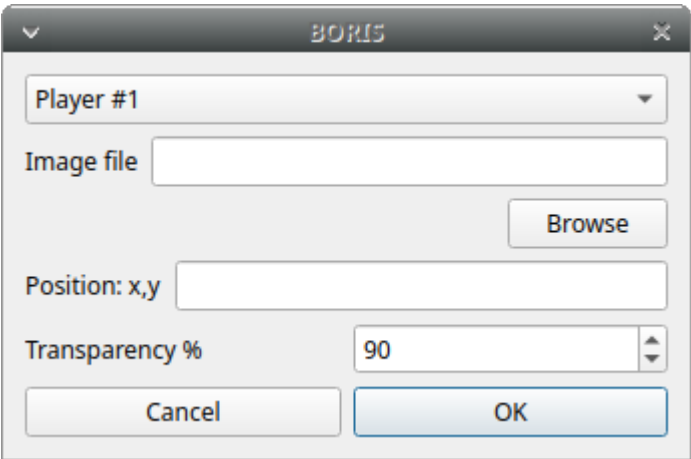
Select to display or hide the subtitles using the menu **Playback > Display subtitles**. The subtitles file must have exactly the same name of the video file except for the extension and be placed in the same directory.



2.8.8 Image overlay on video

Select an image overlay to be displayed on the video **Playback > Image overlay on video > Add**. If the selected image does not have a transparent background the transparency can be set from 0 (full transparency) to 255 (no transparency).

The image must be in PNG format, if the image is smaller than the video resolution the image position can be set from the top-left corner (x: horizontally, y: vertically).

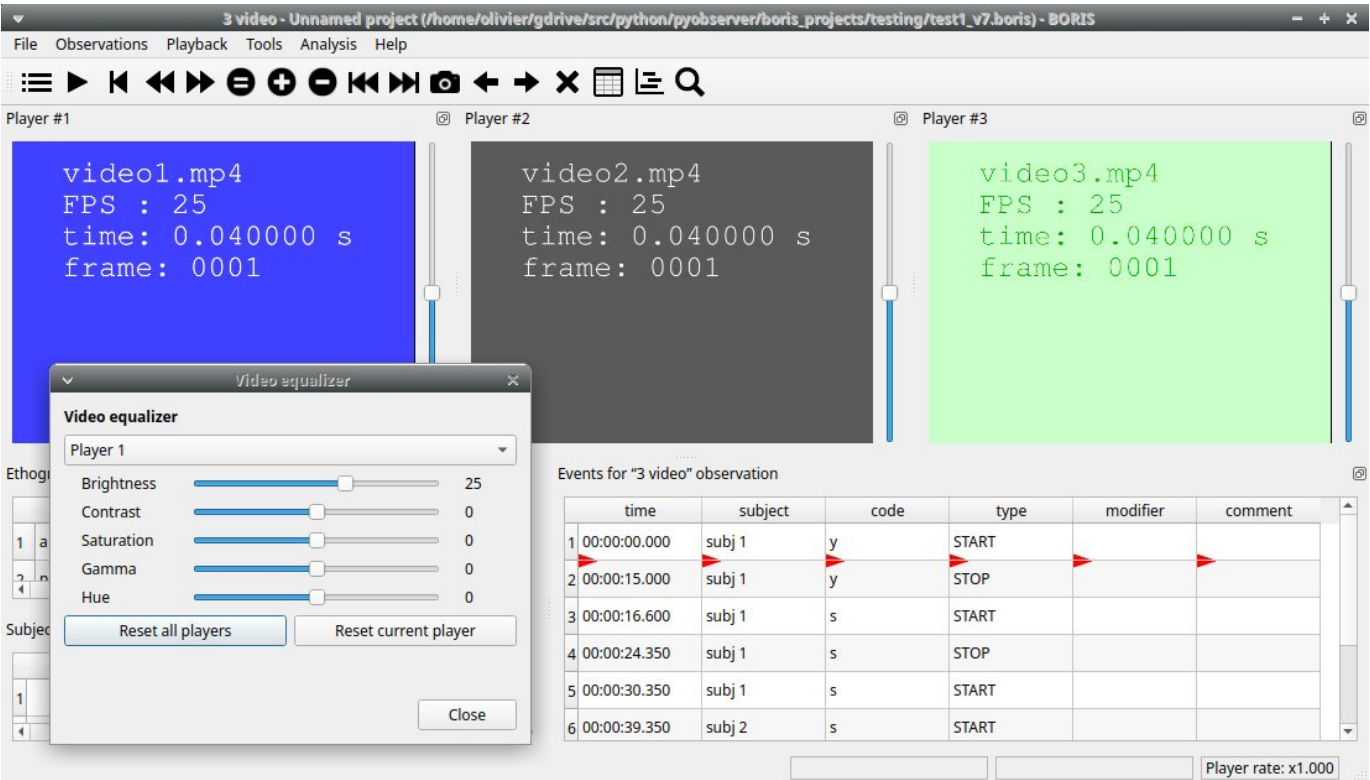


Select > Playback > Image overlay on video > Remove to remove the image overlay.

2.8.9 Video equalizer

Playback > Video equalizer

Using this function the **brightness**, the **contrast**, the **saturation**, the **gamma** and the **Hue** can be set for each player.



2.9 Tools

2.9.1 Media file

MEDIA FILE INFORMATION

Obtain information about your media file like duration, resolution, FPS, bitrate ...

RE-ENCODE / RESIZE VIDEO

BORIS can re-encode and resize your video files in order to reduce the size of the files and have a smooth coding (specially with two video files playing together). The re-encoding and resizing operations are done with the embedded ffmpeg program with high quality parameters (bitrate 2000k).

Select the files you want re-encode and resize and select the horizontal resolution in pixels (the default is 1024). The aspect ratio will be maintained.

You can continue to use BORIS during the re-encoding/resizing operation.

The re-encoded/resized video files are renamed by adding the re-encoded.avi extension to the original files.

ROTATE VIDEO

BORIS can rotate your video files in order to code them using the right view. The rotating operation is done with the embedded ffmpeg program using the same quality parameters as the original video.

Select the files you want to rotate and select the rotation angle: **Rotate 90 clockwise**, **Rotate 90 counter clockwise**, or **Rotate 180**.

The aspect ratio will be maintained.

You can continue to use BORIS during the rotation operation.

The rotated video files are renamed by adding the **rotated\<ANGLE>** to the original file name.

MERGE MEDIA FILES

Use this function to concatenate various media files together.

CREATE VIDEO SPECTROGRAM

Create a video with the spectrogram of the audio track

2.9.2 Plot events in real-time

This function can be activated with **Tools > Plot event in real time**.

The screenshot displays the Video1 software interface. The main window is titled "Video1 - test 5 - BORIS". It features a menu bar (File, Observations, Playback, Tools, Analysis, Help) and a toolbar with various playback controls. The video player shows a black frame with a blue overlay displaying video metadata: "video1.mp4", "FPS : 25", "time: 16.880000 s", and "frame: 0422".

An "Events plot" window is open, showing a timeline with three colored bars: an orange bar from approximately 12.44 to 21.64 seconds, a green bar from 21.64 to 36.44 seconds, and another orange bar from 50.28 to 60.00 seconds. The y-axis is labeled with "s1 - b", "s2 - b", and "s1 - a".

The "Ethogram" window shows a table with columns "Key", "Code", and "Type". It lists two events: "1 p p Point event" and "2 a a State event". Below this, the "Subjects" window shows a table with columns "Key" and "Name", listing "1 No focal subject", "2 1 s1", and "3 2 s2".

The "Events for 'video1' observation" window displays a table with columns: "time", "subject", "code", "type", "modifier", and "con". The data is as follows:

time	subject	code	type	modifier	con
1 12.440	s1	a	START	None	
2 21.640	s2	b	START		
3 28.200	s1	a	STOP	None	
4 28.200	s1	b	START		
5 35.200	s1	b	STOP		
6 36.440	s1	a	START	None	
7 50.280	s2	b	STOP		
8 60.000	s1	a	STOP	None	

The interface also includes a "Player #1" section, a "Player paused" status, and a "Player rate: x1.000" indicator.

2.9.3 Geometric measurements

Several geometric measurements are available: **distances**, **areas**, and **angles** can be measured, and **point positions** can be recorded.

Click **Tools > Geometric measurements** to activate the measurement tools.

Geometric measurements

☐ Point (left click)
 ☐ Polyline (left click for vertices, right click to finish)
 ☐ Polygon (left click for Polygon vertices, right click to close the polygon)
 ☐ Angle (vertex: left click, segments: right click)
 ☐ Oriented angle (vertex: left click, segments: right click)
 ☒ Measurements are persistent

Choose color of marks

Scale

Reference

Pixels

1

1

Player	media file name	Time	Frame index	pe of measureme	x

Clear measurements

Save current picture

Save all pictures

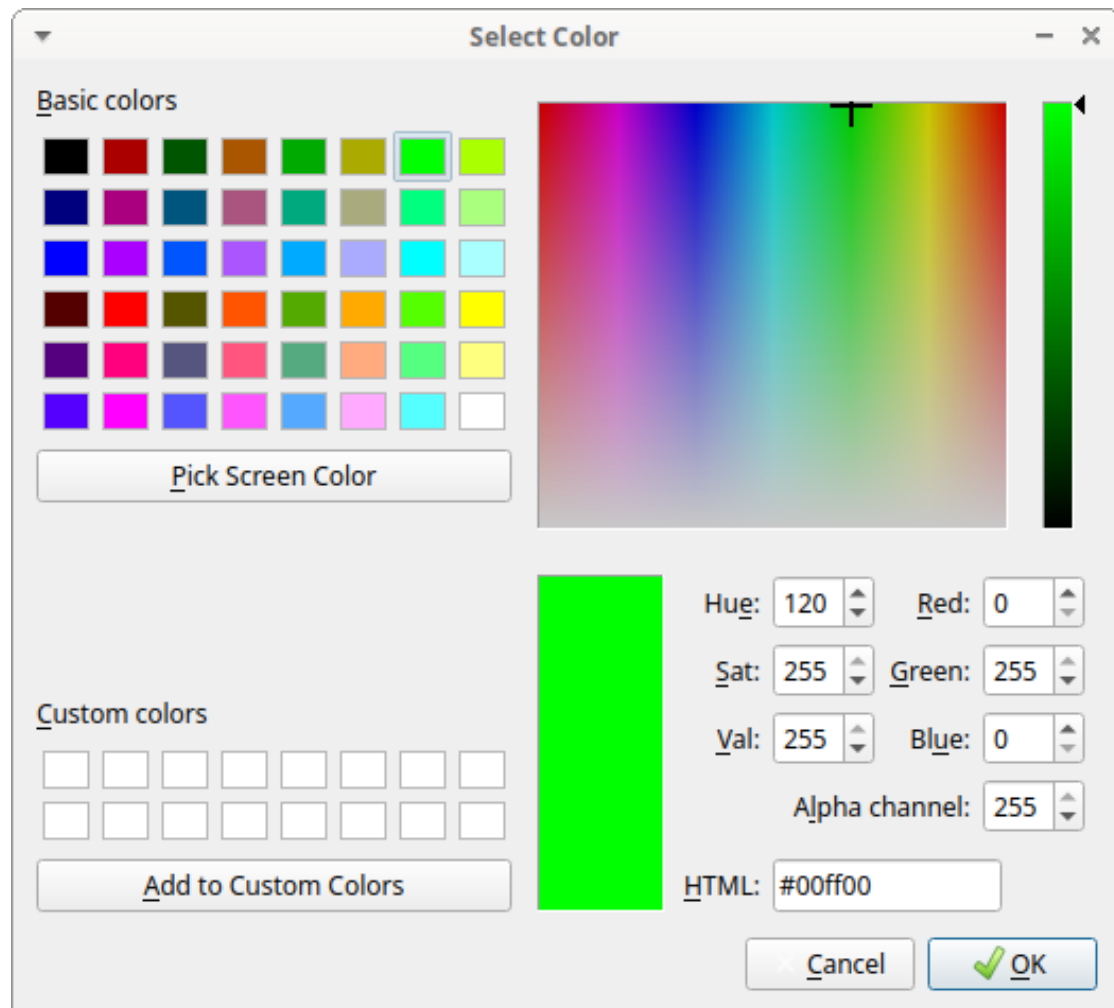
Save results

Close

The geometric measurements window

Mark color

Use the **Choose color of marks** button to select a color. All marks will be drawn with the selected color. The color transparency can be set using the **Alpha channel** value (0 for 100% transparent, 255 for a solid color).



The color selection window

Setting the scale

For distance and area measurements, you can set a scale so that the results are expressed in real units such as centimeters or meters.

Scale	
Reference	Pixels
1	1

Setting the scale

1. Measure a reference object with a known size on the frame (with the distance tool; see the next section for details) and enter the pixel distance in the **Pixel** text box.
2. Set the real size of the reference object in the **Reference** text box (must be a number without unit).

Point

Select the **Point** radio button. Click the left mouse button on the video/image to record the position of the clicked pixel.

Geometric measurements

☒ Point (left click)
☐ Polyline (left click for vertices, right click to finish)
☐ Polygon (left click for Polygon vertices, right click to close the polygon)
☐ Angle (vertex: left click, segments: right click)
☐ Oriented angle (vertex: left click, segments: right click)
☒ Measurements are persistent Choose color of marks

Scale

Reference: Pixels:

Player	media file name	Time	Frame index	Type of measurement	x	y	Distance	Area	Angle	Coordinates
1	video1.mp4	0.000	0	Point	193	303	NA	NA	NA	[(193, 303)]
1	video1.mp4	0.000	0	Point	324	296	NA	NA	NA	[(324, 296)]
1	video1.mp4	0.000	0	Point	442	265	NA	NA	NA	[(442, 265)]
1	video1.mp4	0.000	0	Point	447	376	NA	NA	NA	[(447, 376)]
1	video1.mp4	0.000	0	Point	262	395	NA	NA	NA	[(262, 395)]
1	video1.mp4	0.000	0	Point	101	269	NA	NA	NA	[(101, 269)]

Distance measurements

Select the **Distance** radio button. Click the left mouse button on the frame bitmap to set the start of the segment that will be measured. A circle with a cross will be drawn. Click the right mouse button to set the end. A red circle with a cross will be drawn. The distance between the two selected points will be available in the text area of the **Measurements window**.

Geometric measurements

☐ Point (left click)
☒ Polyline (left click for vertices, right click to finish)
☐ Polygon (left click for Polygon vertices, right click to close the polygon)
☐ Angle (vertex: left click, segments: right click)
☐ Oriented angle (vertex: left click, segments: right click)
☒ Measurements are persistent Choose color of marks

Scale

Reference: Pixels:

Player	media file name	Time	Frame index	Type of measurement	x	y	Distance	Area	Angle	Coordinates
1	video1.mp4	0.240	6	Polyline	NA	NA	417.3	NA	NA	[(56, 328), (472, 295)]
1	video1.mp4	0.240	6	Polyline	NA	NA	414.7	NA	NA	[(83, 392), (496, 430)]
1	video1.mp4	0.240	6	Polyline	NA	NA	101.2	NA	NA	[(531, 196), (451, 258)]

Area measurements

Select the **Area** radio button. Click the left mouse button on the frame bitmap to set the area vertices. Circles with a cross will be drawn. Click the right mouse button to close the area. The area of the drawn polygon will be available in the text area of the **Measurements window**.

Angle measurements

Select the **Angle** radio button. Click the left mouse button on the frame bitmap to set the angle vertex. A red circle with a cross will be drawn. Click the right mouse button to set the two segments. Circles with a cross will be drawn. The angle between the two drawn segments will be available in the text area of the **Measurements window**.

Persistent measurements

If the **Measurements are persistent** checkbox is selected, the measurement overlays will remain available on all frames; otherwise, they will be cleared between frames.

The marks selected on other frames will be drawn in red.

2.9.4 Coding pad

During an observation, a coding pad with the available behaviors can be displayed (**Tools > Coding pad**). This **Coding pad** allows you to code using a touch screen or by clicking the buttons. When the **Coding pad** is displayed, you can still code using the keyboard or the ethogram.



The button size can be increased or decreased.

Button colors can be set per behavior, per behavioral category, or disabled entirely.

See the drop-down list in the upper-left corner of the Coding pad window.

2.9.5 Subjects pad

A pad with all defined subjects, or only filtered subjects, can be displayed during the observation (**Tools > Subjects pad**). This **Subjects pad** allows you to select the focal subject using a touch screen or by clicking the buttons. When the **Subjects pad** is displayed, you can continue to select the focal subject using the keyboard or the subjects list.



2.9.6 Transitions flow diagram

BORIS can generate DOT scripts and flow diagrams from the transitions matrices (See Observations > Create transition matrix for obtaining the transitions matrices).

DOT script (Graphviz language)

Tools > Transitions flow diagram > Create transitions DOT script

Choose one or more transition matrix files and BORIS will create the corresponding DOT script file(s).

The DOT script files can then be used with [Graphviz](#) (Graph Visualization Software) or [WebGraphviz](#) (Graphviz in the Browser) to generate transition flow diagrams.

See [DOT \(graph description language\)](#) for details.

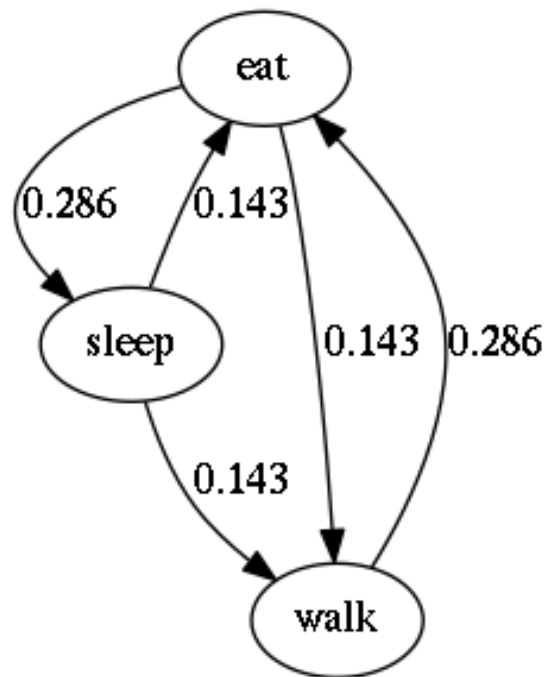
Flow diagram

If [Graphviz](#) (Graph Visualization Software) is installed on your system (and the **dot** program available in the path), BORIS can generate a flow diagram (PNG format) from a transition matrix file.

Tools > Transitions flow diagram > Create transitions flow diagram

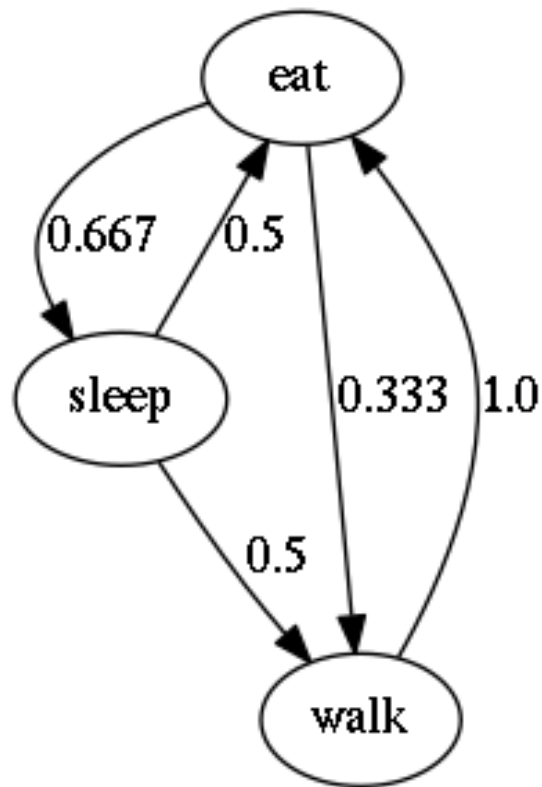
Choose one or more transition matrix files and BORIS will create the corresponding flow diagrams.

Flow diagram of frequencies of transitions



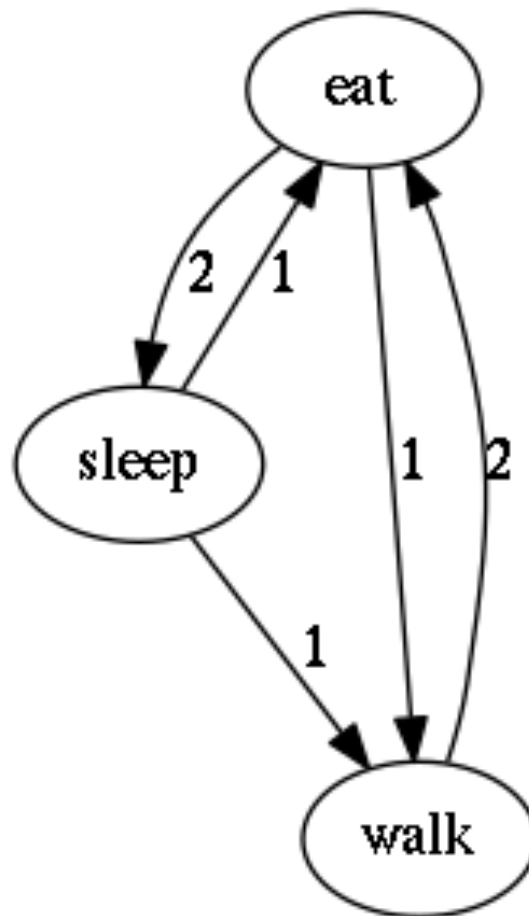
Frequencies of total transitions (the frequencies are plotted on the edges)

Flow diagram of frequencies of transitions after behavior



Frequencies of transitions after behavior (the frequencies are plotted on the edges)

Flow diagram of number of transitions



Number of transitions (the frequencies are plotted on the edges)

2.10 Coding map

A coding map is a bitmap image with user-defined clickable areas that will help to code for behaviors or modifiers for a behavior.

2 types of coding maps are available:

- Behaviors coding map
- Modifiers coding map

2.10.1 The Behaviors coding map

BORIS allows creating a **Behaviors coding map** using the **Map creator** tool (**Tools > Create a coding map > for behaviors**).

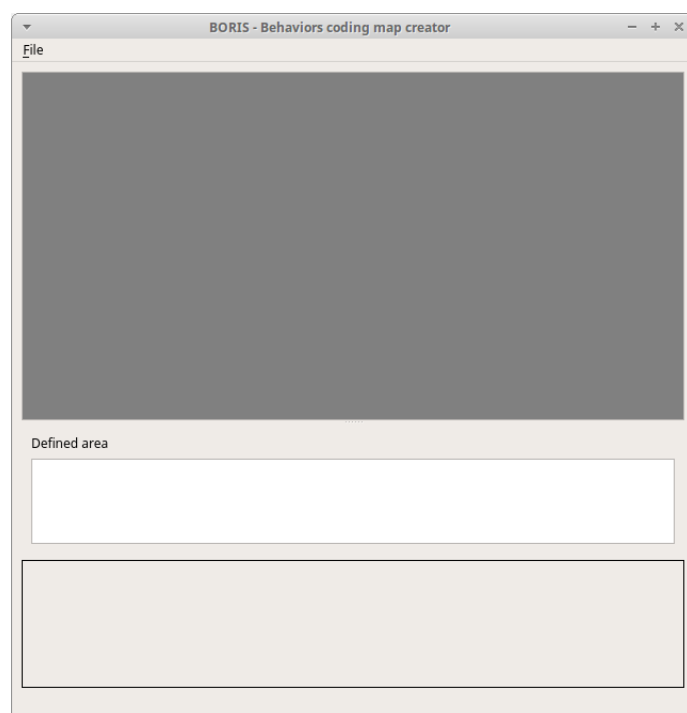
A **Behaviors coding map** can be created only if you have defined behaviors in your ethogram.

Creating a Behaviors coding map

To create a new **Behaviors coding map** launch the **Behaviors coding map creator**

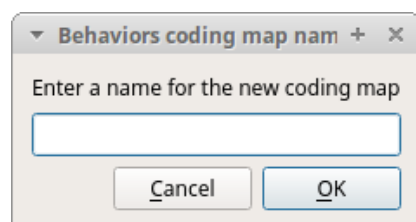
Tools > Create a coding map > for behaviors).

A new window will open



File > New behaviors coding map

Enter a name for the new **Behaviors coding map**

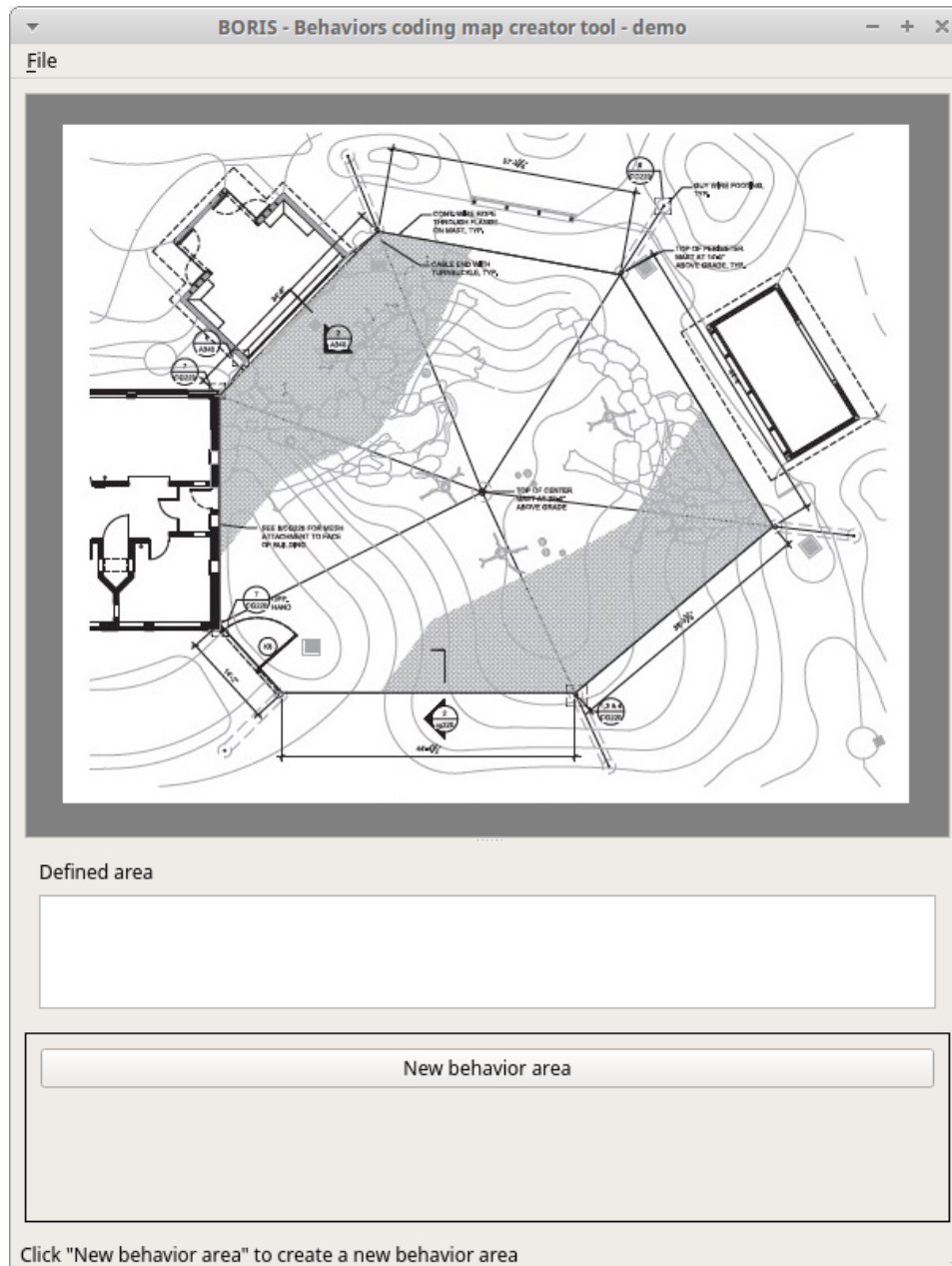


Loading a bitmap for a behaviors coding map

Click the **Load bitmap** button in the bottom of the window and select a bitmap image (PNG and JPEG formats are accepted).

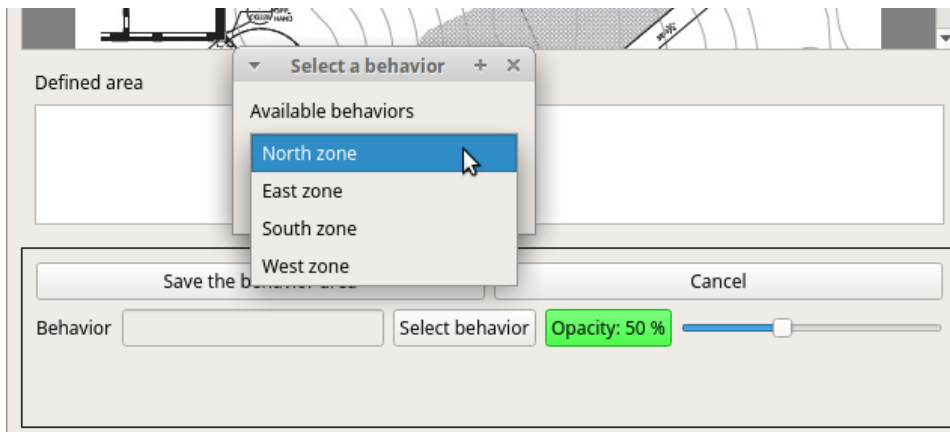
If the size of your bitmap image is bigger than 640 x 640 pixels BORIS will resize it to 640 x 640 pixels keeping the aspect ratio and store the resized version in the coding map file.

The bitmap will be displayed



Adding areas corresponding the behaviors

Click the **New behavior area** button in the bottom of the window and select a behavior by clicking on the **Select behavior** button.



The available behaviors are taken from the ethogram of the current project.

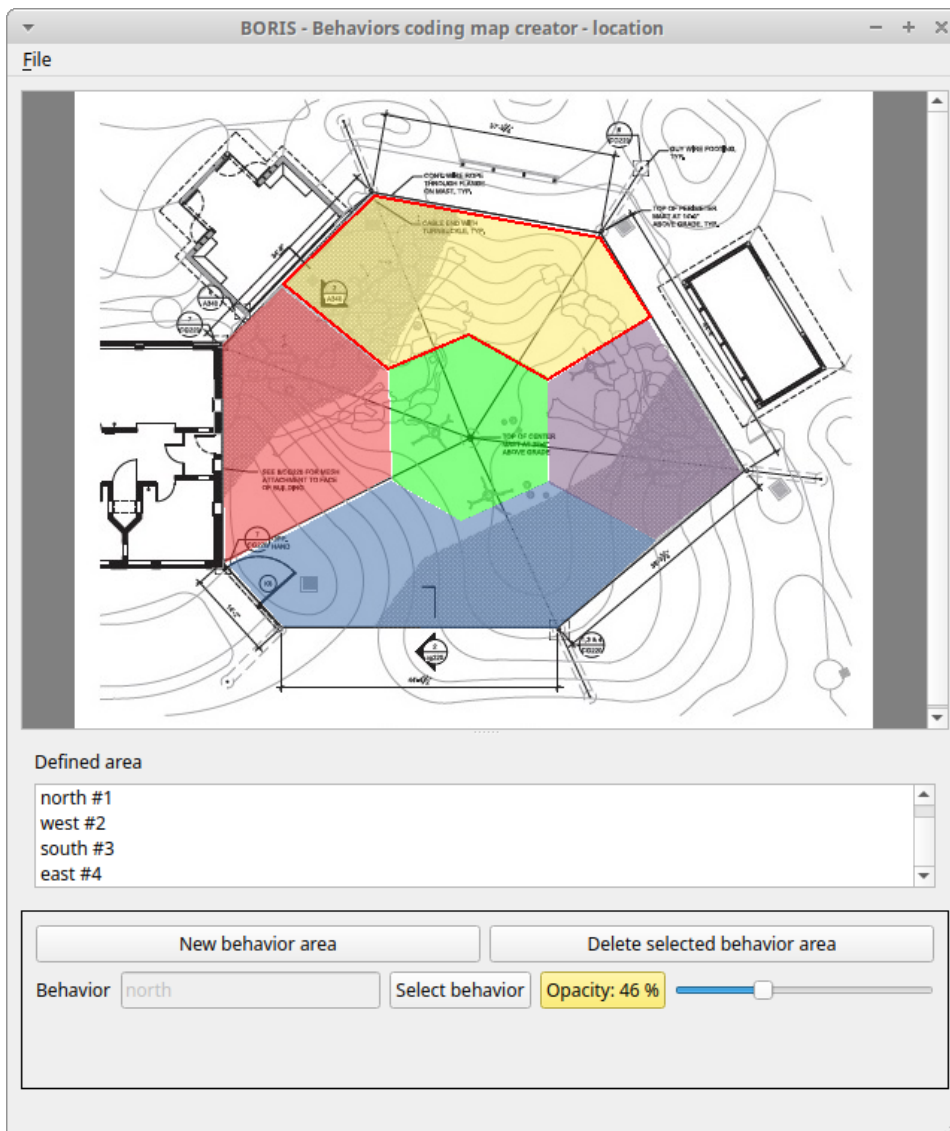
Click on the bitmap to define the vertex on the area that will code the selected behavior. Close the area by clicking again on the first point.

The color of the new area can be changed using the **Opacity** button. The opacity can be changed (from 0 to 100%) using the slider.

Save the behavior area by clicking on the **Save the behavior area** button

The area will be added to the **Defined area** list

You can add more area and also add more than one area for a same behavior. Two or more areas can overlap. In this case all corresponding behaviors will be triggered.



Add the Behaviors coding map to the current project

File > Add coding map to project

The coding map will be added to the current project

You can add a **Behaviors coding map** to the current project from a file containing the coding map:

(File > Edit project > Behaviors coding map > Add a behaviors coding map)

Saving the Behaviors coding map

Saving the **Behaviors coding map** will create a file containing the **Behaviors coding map** including the bitmap image.

File > Save the current Behaviors coding map

The file containing the **Behaviors coding map** can be then reloaded in the **Behaviors coding map creator** or added to a BORIS project (File > Edit project > Behaviors coding map > Add a behaviors coding map)

2.10.2 The Modifiers coding map

BORIS allows creating a modifiers coding map using the **Modifiers Map creator** tool (**Tools > Create a coding map > for modifiers**.) Clickable areas may correspond to specific modifiers that can be meaningful for the behavioral coding. Facial expression is the case we thought to when developing this function.

Creating a modifiers coding map

Loading a bitmap for a modifiers coding map

To create a new **Modifiers coding map**, launch the **Modifiers Map creator** tool (**Tools > Create a coding map > for modifiers**). The BORIS main window will be replaced by the **Modifiers Map creator** window. Click on **Modifiers Map creator > New Modifiers map** and enter a name for the new map in the edit box. You have to load a bitmap image (JPEG or PNG) using the **Load bitmap** button. The loaded image will be displayed.



If the size of your bitmap image is bigger than 640 x 640 pixels BORIS will resize it to 640 x 640 pixels keeping the aspect ratio and store the resized version in the coding map file.

Adding areas corresponding to the modifiers

To create clickable areas on a coding map, you have to click on the **New area** button and enter an **Area code** in the edit box. The new area can now be defined by clicking on the image. The drawing tool allows defining a irregular polygon (a plane shape with straight sides, which does not have all sides equal and all angles equal) by clicking to determine subsequent vertices. It can be convex or concave. Straight sides must not cross each other. Once selected an area can be deleted using the **Delete area** button. When an area is closed and its name has been defined in the **Area code** field, it can be saved by using the **Save area** button. The areas can partially overlap each other. See the **Using a Coding map** section for more details. Once

all areas are added the entire map can be saved using the **Save map** option menu (**Map creator > Save map**). The map is now saved in its own file (.boris_map) which is NOT part of the BORIS project. A map can be edited at anytime by opening the map file from the **Open map** menu option (**Map creator > Open map**).

Adding a modifiers coding map to your project

Creating a Coding map is not automatically adding the map to your project. The Coding map have to be added to your project by selecting the corresponding **Behavior type (Point event with coding map, State event with coding map)**. BORIS will ask to select the file name containing the coding map (.boris_map) and load the coding map in the project. The coding map name will appear in the **Coding map** column and will be saved in the BORIS project file.



Important

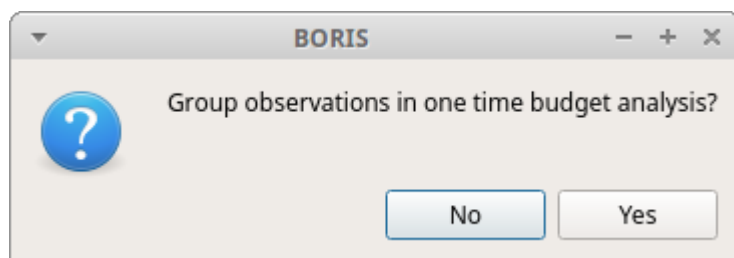
If you later modify your coding map you must reload the new version in your BORIS project.

2.11 Analysis and plot

2.11.1 Time budget analysis

The **Time budget analysis** can be performed by behavior (including or excluding modifiers) or by behavioral category. Choose the option from the **Analysis** menu.

The **Time budget analysis** can be performed on one or more observations. If you select more than one observation, you can choose either a global time budget analysis that includes all selected observations or a separate time budget analysis for each observation.



Choose **Yes** to group all observations into a single time budget analysis.

The **Analysis > Time budget** menu option shows the time budget for the events of the selected observations. Select the subjects and behaviors you want to include in the time budget analysis:

Select subjects and behaviors

Subjects

Select all Unselect all Reverse selection

- ☒ Nina
- ☒ Himal
- ☒ Sharky
- ☒ Nautilus

Behaviors

Select all Unselect all Reverse selection

physiology

- ☒ Eat
- ☒ Defecate
- ☒ Drink
- ☒ Rest
- ☒ Urinate
- ☒ Vomit

social

- ☒ Play on the ground
- ☐ Play in the water

reproduction

- ☐ Breed

No category

- ☒ Tear

☐ Include modifiers ☐ Exclude behaviors without events

Time interval

☒ Observed events ☐ User defined ☐ Media file(s) duration

Cancel OK

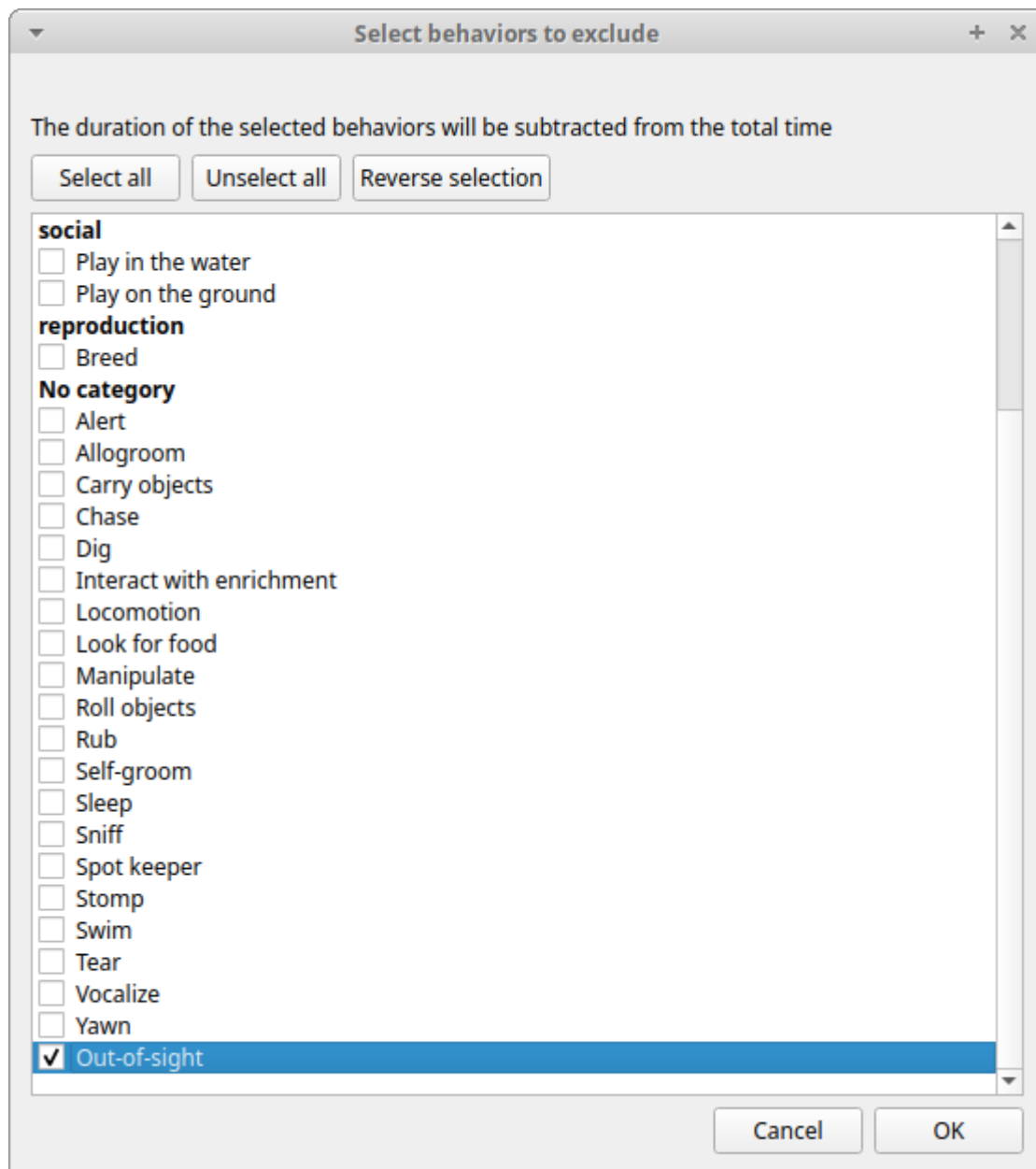
All behaviors can be selected or deselected by clicking the Category name (in bold).

You can choose whether to include behavior modifiers in the time budget analysis and to exclude behaviors without coded events.

The time budget analysis can be restricted to part of the observation:

- Full observation(s): the analysis will be made on the full observation length.
- Limit to time interval: use the **Start time** and **End time** boxes to select starting time and ending time.
- Limit to observed events: the analysis will be made from the first observed event to the last observed event.

The last dialog window allows you to subtract the duration of one or more behaviors from the total duration of the observation. This can be useful, for example, if you have defined an "out-of-sight" code in your ethogram:



For each subject and behavior, the results include the **total number of occurrences**, the **total duration** (for behaviors defined as state events), the **mean duration** (for behaviors defined as state events), the **standard deviation of duration**, the **mean inter-event interval**, the **standard deviation of the inter-event interval**, and the **percentage of the total observation duration**. All durations are expressed in seconds (s).

Time budget

Selected observations

0001_a
0001_b
0002
0003
0004
0005

Total observation length: 86632.554

	Subject	Behavior	Modifiers	total number of occurrence	Total duration (s)	Duration mean (s)	Duration std dev	inter-event intervals mean (s)	inter-event intervals std dev	% of total length
30	Nina	Locomotion		2458	5725.161	2.329	15.832	12.350	19.784	6.6
31	Nina	Swim		165	921.317	5.584	7.632	14.566	22.372	1.1
32	Nina	Eat		242	3232.485	13.357	22.646	8.847	17.604	3.7
33	Nina	Defecate		77	485.087	6.300	1.880	8.778	12.055	0.6
34	Nina	Drink		21	71.319	3.396	2.897	4.176	6.215	0.1
35	Nina	Rest		484	2645.019	5.465	7.789	12.899	16.952	3.1
36	Nina	Urinate		91	528.810	5.811	2.397	4.485	6.334	0.6
37	Nina	Vomit		0	0.000	NA	NA	NA	NA	0.0
38	Nina	Breed		27	2367.218	87.675	174.943	7.380	2.951	2.7
39	Nina	Play on the ground		111	1808.718	16.295	19.517	3.882	6.464	2.1
40	Nina	Play in the water		42	407.223	9.696	9.037	5.925	5.593	0.5
41	Nina	Tear		11	79.060	7.187	5.410	29.728	38.196	0.1

Save results Close

Results of the time budget analysis

The time budget results can be saved in various formats for further analysis:

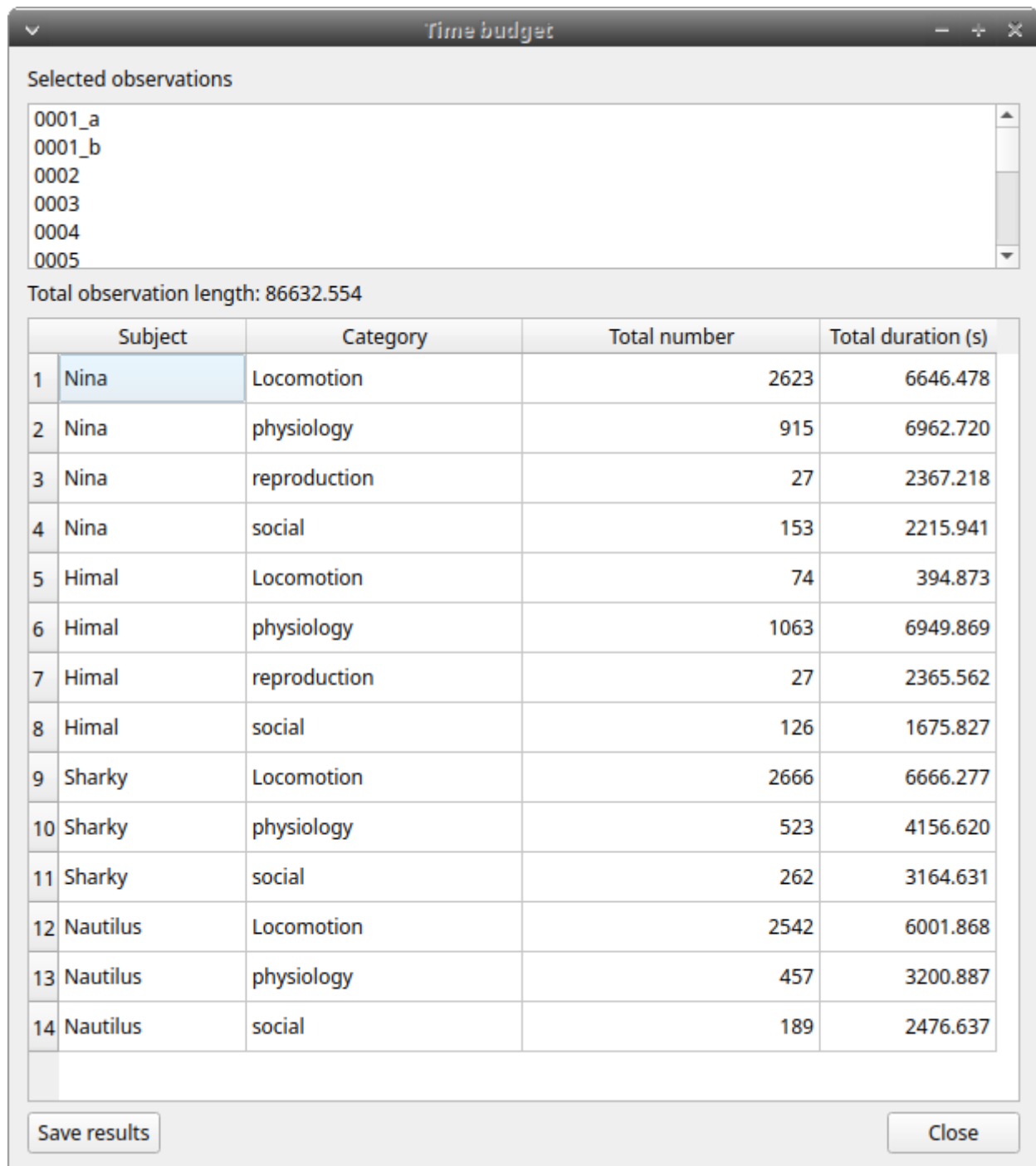
- Plain text in tabular format
 - **Tab Separated Values** (TSV)
 - **Comma Separated Values** (CSV)
 - **Hyper Text Markup language** (HTML)
- Spreadsheet files
 - **OpenDocument** (ODS)
 - **Microsoft Excel** (XLSX, XLS)
- **Pandas dataframe** (to be loaded in Python with the [pickle module](#))
- **R dataframe** (to be loaded in R with [readRDS function](#))

Important

If a STATE behavior has an odd number of coded events, BORIS will report **UNPAIRED** instead of results.

2.11.2 Time budget by behavioral category

The **Time budget by behavioral category** is similar to the **Time budget analysis** except that the behaviors are grouped into **behavioral categories**.



Results of a time budget by behavioral category analysis

2.11.3 Synthetic time budget

The synthetic time budget is similar to the time budget analysis, but with fewer parameters and a different organization of results. Results from all selected observations are organized in columns on a single page. Two parameters are currently provided: **number of occurrences** and **total duration** (for behaviors defined as state events).

	A	B	C	D	E	F	G	H	I	J	
1			Nina	Nina	Nina	Nina	Nina	Nina	Nina	Nina	Nina
2			Alert	Alert	Allogroom	Allogroom	Breed	Breed	Carry objects	Carry objects	Chas
3		Total length (s)	Total duration	Number of occurrences	Total duration	Number of occurrences	Total duration	Number of occurrences	Total duration	Number of occurrences	Total
40	0037	32.160									
41	0038	86.880	1.675	1							
42	0039	96.960	13.471	2							
43	0040	335.520	86.615	12	15.84	2					
44	0041	130.560	9.069	2	5.85	1					
45	0042	36.960	24.298	3							
46	0043	107.040	42.928	5							
47	0044	109.920	11.769	4	12.471	2					
48	0045	102.240	72.857	6							
49	0046	84.960	44.459	6							
50	0047	77.280	68.929	2							
51	0048	23.520	3.1	1	10.797	1					
52	0049	34.080	9.897	2	8.431	1					
53	0050	23.040									
54	0051	44.160	1.079	1							
55	0052	18.240									
56	0053	39.840									
57	0054	35.040									
58	0055	256.800	61.321	15	19.73	1					
59	0056	26.400	15.241	5							
60	0057	45.120	6.887	3	10.984	2					
61	0058	45.120	2.647	1							
62	0059	41.760									
63	0060	292.800	1.543	2							
64	0061	25.920									

All duration times are expressed in seconds (s).

The time budget results can be saved in various formats for further analysis:

- Plain text in tabular format
 - **Tab Separated Values (TSV)**
 - **Comma Separated Values (CSV)**
 - **Hyper Text Markup language (HTML)**
- Spreadsheet files
 - **OpenDocument (ODS)**
 - **Microsoft Excel (XLSX, XLS)**

2.11.4 Synthetic time budget with time bin

The **synthetic time budget with time bin** is similar to the **Synthetic time budget**, but the results are divided into time bins.

Analysis > Synthetic time budget with time bin

Choose a time bin size (in seconds)

Select subjects and behaviors

Subjects

Select all Unselect all Reverse selection

- ☒ Nina
- ☒ Himal
- ☒ Sharky
- ☒ Nautilus

Behaviors

Select all Unselect all Reverse selection

Locomotion

- ☒ Locomotion
- ☒ Swim

physiology

- ☐ Eat
- ☐ Defecate
- ☒ Drink
- ☒ Rest
- ☐ Urinate
- ☐ Vomit

reproduction

- ☐ Breed

☐ Include modifiers

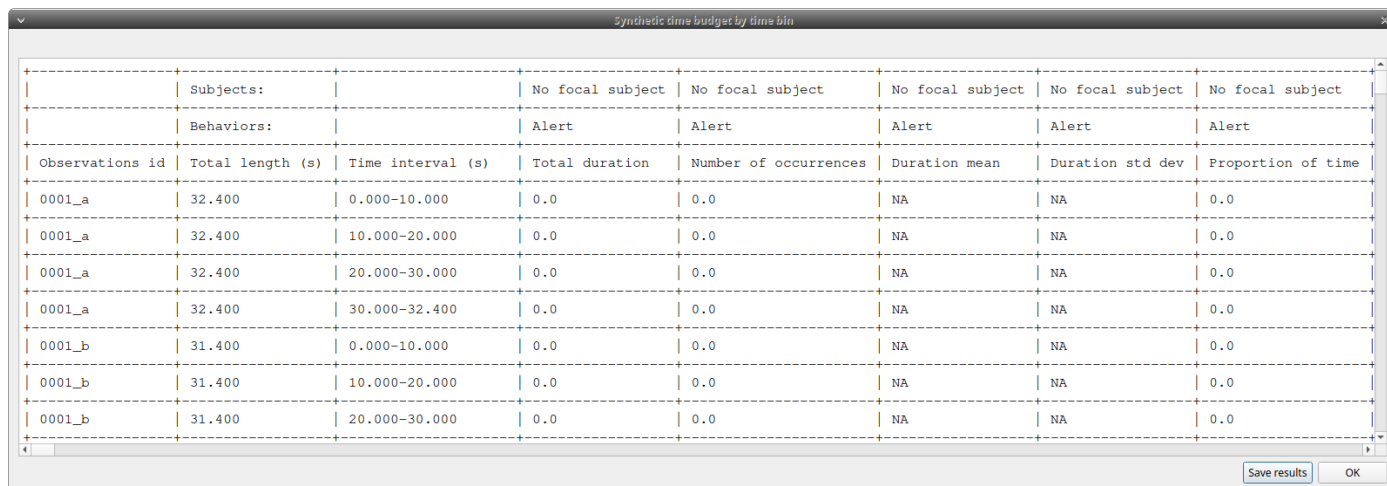
Time bin size (s) **< =**

Time interval

☒ Observed events ☐ User defined ☐ Media file(s) duration

Cancel OK

Time bin size of 10 seconds



Synthetic time budget by time bin							
Subjects:		No focal subject					
Behaviors:		Alert					
Observations id	Total length (s)	Time interval (s)	Total duration	Number of occurrences	Duration mean	Duration std dev	Proportion of time
0001_a	32.400	0.000-10.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	10.000-20.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	20.000-30.000	0.0	0.0	NA	NA	0.0
0001_a	32.400	30.000-32.400	0.0	0.0	NA	NA	0.0
0001_b	31.400	0.000-10.000	0.0	0.0	NA	NA	0.0
0001_b	31.400	10.000-20.000	0.0	0.0	NA	NA	0.0
0001_b	31.400	20.000-30.000	0.0	0.0	NA	NA	0.0

Results of a Synthetic time budget with time bin of 10 seconds

The **time budget with time bin** results can be saved in various formats for further analysis:

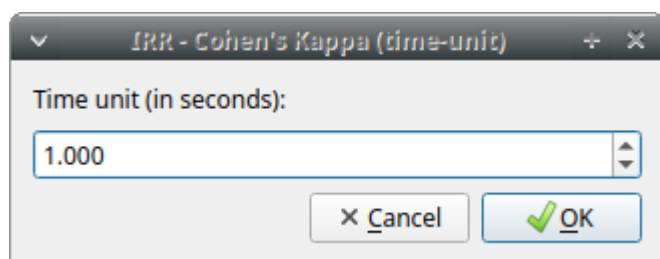
- Plain text in tabular format
 - **Tab Separated Values** (TSV)
 - **Comma Separated Values** (CSV)
 - **Hyper Text Markup language** (HTML)
- Spreadsheet files
 - **OpenDocument** (ODS)
 - **Microsoft Excel** (XLSX, XLS)

2.11.5 Inter-rater reliability

The Cohen's kappa coefficient can be calculated (**Analysis > Inter-rater reliability > Cohen's kappa**).

[Cohen's kappa on Wikipedia](#)

After selecting 2 observations and a time window (in seconds) for the analysis (the default value is 10 seconds), Cohen's kappa will be displayed in the results window.



IRR - Cohen's Kappa (time-unit)

Time unit (in seconds):

1.000

Cancel OK

Implementation of IRR Cohen's Kappa in BORIS

If a time window of n seconds is set, the 2 selected observations will be checked every n seconds for agreement or disagreement, from the first event to the last event of the 2 observations. In the case of a point event, the presence of a corresponding event in the other observation will be verified using a time window of n seconds centered on the point event.

An IRR Cohen's Kappa analysis is available in the GSEQ program (<http://www2.gsu.edu/~psyrab/gseq>). For this the coded events can be exported as aggregated events in SDIS format. See [export aggregated events](#).

2.11.6 Similarities

Needleman-Wunsch similarity

2.11.7 Co-occurrence

This function allows you to determine the co-occurrence of 2 behaviors.

2.11.8 Advanced event filtering

This function allows filtering events from one or more observations by selecting subjects, behaviors and logical operators.

To use the filter, select a subject, select a behavior and click on the button with the green arrow on the side of the behaviors list. The tuple subject/behavior will be added in the **filter text edit**. A complex filter query can be constructed by adding parenthesis and logical operator & (AND) or | (OR) for combining various subjects and behaviors.

Example of a summarized output showing the occurrences of Himal resting while Nautilus in alert:

The screenshot shows the 'Advanced event filtering' window. At the top, the filter text is '"Himal|Rest" & "Nautilus|Alert"'. Below this, there are three lists: 'Subjects' (Himal, Nautilus, Nina, Sharky), 'Behaviors' (Alert, Allogroom, Breed, Carry objects, Chase, Defecate, Dig, Drink, Eat, Interact with enrichment, Locomotion, Look for food), and 'Logical operators' (AND, OR). The 'Summary' tab is selected. Below the lists, a table shows the results of the filter query.

Results (197 observations)

	Observation id	Number of occurrences	Total duration	Mean	Std Dev
1	0015	1	3.351	3.351	NA
2	0019	2	8.581	4.29	2.461
3	0028	1	3.905	3.905	NA
4	0030	1	8.231	8.231	NA
5	0032	1	14.245	14.245	NA
6	0043	1	9.219	9.219	NA
7	0048	1	2.916	2.916	NA

Buttons at the bottom: Save results, Close.

Example of a detailed output showing the overlapping intervals while Himal rests and Nautilus is in alert:

Advanced event filtering

Filter

"Himal | Rest" & "Nautilus | Alert" Filter events Clear

☐ Summary ☒ Details

Subjects

- Himal
- Nautilus**
- Nina
- Sharky

Behaviors

- Alert**
- Allogroom
- Breed
- Carry objects
- Chase
- Defecate
- Dig
- Drink
- Eat
- Interact with enrichment
- Locomotion
- Look for food

Logical operators

- AND**
- OR

Results (146 events)

	Observation id	Comment	Start time	Stop time	Duration
1	0019		3.559	6.088	2.529
2	0019		14.907	17.457	2.550
3	0028		0.0	3.905	3.905
4	0048		18.333	18.627	0.294
5	0054		28.77	30.019	1.249
6	0138		174.796	178.699	3.903
7	0138		180.77	181.107	0.337

Save results Close

The same subject can be used multiple times in the query with OR or AND (in the case of non-exclusive behaviors):

Advanced event filtering

Filter

"Himal|Drink" | "Himal|Eat" Filter events Clear

☐ Summary ☒ Details

Subjects

- Himal
- Nautilus
- Nina
- Sharky

Behaviors

- Carry objects
- Chase
- Defecate
- Dig
- Drink
- Eat
- Interact with enrichment
- Locomotion
- Look for food
- Manipulate
- Play in the water

Logical operators

- AND
- OR

Results (188 events)

	Observation id	Comment	Start time	Stop time	Duration
1	0004		49.275	50.408	1.133
2	0009		5.579	10.48	4.901
3	0009		18.683	40.19	21.507
4	0013		24.324	30.869	6.545
5	0055		3.65	5.326	1.676
6	0055		21.362	29.117	7.755
7	0056		22.885	26.37	3.485

Save results Close

An unlimited number of conditions can be used:

Advanced event filtering

Filter

"Himal|Rest" & "Nautilus|Rest" & "Nina|Rest" & "Sharky|Rest"

Filter events

Clear

Summary

Details

Subjects

Himal

Nautilus

Nina

Sharky

Behaviors

Rest

Roll objects

Rub

Self-groom

Sleep

Sniff

Spot keeper

Stomp

Swim

Tear

Urinate

Yawn

Logical operators

AND

OR

Results (2 observations)

	Observation id	Number of occurrences	Total duration	Mean	Std Dev
1	0999	1	0.403	0.403	NA
2	1000	2	2.769	1.384	1.592

Save results

Close

Parentheses can be used to group logical conditions into blocks:

Advanced event filtering

Filter: "Himal|Alert" & ("Nautilus|Eat" | "Nina|Eat" & "Sharky|Eat") Filter events Clear

☐ Summary ☒ Details

Subjects

- Himal
- Nautilus
- Nina
- Sharky**

Behaviors

- Carry objects
- Chase
- Defecate
- Dig
- Drink
- Eat**
- Interact with enrichment
- Locomotion
- Look for food
- Manipulate
- Play in the water

Logical operators

- AND
- OR**

Results (30 events)

	Observation id	Comment	Start time	Stop time	Duration
1	0055		8.689	12.435	3.746
2	0055		19.778	21.361	1.583
3	0055		29.118	30.692	1.574
4	0055		66.079	69.591	3.512
5	0055		199.744	200.81	1.066
6	0239		149.63	152.719	3.089
7	0239		163.807	164.859	1.052

Save results Close

The results can be saved in a Tab Separated Values (TSV) file using the **Save results** button. Other formats may be added in the future.

2.11.9 Latency

The latency will analyze the time between one or more markers (arbitrary behaviors(s)) and other behaviors.

2.11.10 Plugins

BORIS can load Python and R analysis plugins. See [Analysis plugins](#) for plugin configuration, loading order, and plugin authoring details.

2.12 Analysis plugins

Starting from version 9, BORIS can load analysis plugins to process coded events. Plugins can be written in **Python** or **R**.

Important

The plugin interface is intended for custom analyses and may change between BORIS versions.

2.12.1 Plugin sources

BORIS loads plugins from two sources.

Official BORIS plugins

Official plugins are loaded first. BORIS looks in the directory configured in **Preferences > Analysis plugins > Official BORIS plugins**.

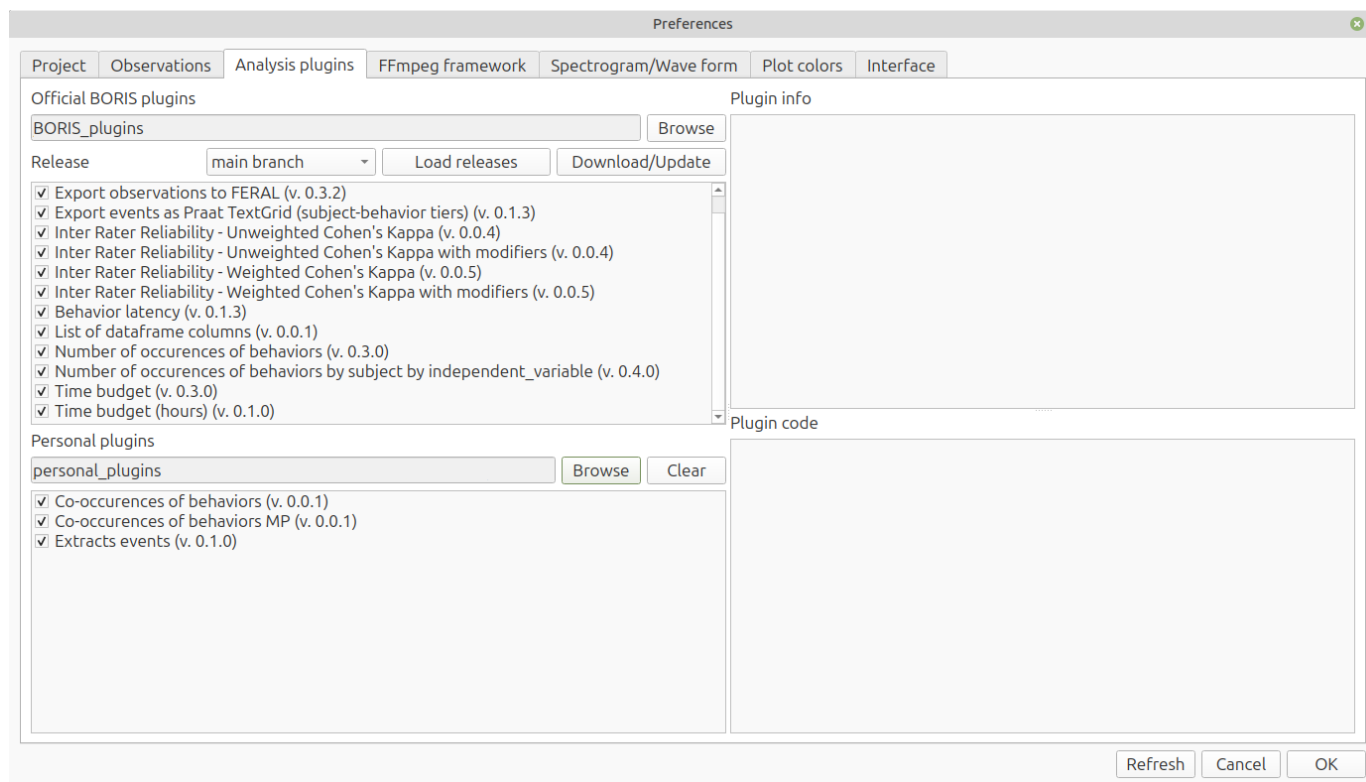
Personal plugins

Personal plugins are loaded from the directory selected in **Preferences > Analysis plugins > Personal plugins**. BORIS loads Python files (`.py`) and R files (`.R`) found directly in that directory. Files whose name starts with `_` are ignored.

If an official plugin and a personal plugin use the same plugin name, the official plugin is kept and the duplicate personal plugin is not loaded.

2.12.2 Managing plugins in Preferences

Open **File > Preferences > Analysis plugins**.



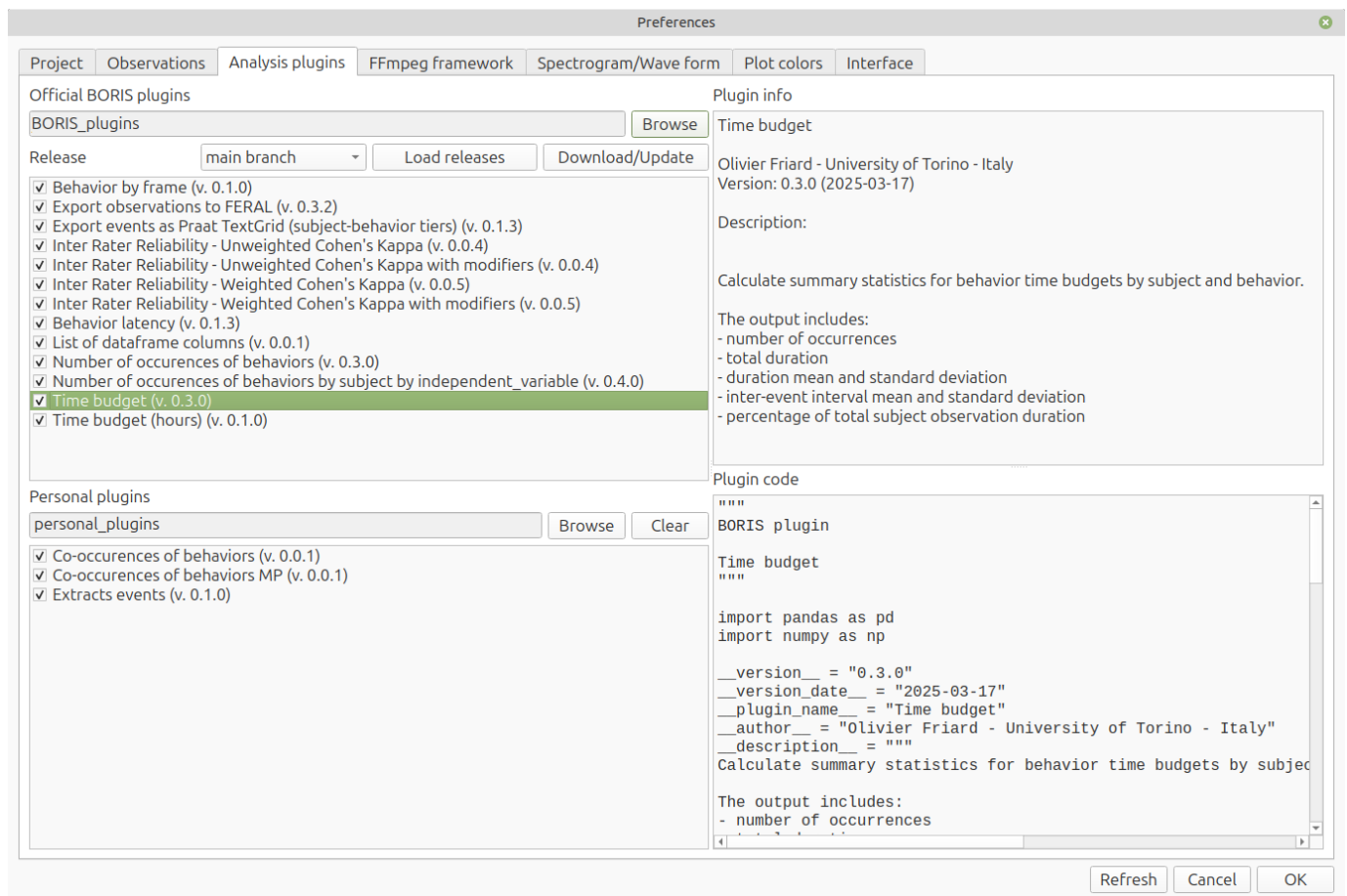
The **Official BORIS plugins** section contains:

- the directory currently used for the official plugin repository;
- a **Browse** button to select another official plugin repository directory;
- a **Release** drop-down list to select the release of the BORIS plugins
- a **Load releases** button to update the available releases
- a **Download/Update** button to download or update the official BORIS plugins directory from the BORIS plugins GitHub repository (https://github.com/olivierfriard/BORIS_plugins)[https://github.com/olivierfriard/BORIS_plugins];
- a list of official plugins with check boxes.

The **Personal plugins** section contains:

- the directory currently used for personal plugins;
- a **Browse** button to select a personal plugins directory;
- a **Clear** button to remove the saved personal plugin path without deleting the directory or its files;
- a list of personal plugins with check boxes.

Plugin lists show the plugin version next to the plugin name when a version is available. Click a plugin in the list to display its metadata and source code.



Changes are saved when you click **OK** in the Preferences window.

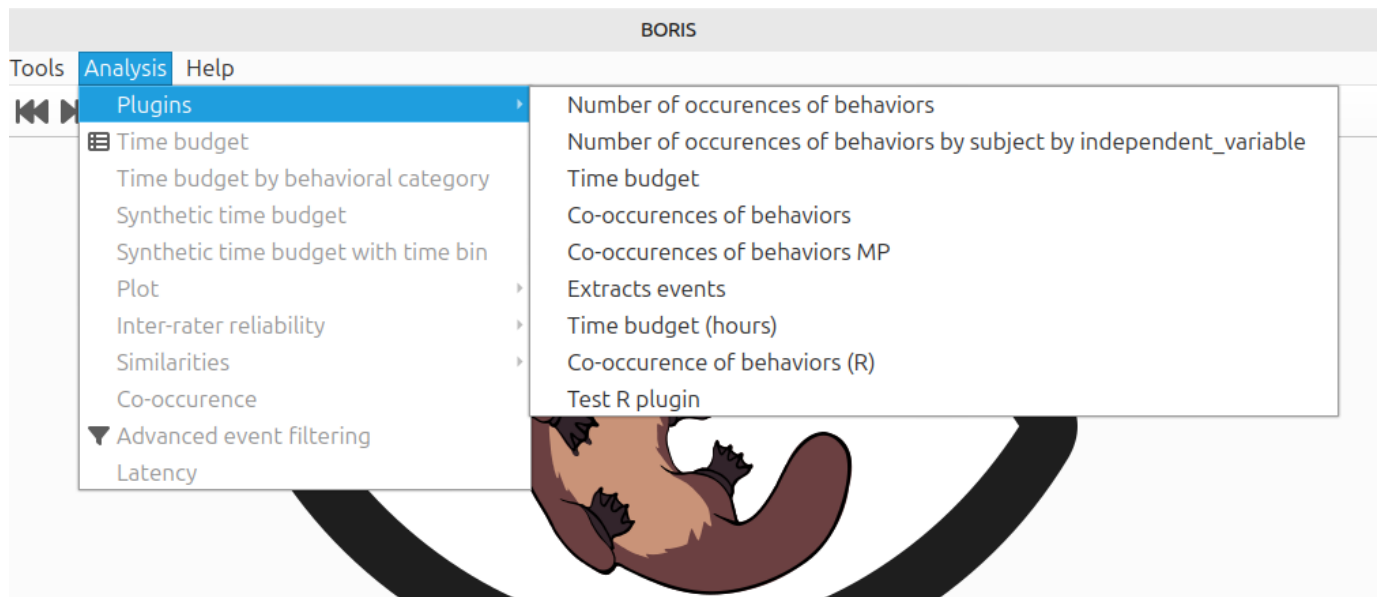
2.12.3 Downloading plugins from the official BORIS plugins GitHub repository

To select the BORIS plugin release you want to use, click **Load releases**, choose the release number from the **Release** drop-down list, and then click **Download/Update**.

To support reproducible analyses, all BORIS plugin releases will remain available.

2.12.4 Running a plugin

To run a plugin, select **Analysis > Plugins**, then choose the plugin.



BORIS lists official plugins first. Personal plugins are listed after a separator.

When a plugin is started, BORIS asks you to select:

- the observations to analyze;
- the subjects and behaviors to include;
- the time interval or observation interval to use.

The plugin result is displayed in a result window.

Number of occurrences of behaviors v. 0.1.0 (2024-11-14)			
Number of occurrences of behaviors v. 0.1.0 (2024-11-14)			
	Subject	Behavior	number of occurrences
0	Himal	Alert	2962
1	Himal	Allogroom	114
2	Himal	Breed	27
3	Himal	Carry objects	174
4	Himal	Chase	4
5	Himal	Defecate	77
6	Himal	Dig	176
7	Himal	Drink	33
8	Himal	Eat	155
9	Himal	Interact with enrichment	26
10	Himal	Locomotion	2329
11	Himal	Look for food	26
12	Himal	Manipulate	911
13	Himal	Play in the water	15

Save results Close

Plugin results can be saved from the result window in the available export formats.

2.12.5 Data passed to plugins

For Python plugins, BORIS inspects the arguments of the plugin `run` function and passes only the requested objects.

The following arguments are supported:

`df`

A Pandas DataFrame containing the selected and filtered events. The parameter must be annotated as `pd.DataFrame` or as a union that includes `pd.DataFrame`, for example `pd.DataFrame | None`.

`project`

A copy of the BORIS project dictionary containing only the selected observations. The parameter must be annotated as `dict` or as a union that includes `dict`, for example `dict | None`.

`parameters`

A dictionary with the analysis selections made by the user, including selected subjects, selected behaviors, and time filtering options. The parameter must be annotated as `dict` or as a union that includes `dict`, for example `dict | None`.

String annotations are also accepted. This means that plugins using `from __future__ import annotations` can use the same signatures.

Example:

```
import pandas as pd

def run(
```

```

df: pd.DataFrame | None = None,
project: dict | None = None,
parameters: dict | None = None,
) -> pd.DataFrame:
    if df is None:
        return pd.DataFrame()

    return df.groupby(["Subject", "Behavior"]).size().reset_index(name="count")

```

The DataFrame passed to the plugin includes columns such as:

```

Observation id
Observation date
Description
Observation type
Observation interval start
Observation interval stop
Source
Media duration (s)
FPS (frame/s)
independent variable #1
independent variable #2
...
Subject
Observation duration by subject by observation
Behavior
Behavioral category
(BEHAVIOR1, Modifiers set #1)
(BEHAVIOR1, Modifiers set #2)
(BEHAVIOR2, Modifiers set #1)
...
Behavior type
Start (s)
Stop (s)
Duration (s)
Comment start
Comment stop

```

The DataFrame also includes one column for each independent variable and one column for each behavior modifier set defined in the project.

2.12.6 Python plugin structure

A Python plugin is a `.py` file that defines a `run` function and plugin metadata.

The plugin metadata are:

```

__plugin_name__ = "PLUGIN NAME"
__version__ = "x.y.z"
__version_date__ = "YYYY-MM-DD"
__author__ = "AUTHOR - INSTITUTION"
__description__ = "Short plugin description"

```

You can optionally define the `__require_boris_version__` constant. This ensures that the plugin runs only with the allowed BORIS versions:

```
__require_boris_version__ = ">= 9.12"
```

The `run` function can return:

- a string;
- a Pandas DataFrame;
- a tuple of results.

Each result can optionally be returned as `(title, payload)`, where `payload` is a string or a Pandas DataFrame. For a single titled result, return it inside a tuple of results:

```
return (("Occurrences", result_df),)
```

For multiple results:

```

return (
    ("Summary", summary_text),
    ("Table", result_df),
)

```

Example of a Python plugin:

```
"""
BORIS plugin

Number of occurrences of behaviors.
"""

import pandas as pd

__version__ = "0.3.0"
__version_date__ = "2025-03-17"
__plugin_name__ = "Number of occurrences of behaviors"
__author__ = "Olivier Friard - University of Torino - Italy"
__description__ = "Count behavior occurrences by subject."

def run(df: pd.DataFrame) -> pd.DataFrame:
    return df.groupby(["Subject", "Behavior"]).size().reset_index(name="number of occurrences")
```

If you want a plugin to be usable outside BORIS, avoid importing BORIS internal modules such as `boris.config`. Use the `DataFrame` columns, the `project` dictionary, and the `parameters` dictionary passed to `run`.

2.12.7 R plugin structure

An R plugin is an `.R` file that defines a `run` function and plugin metadata. BORIS passes the selected and filtered events as an R data frame.

The plugin metadata are:

```
plugin_name <- "PLUGIN NAME"
version <- "x.y.z"
version_date <- "YYYY-MM-DD"
author <- "AUTHOR - INSTITUTION"
description <- "Short plugin description"
```

Example:

```
plugin_name <- "Number of occurrences of behaviors (R)"
version <- "0.0.1"
version_date <- "2026-05-29"
author <- "Olivier Friard - University of Torino - Italy"
description <- "Count behavior occurrences by subject."

run <- function(df) {
  aggregate(df$Behavior, by = list(Subject = df$Subject, Behavior = df$Behavior), FUN = length)
}
```

R plugins require the `rpy2` Python module.

2.13 Plot

2.13.1 Plot events

Recorded events can be plotted along a time axis.

Analysis > Plot > Plot events

Select the **observations** you want to plot. If more than one observation is selected, BORIS will ask you to choose a directory in which to save the plots.

Select observations for plotting events

1443 observations

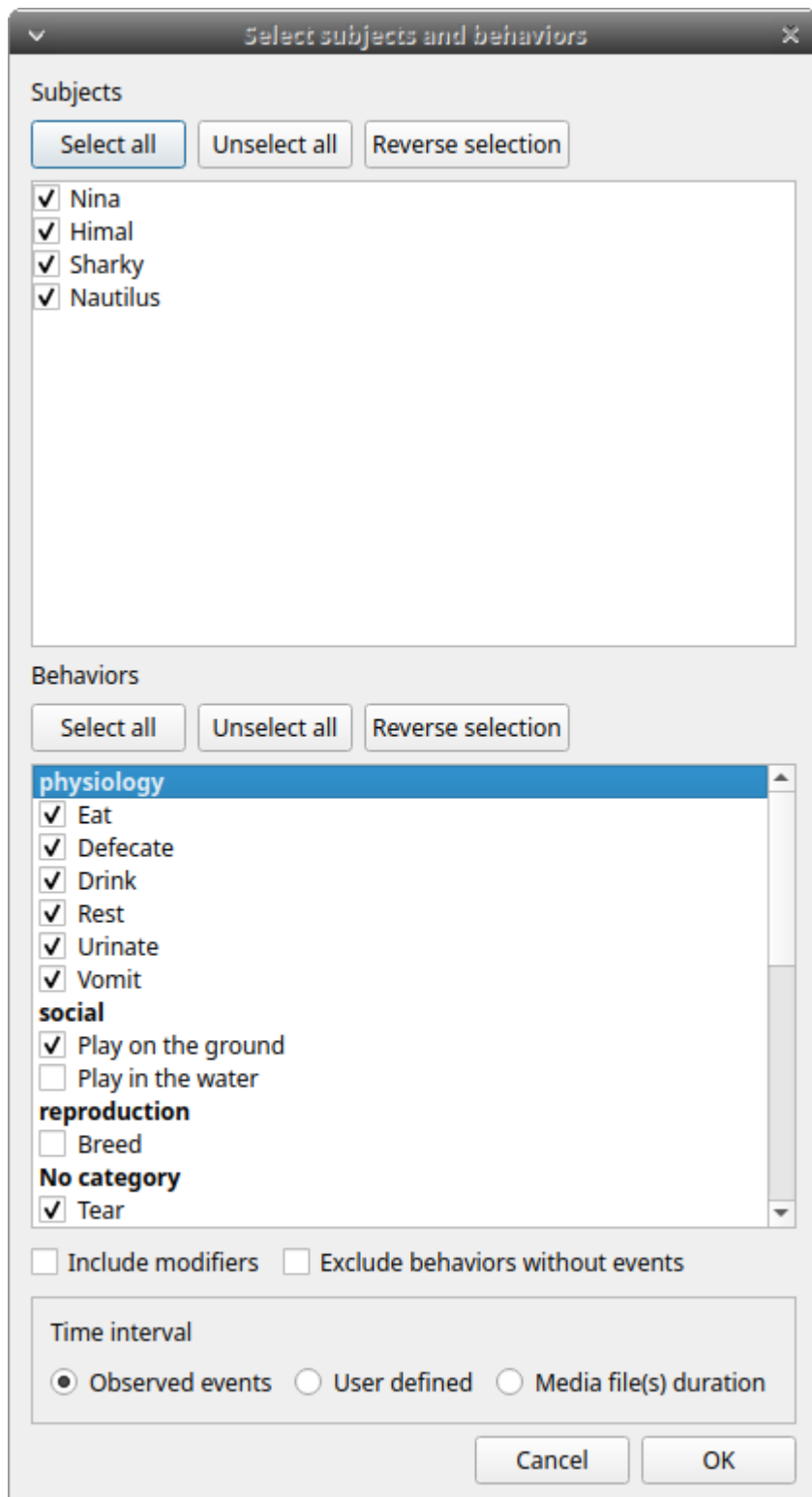
id contains

	id	date	description	subjects	observation duration	exhaustivity %	media	Location	Weather	temperature	Visitors
1	0001_a	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	100.0	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
2	0001_b	2016-05-17 00:00:31	Vegetation	Himal, Nautilus	32.400	98.5	#1: 20160517162539.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
3	0002	2016-05-17 00:00:24	Vegetation	Sharky, Himal, Nautilus	34.470	89.6	#1: 20160517162540.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
4	0003	2016-05-17 00:00:05	Vegetation	Sharky, Himal, Nautilus, Nina	22.500	87.1	#1: 20160517162641.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
5	0004	2016-05-17 00:00:59.000	Central trunks	Sharky, Himal, Nautilus, Nina	50.697	72.9	#1: video1.mp4	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
6	0005	2016-05-17 00:00:49	In the pool	Sharky, Nautilus	40.770	100.0	#1: 20160517163131.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
7	0006	2016-05-17 00:00:42	In the pool	Sharky, Himal, Nautilus	61.830	73.3	#1: 20160517163231.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	24	1046
8	0007	2016-05-17 00:00:13	In the pool	Sharky, Himal, Nina	124.986	41.3	#1: 20160517163347.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
9	0008	2016-05-17 00:00:17	In the pool	Sharky, Himal, Nautilus	64.800	85.1	#1: 20160517163743.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
10	0009	2016-05-17 00:00:10	In the pool	Sharky, Himal, Nina	46.277	84.6	#1: 20160517163927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	25	1046
11	0010	2016-05-17 00:00:57	Area near the glass window	Sharky, Himal, Nautilus	16.779	81.1	#1: 20160517164021.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
12	0011	2016-05-17 00:00:50	Area near the glass window	Sharky, Himal, Nautilus, Nina	25.101	49.1	#1: 20160517164106.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
13	0012	2016-05-17 00:00:45	Area near the glass window	Sharky, Nautilus, Nina	78.210	62.4	#1: 20160517164204.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	26	1046
14	0013	2016-05-17 00:00:25	Central trunks	Sharky, Himal, Nautilus, Nina	63.631	66.2	#1: 20160517164715.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
15	0014	2016-05-17 00:00:52	In the pool	Sharky, Himal, Nautilus, Nina	242.596	53.5	#1: 20160517164927.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046
16	0015	2016-05-17 00:00:18	Central trunks	Sharky, Himal, Nautilus, Nina	111.302	49.7	#1: 20160517165631.m2ts	44° 55' 59" N - 7° 25' 18" E	sun	16.0	1046

Select all Unselect all Cancel OK

Select the observations to plot

The **subjects** and **behaviors** to include in the plot can be selected in the following window:



You can choose to include or not the behavior modifiers (if any) and to exclude behaviors without coded events.

The time interval can be selected (see time budget).

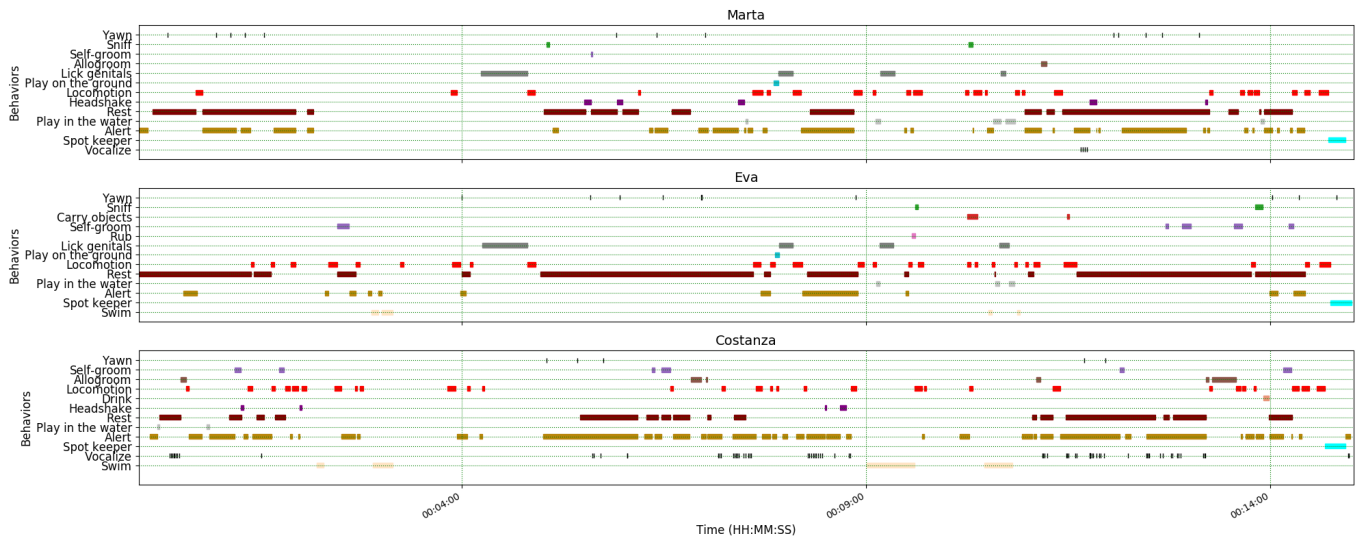
The plot can be exported in various bitmap formats (PNG, JPG, TIFF) or vector graphics formats (SVG, PDF, EPS, PS). The SVG format can be further edited with the [Inkscape vector graphics editor](#).

Important

If a STATE behavior has an odd number of coded events, you will see this error message: "The STATE behavior XXX is not paired"

This function creates one plot by subject on one figure.

Behavior colors can be customized. See [Plot colors](#).

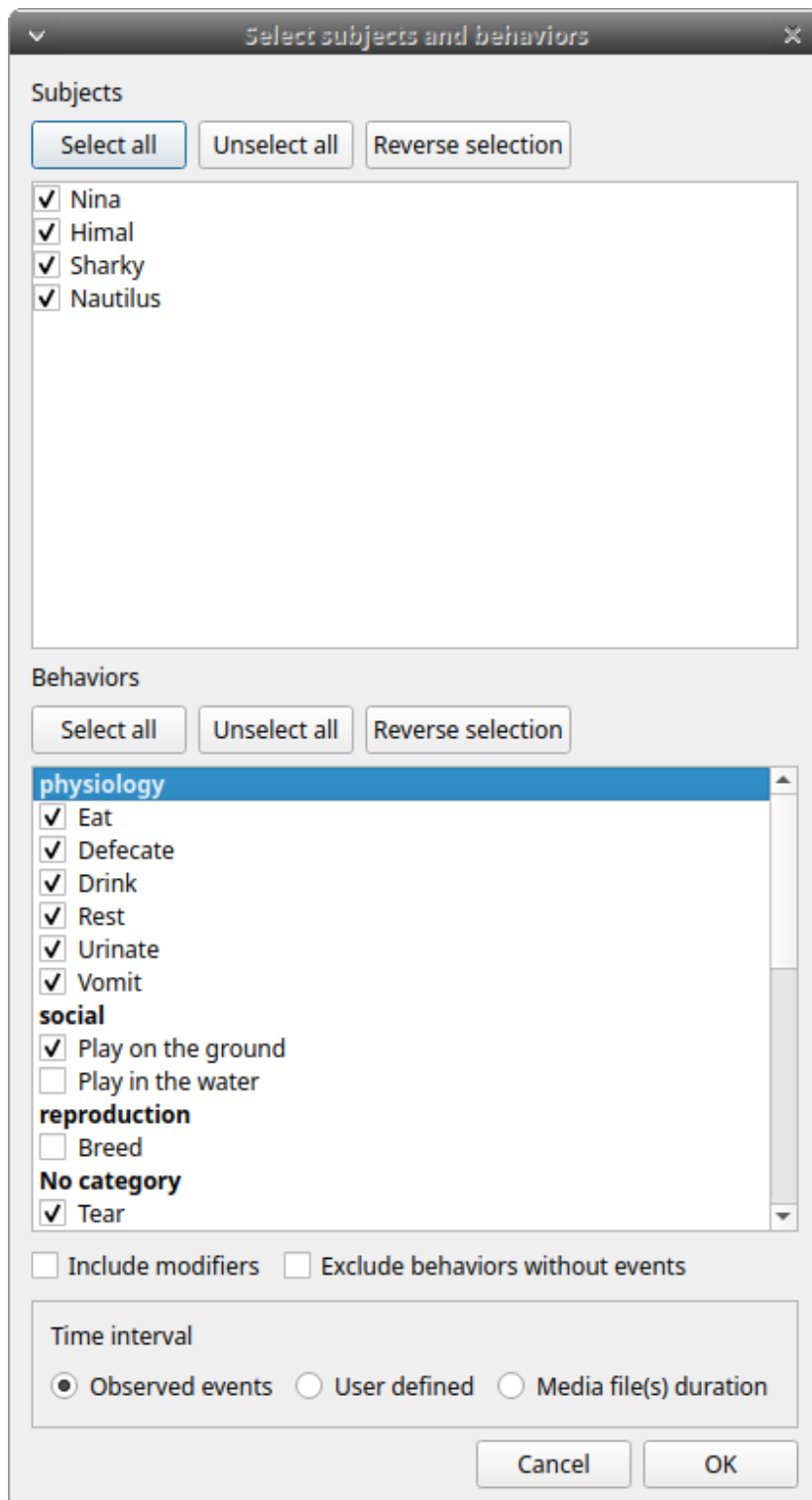


2.13.2 Plot time budget

The duration and number of occurrences can be plotted for each subject and behavior.

Analysis > Plot > Plot time budget

The subjects and behaviors to include in the plot can be selected in the following window:



Behavior modifiers cannot currently be included in the plot.

The time interval can be selected (see time budget).

The plot can be exported in various bitmap formats (PNG, JPG, TIFF) or vector graphics formats (SVG, PDF, EPS, PS). The SVG format can be further edited with the [Inkscape vector graphics editor](#).

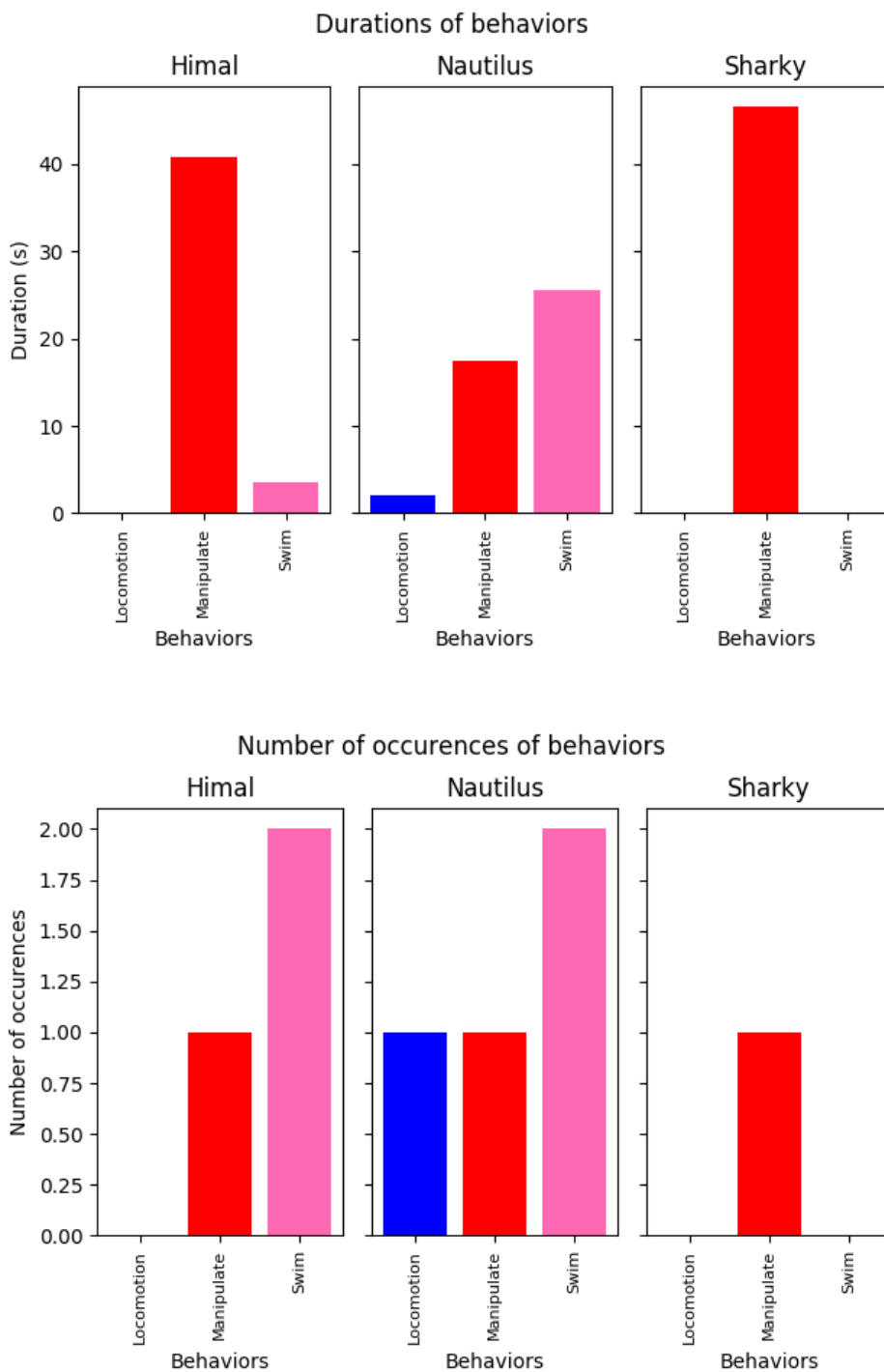
⚠ Important

If a STATE behavior has an odd number of coded events, you will see this error message: "The STATE behavior XXX is not paired"

This function creates 2 plots including all subjects for each observation:

- a plot of the behavior durations for the behaviors defined as STATE event.
- a plot of the number of occurrences for all the behaviors.

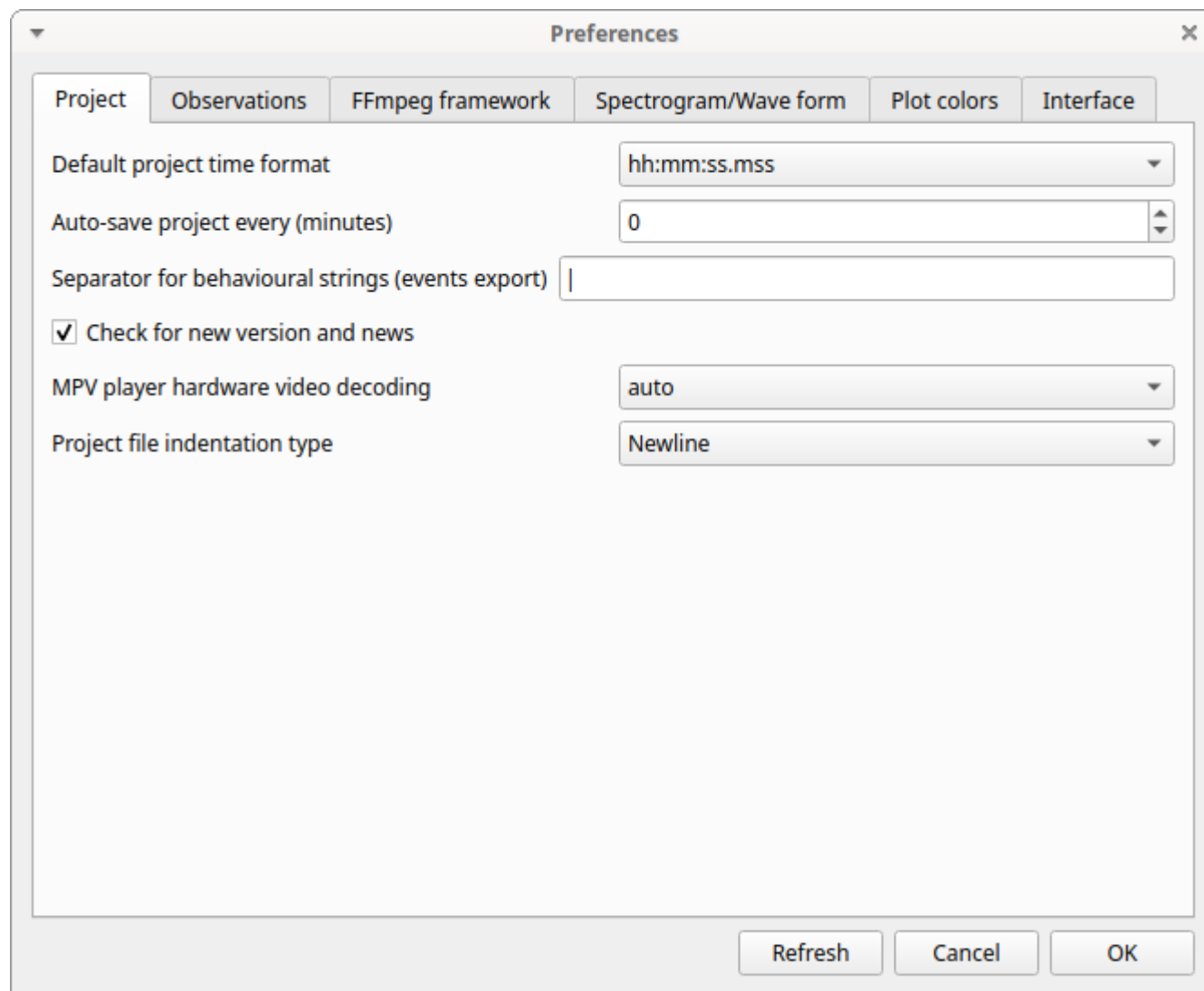
Behavior colors can be customized. See [Plot colors](#).



2.14 Preferences

You can customize BORIS in the Preferences window (**File > Preferences** or the **Preferences** button in the toolbar).

2.14.1 Project preferences



The screenshot shows the 'Preferences' dialog box with the 'Project' tab selected. The dialog has a title bar with a close button (X). Below the title bar are six tabs: 'Project', 'Observations', 'FFmpeg framework', 'Spectrogram/Wave form', 'Plot colors', and 'Interface'. The 'Project' tab is active, showing the following settings:

- Default project time format:** A dropdown menu showing 'hh:mm:ss.mss'.
- Auto-save project every (minutes):** A numeric input field showing '0'.
- Separator for behavioural strings (events export):** A text input field showing a single vertical bar '|'.
- Check for new version and news:** A checked checkbox.
- MPV player hardware video decoding:** A dropdown menu showing 'auto'.
- Project file indentation type:** A dropdown menu showing 'Newline'.

At the bottom right of the dialog are three buttons: 'Refresh', 'Cancel', and 'OK'.

Default project time format

This option allows you to choose the format used to display time in the project. Time is always stored internally in seconds with a precision of 3 decimal places.

Auto-save project every (minutes)

If set, BORIS will save your project automatically every n minutes. A value of 0 disables automatic backup. The project will be saved only if it has already been saved and an observation is open.

Separator for behavioural strings

Character or string used to separate behaviors when exporting events as behavioral strings. See also Behatrix.

Check for new version

Checks for a new version on the BORIS website every 15 days (internet access required).

MPV hardware video decoding

If you experience problems with the embedded MPV player, try changing this value.

Project file indentation type

The BORIS project file is encoded in JSON format. Choose the indentation style for the project file between:

- None
- Newline
- Tab
- 2 spaces
- 4 spaces

Refresh button

Resets the configuration to the default values. BORIS will then close.

2.14.2 Observations

Preferences

Project Observations FFmpeg framework Spectrogram/Wave form Plot colors Interface

Fast forward/backward value (seconds) 10

☐ Adapt the fast forward/backward jump to playback speed

Playback speed step value 0.1

Time offset for video/audio reposition (seconds) 0

☐ Play sound when a key is pressed

☐ Close the same current event independently of modifiers

Beep every (seconds) 0

☐ Display subtitles

☐ Tracking cursor above current event

☐ Alert if focal subject is not set

☒ Pause media before "Add event" command

Refresh Cancel OK

Fast forward/backward value (seconds)

This option allows the user to customize the amount of time for "jumping" forward or backward in media.

Adapt the fast for/backward jump to playback speed

The jump value will be adapted to the playback speed.

Playback speed step value

This value indicates how much the speed will increase or decrease after pressing the *change playback speed* buttons.

Time offset for media reposition (seconds)

This value indicates the time offset used when repositioning the media after double-clicking an event row in the *Events* table. For example, **-4** means that after a double-click the media will be repositioned 4 seconds before the recorded event.

Play sound when a key is pressed

Plays a sound after each keypress event.

Close the same current event independently of modifiers

Stops the current behavior regardless of modifiers.

Display subtitles

Displays or hides subtitles. If subtitles are stored in a separate file, the subtitle file must have the same base name as the video file and use the `.srt` extension.

Tracking cursor above current event

Positions the tracking cursor above the current event in the events list table.

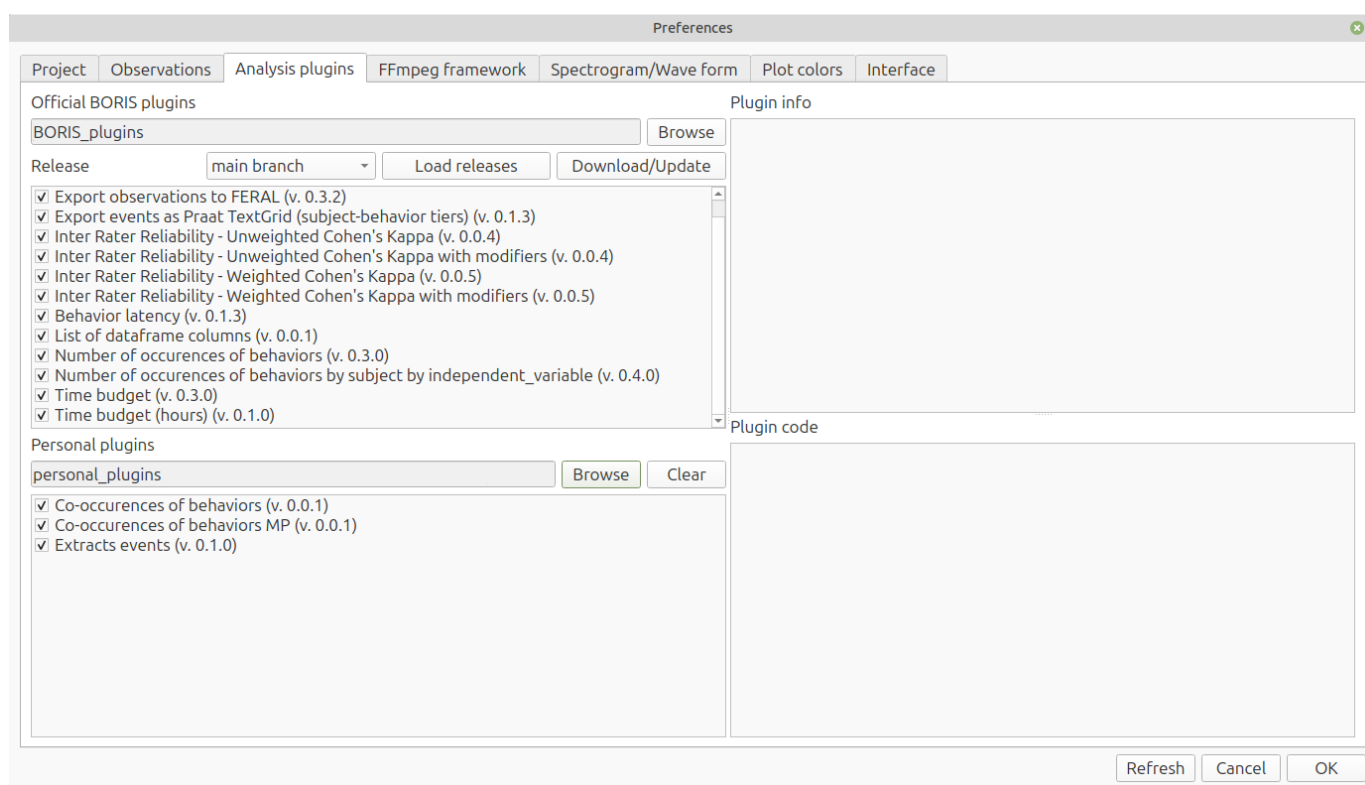
Alert if focal subject is not set

If this option is enabled, BORIS will show an alert box if no focal subject is selected.

Pause media before "Add event" command

Pauses the media before manually adding an event.

2.14.3 Analysis plugins



The **Analysis plugins** tab lets you configure official BORIS plugins and personal plugins. The plugin lists show the plugin version when available.

Official BORIS plugins

Select, download, update, enable, or disable official BORIS plugins.

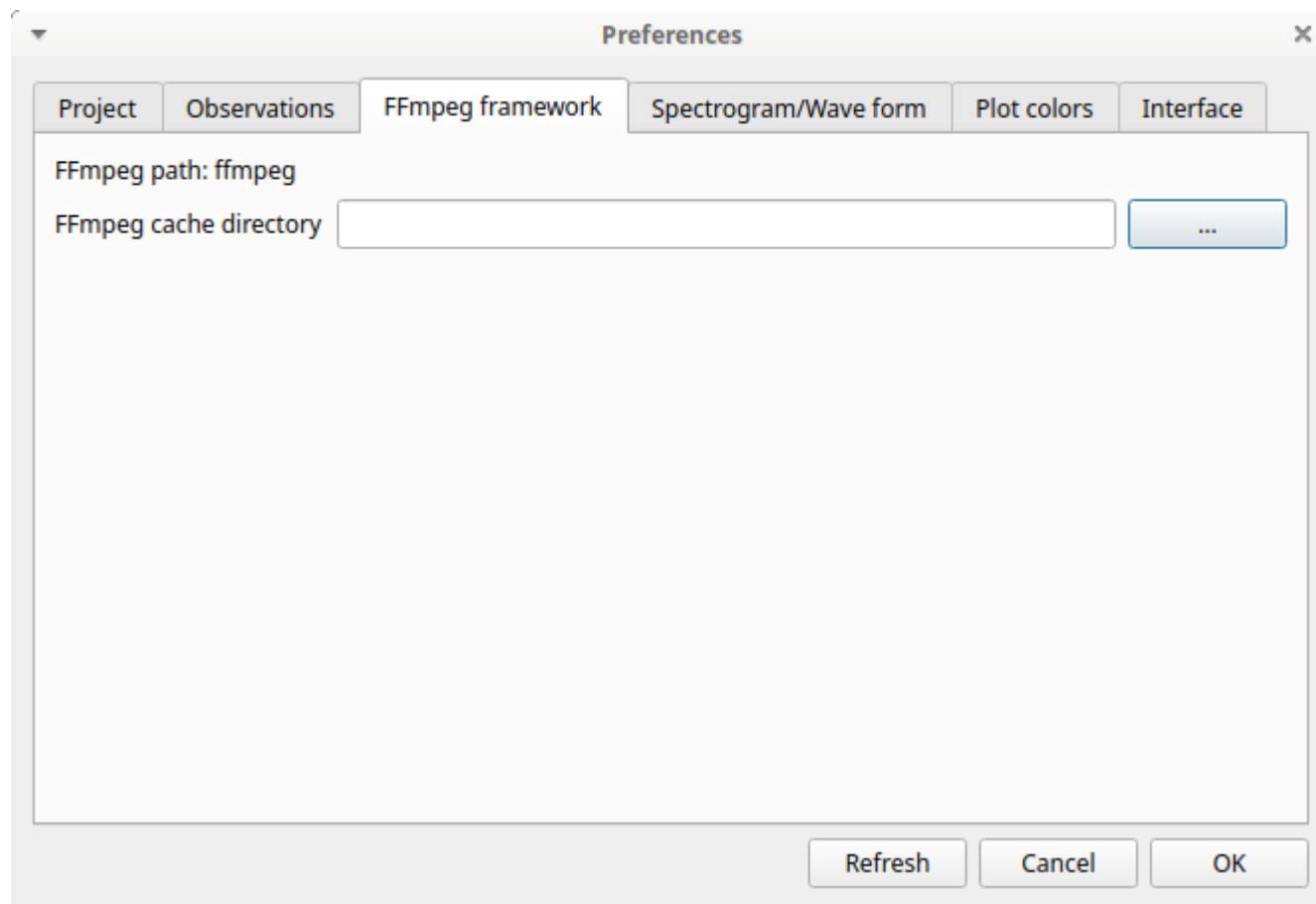
Personal plugins directory

Select the directory that contains your personal plugins.

Clear

Remove the saved personal plugins path without deleting the directory or any plugin file.

See [Analysis plugins](#) for details about plugin loading, writing plugins, and running plugins.

2.14.4 FFmpeg framework

The path to the `ffmpeg` executable is displayed. The FFmpeg executable is included with BORIS for Windows. The FFmpeg framework is required to run BORIS.

FFmpeg cache directory

Indicates the directory that will be used as the image cache for frame-by-frame mode and spectrogram visualization. If you do not specify a path, BORIS will use the default temporary directory of your system.

2.14.5 Spectrogram / wave form

Preferences

Project Observations Analysis plugins FFmpeg framework **Spectrogram/Wave form** Plot colors Interface

Spectrogram

Reset to default values

Color map **viridis**

Default time interval (s) **10**

Window type **hanning**

NFFT **256**

noverlap **128**

☒ Use vmin/vmax

vmin **-100** dBFS

vmax **-20** dBFS

Refresh Cancel OK

Color map

Selects the color map used to display the generated spectrogram. See [Matplotlib colormaps](#) for details.

Default time interval

Selects the time interval, in seconds, used to display the spectrogram and waveform.

Window type

Selects the window type: Hanning, Hamming, or Blackmanharris.

NFFT size

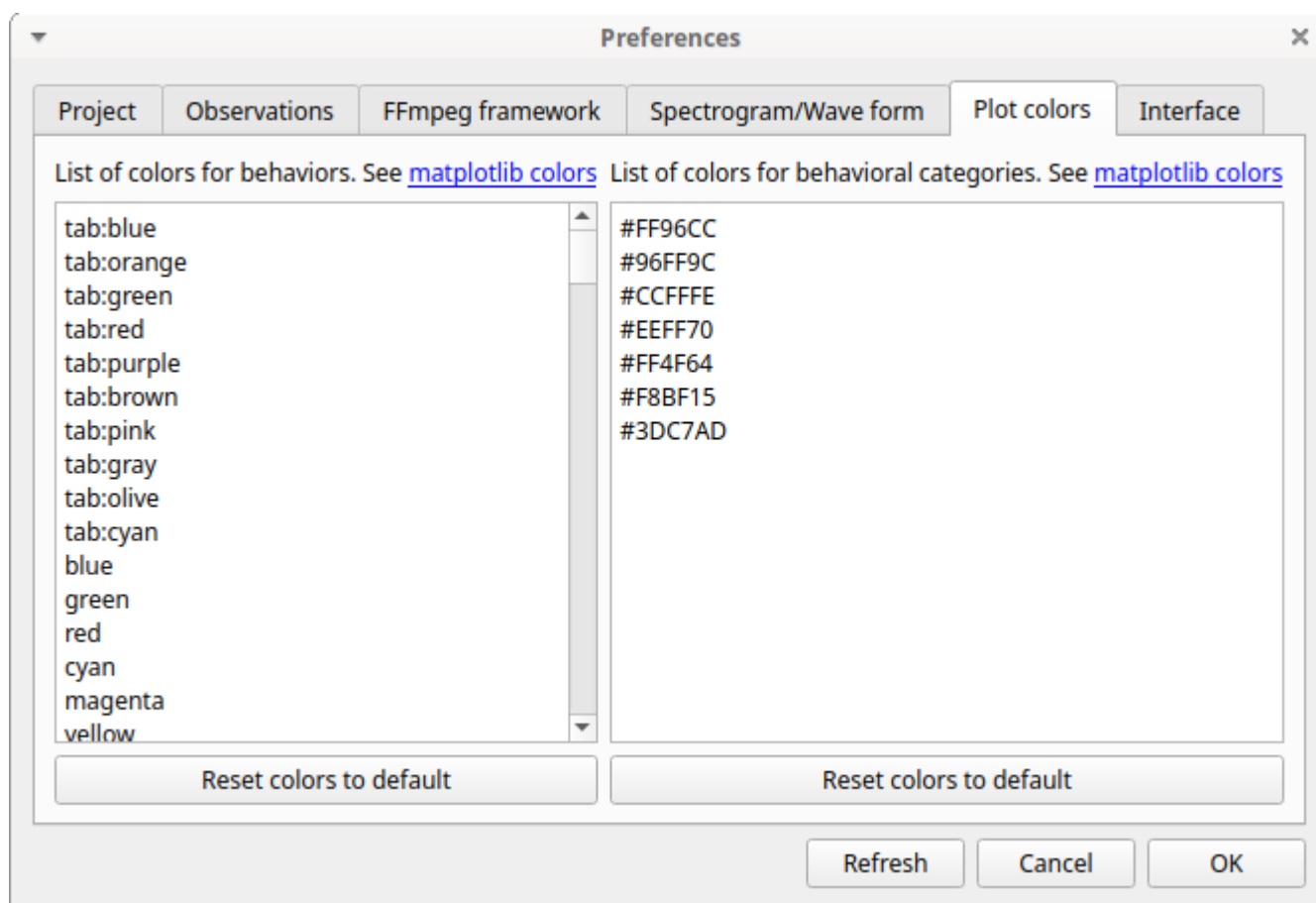
noverlap

Use vmin/vmax

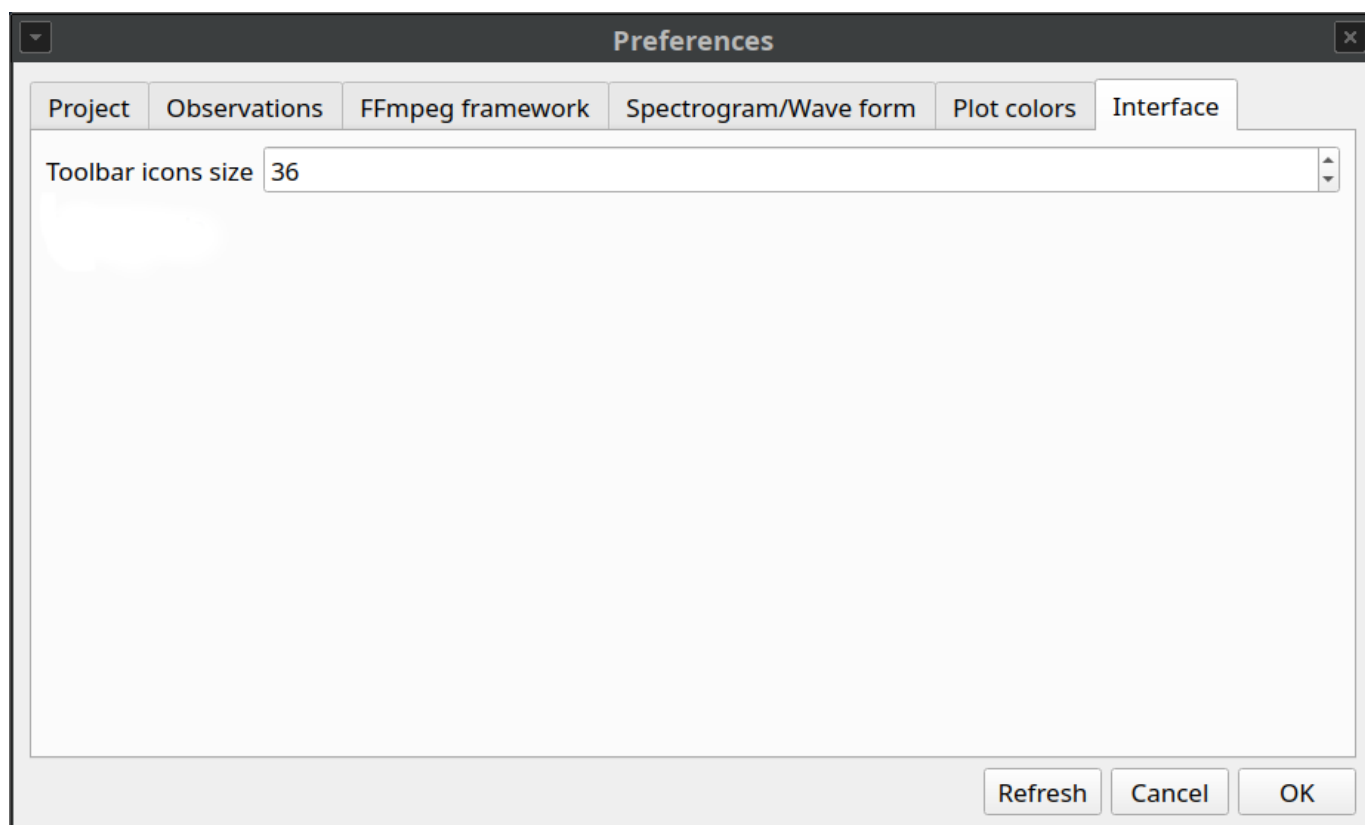
2.14.6 Plot colors

Behavior colors used in the plotting functions can be customized. The first color will be associated to the first behavior in your ethogram, the second color to the second behavior and so on. Various color formats can be used to specify a color: **named color** or **hex RGB** (like #0F0F0F). See https://matplotlib.org/api/colors_api.html and https://matplotlib.org/examples/color/named_colors.html for details.

The **reset colors to default** button will reload the default colors.



2.14.7 Interface



Toolbar icons size

Sets the size of the toolbar icons in pixels.

2.15 Various

2.15.1 Removing path of media files

With BORIS, you can choose to store the full path of media/data files in the project file (for example: `/home/user/Video/video_n1.mp4` or `c:\Users\user\Documents\video1.avi`).

If you want to move your project to a different computer, or if you want to move your media/data files, you may choose not to store the full path. To do this, you can add media/data files using relative paths (see the **Add media files** section). You can also remove the full path of your media/data files from all observations in the current project (**File > Remove path from media files**). Please note that this operation is irreversible. After removal, the full path of your media will be lost and cannot be recovered.

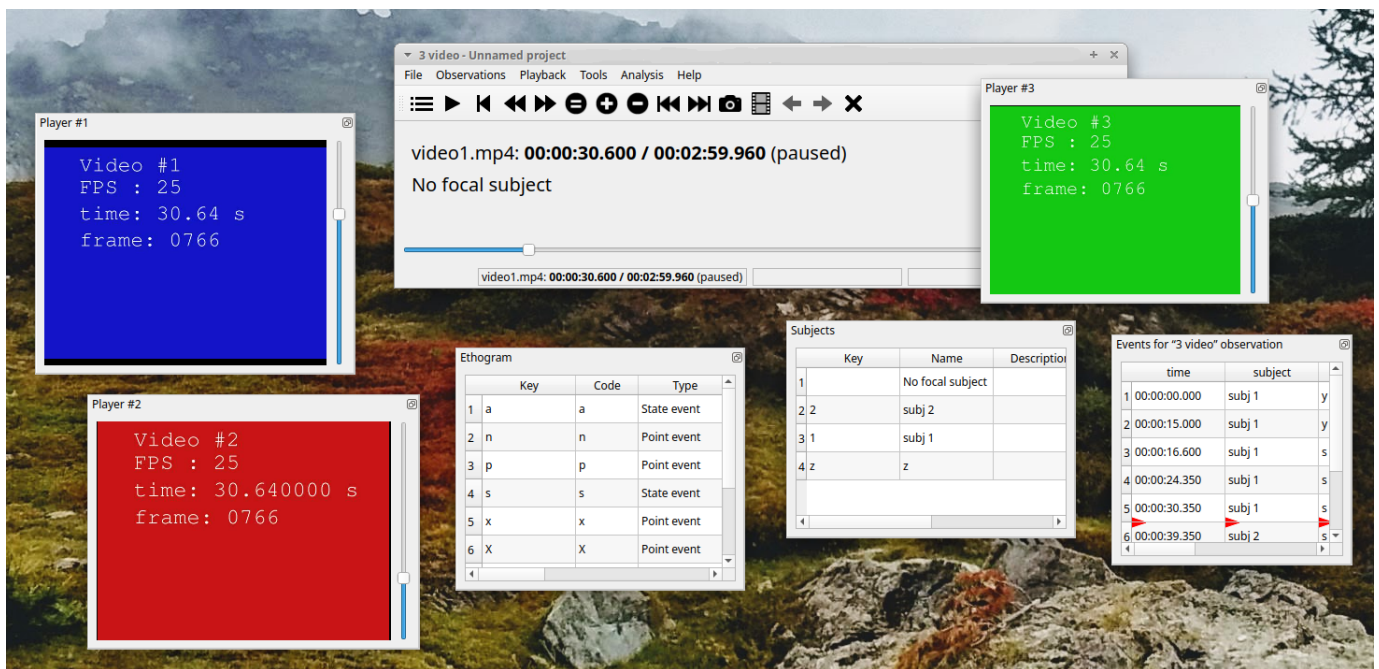
If you choose not to store the full path of media/data files, the path of the media/data files must contain the path of your BORIS project file.

Example: if your BORIS project file is saved in `/home/user/projects/test.project` your media/data files can be saved in the `/home/user/projects/videos` directory but **NOT** in the `/home/user/videos` directory.

2.15.2 Docking / undocking graphical elements

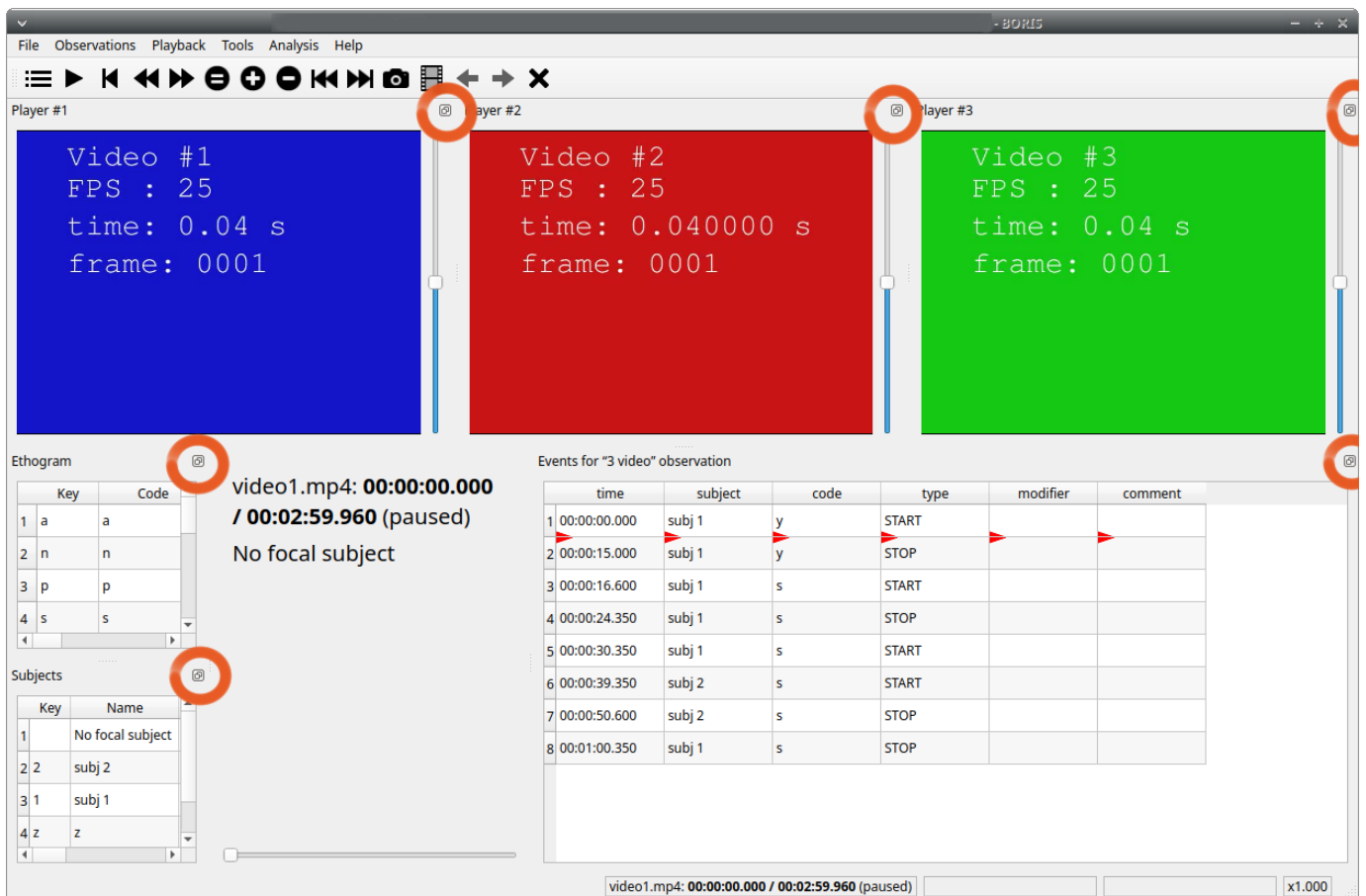
All elements, including all media players, can be undocked from the main window and positioned wherever you prefer (e.g. they can be on the same desktop over one or many screens).

The position of the various widgets is saved in the [configuration file](#) at the end of the work session.

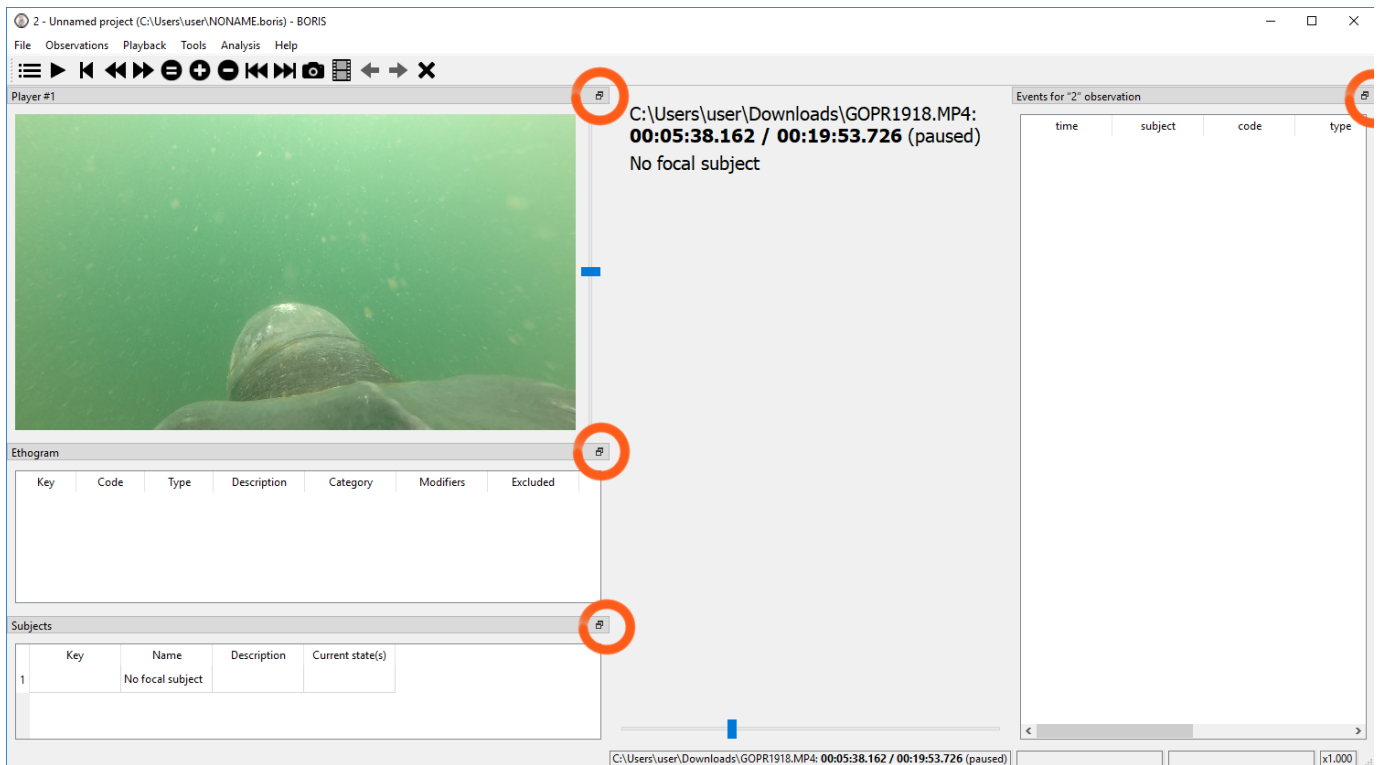


Clicking the icon in the top-right corner of the widget (on macOS, the icon is located in the top-left corner) undocks the widget so that it can be repositioned in another docking area or moved outside the main window. Double-clicking the top bar of the widget docks it back into the main window.

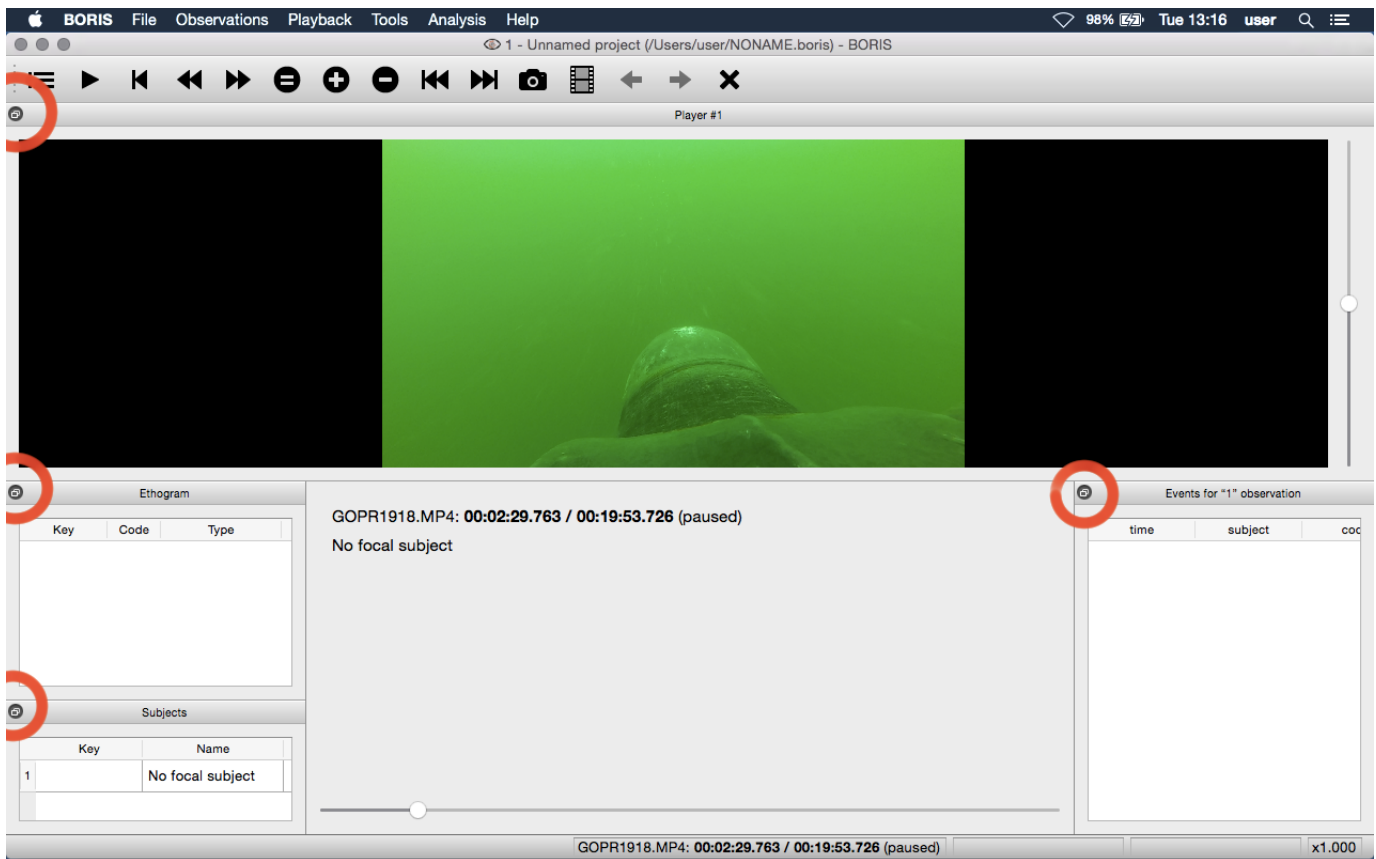
For Linux:



For Microsoft-Windows:



For MacOS:



If you prefer not to move dock widgets, you can lock them in the main window by enabling **Lock dockwidgets** (see **Tools > Lock dockwidgets**). All dock widgets will be docked in the main window and locked on it except the player dockwidgets.

2.15.3 Configuration files

BORIS saves the configuration (user preferences, window positions, widget positions) in a configuration file named **.boris**, stored in the home directory of the current user:

```
for Linux:
/home/USERNAME/.boris

for Microsoft-Windows:
C:\Users\USERNAME\.boris

for MacOS:
/Users/USERNAME/.boris
```

If you have trouble using BORIS, try closing the program, deleting this file and relaunch BORIS.

The **recent projects list** is saved in the **.boris_recent_projects** file in the home directory of the current user.

2.15.4 Lock the dockwidgets

The dockwidgets (except the player dockwidgets) can be locked on the main window (See **Tools > Lock dockwidgets**).

2.15.5 Valid keys for triggering behavior

BORIS distinguishes between lowercase and uppercase characters.

- keys from a to z
- keys from A to Z
- keys from 0 to 9
- function keys from F1 to F12
- à é è ù ì ç
- !"£\$%&/'()=?^[]{}@|§°#

3. Community

3.1 Acknowledgement

The authors would like to thank all users who report bugs and request new features for their valuable help.

3.2 Citing BORIS

If you use BORIS in a publication, please cite:

```
Friard, O. and Gamba, M.,  
BORIS: a free, versatile open-source event-logging software for video/audio coding and live observations.  
(2016) Methods in Ecology and Evolution, 7: 1325-1330.
```

DOI: [10.1111/2041-210X.12584](https://doi.org/10.1111/2041-210X.12584)

You can also send us a [nice postcard](#).

Please consider giving a star to the [BORIS GitHub repository](#).

3.3 Bug Reports and Feature Requests

Please report any bugs you find in the latest BORIS version through the GitHub repository.

Before reporting a bug, please:

- Check the Frequently Asked Questions (FAQ) section.
- Check whether the issue has already been reported on GitHub.
- Delete the configuration file and try again (see [configuration file](#)).

Remember to indicate:

- your operating system
- your operating system version
- the computer you are using (model, RAM, etc.)
- the BORIS version you are using
- information about the media file you are coding, if any. See [Tools > Media file information](#)

Provide all the information needed to reproduce the bug, such as a detailed procedure or a screen recording.

In case of a crash, please send the `boris_error.log` file generated in your home directory immediately after the crash and before restarting BORIS:

```
Linux:  
/home/YOUR_PROFILE_NAME/boris_error.log  
  
Microsoft-Windows:  
c:\Users\YOUR_PROFILE_NAME\boris_error.log
```

Note

If the bug you reported is fixed, please remember to close the issue.